



COPENHAGEN BUSINESS SCHOOL

MSc Programmes at Copenhagen Business School

Firm Frictions, Adjustment Dynamics, and Cross-Sectional Returns

Master's Thesis 2026

Authors: Jacob Korsgaard and Axel Ulveman

Supervisor: Niels Joachim Gormsen

Department: Department of Finance

Number of Pages: 111

Character Count: 270,261/273,000

Course-codes: CFSA45357 and COEC45382

May 15, 2026

Firm Frictions, Adjustment Dynamics, and Cross-Sectional Returns

Jacob Korsgaard[†] Axel Ulveman^{††}

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Abstract

We examine whether the Fama–French five-factor model partly captures the economic mechanisms underlying the q -theory of investment. To do so, we replicate the Fama–French factors and introduce an adjustment-cost dimension through a slow-minus-quick (SMQ) factor constructed from estimates of firms' speed of adjustment. We find that SMQ does not generate a clear unconditional risk premium, but earns a positive and statistically significant alpha relative to the five-factor model. In this regard, the pricing relevance of SMQ appears to be concentrated primarily among small firms, where slow-moving firms exhibit factor exposures consistent with weaker operating profitability. We interpret this as suggesting that small firms are more likely to be early on their saddle path, where the ability to convert growth opportunities into realized operating profitability is especially important. By contrast, among larger firms, the speed of adjustment appears to play a weaker role, potentially because existing productive capital already supports stable operating profitability. Overall, adjustment costs contain information not fully captured by the five-factor model and appear more informative as a conditional firm characteristic than as a standalone unconditional risk premium.¹

Keywords: Production-based asset pricing, q -theory of investment, Fama-French Five-Factor Model, adjustment costs factor, speed of adjustment

[†]Student ID: S153225. Graduate programme: MSc in Applied Economics and Finance.

^{††}Student ID: S153098. Graduate programme: MSc in Finance and Strategic Management.

¹To preserve the originality of the contributions developed herein, the ideas and results presented in this paper are based on Korsgaard and Ulveman (2026), Korsgaard (2026), and Ulveman (2026).

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1 Introduction

The Fama–French Five-Factor Model is one of the most influential empirical asset pricing models in the finance literature. Yet, while the model performs strongly in explaining the cross-section of stock returns, its broader economic interpretation remains only partly settled. Precisely because of this combination of empirical success and limited theoretical clarity, the model has remained central to debates in empirical asset pricing since Fama and French (1993, 2015) first introduced their three-factor model and later expanded it to the five-factor specification.

One potential perspective arises from the *q-theory of investment*. In this theoretical setup, firm value reflects the discounted value of future operating profits, which in turn depends on the firm’s optimal investment path, guided by valuation signals represented by average q . Since investment and profitability are central to both q -theory of investment and the five-factor model, and because book-to-market may be interpreted as a proxy for the inverse of average q , the model may capture mechanisms broadly consistent with this theoretical intuition. In fact, Fama and French (2015) themselves compare their model to an alternative q -factor model, although they do not directly interpret their model as a q -theoretical expression. Nevertheless, if this link is economically meaningful, one important dimension may still be missing: the *speed of adjustment*. To be specific, firms differ in how easily they can adjust their capital stock in response to economic shocks. Firms with a higher speed of adjustment can modify their capital stock relatively quickly, whereas firms with a lower speed of adjustment face greater frictions in adjusting capital and therefore respond more sluggishly to changing economic conditions. Intuitively, if the five-factor model captures return patterns that are partly consistent with q -theoretic mechanisms, then the speed of adjustment may provide a natural additional dimension through which to study cross-sectional return variation.

For this reason, we introduce the speed of adjustment as an additional dimension and examine whether it provides systematic insight into return patterns beyond the Fama–French five-factor model. Because the adjustment frictions examined in this paper build directly on the q -theory of investment, we argue that evidence of their pricing relevance may provide preliminary empirical support for interpreting the five-factor model through a q -theoretical lens. In doing so, the paper examines whether adjustment frictions help clarify the economic mechanisms potentially underlying the Fama–French factors.

1.1 Research Question

The notion that asset prices may be linked to the q -theory of investment is by no means new and has long been present in the production-based asset-pricing literature. This broad strand of research has explored how investment and firm value are connected to expected returns. However, while these links have received considerable attention, the role of the speed of adjustment remains less clearly understood. This is somewhat surprising, since adjustment costs play a central role in q -theory of investment and directly shape how firms respond to changing economic conditions. As such, the research question is:

To what extent are firms' adjustment frictions related to cross-sectional differences in expected stock returns, and do these frictions provide additional insight into the economic interpretation of the Fama–French Five-Factor Model?

The academic contribution of the paper lies in two dimensions. First, it examines whether the speed of adjustment captures an economically relevant firm characteristic associated with cross-sectional differences in expected returns. Adjustment costs are central because they determine how quickly firms can adjust their capital stock when economic conditions change. Firms that differ in their speed of adjustment may therefore also differ in how their values evolve across states of the economy. Since standard asset-pricing theory links expected returns to how payoffs vary across different states of the economy, this provides an economic motivation for examining whether adjustment frictions are systematically related to cross-sectional return variation. Second, the paper explores whether this perspective can contribute to the broader economic interpretation of the Fama–French Five-Factor model. More specifically, if the profitability, investment, and value factors are broadly consistent with the intuition embedded in q-theory of investment, then the speed of adjustment may help clarify their economic content. This follows because speed of adjustment plays a central role in shaping firms' investment behavior and, over time, their ability to convert growth opportunities into realized operating profitability. In this respect, the contribution is not only to test whether adjustment-related heterogeneity matters empirically, but also to assess whether it adds explanatory power beyond the standard Fama–French five factors.

Taken together, the aim of these contributions is not to claim a complete theoretical model, but rather to offer empirical evidence on whether firm adjustment dynamics provide a useful and economically coherent perspective on the expected-return patterns captured by the Fama–French Five-Factor Model. More broadly, the paper seeks to contribute to the gradual development of a stronger theoretical foundation for the model by examining whether adjustment frictions help explain part of its economic content.

1.2 Scope and Delimitations

Empirical asset-pricing results depend on the setting in which they are examined. For this reason, the sample period, asset universe, and geographical scope all shape the results and their final interpretation. This subsection, therefore, defines the boundaries of the analysis and clarifies the empirical settings.

Time horizon. The empirical analysis is limited to the period from July 1982 to December 2024 and is conducted using monthly return data from CRSP and annual accounting data from Compustat. This time horizon is chosen because it provides a sufficiently long sample for estimating return patterns while also extending the endpoint of the original empirical period studied by Fama and French (2015), allowing the analysis to evaluate the relevant return patterns using more recent data. However, the results should be interpreted in light of the specific macroeconomic conditions characterizing this period. Historically, the sample begins around the transition out of the Great Inflation

and largely covers the Great Moderation, a period characterized by lower macroeconomic volatility and more stable growth. It also includes the Great Financial Crisis and its aftermath, an extended period of historically low interest rates, the COVID-19 crisis, and the subsequent inflation and monetary tightening cycle (Wheelock, 2021). Hence, the documented return patterns reflect the behavior of adjustment-cost-related portfolios under these historical conditions and should not automatically be interpreted as stable relations that necessarily extend to other macroeconomic environments.

Asset universe. The asset universe is restricted to common equities listed on the NYSE, AMEX, and NASDAQ. The analysis therefore does not extend to other asset classes such as bonds, derivatives, commodities, currencies, real estate, or private markets. This delimitation is necessary because the thesis is concerned with equity returns and with a universe that is broadly comparable to the one used in the Fama and French (2015) paper. The focus on listed equities also ensures access to consistent stock return and accounting data through CRSP and Compustat, which are central to the construction of both the Fama–French factors and the adjustment-cost-related SMQ factor. The findings should therefore be interpreted as evidence from publicly listed U.S. equities, not as evidence that necessarily generalizes to international markets, private firms, or other asset classes.

Geographical scope. The geographical scope is likewise restricted to the United States. This choice is made for both practical and academic reasons. From a practical perspective, the U.S. market offers one of the broadest and most consistent sources of firm-level data for stock returns, accounting variables, and factor construction. From an academic perspective, the U.S. is also a natural starting point because much of the empirical asset-pricing literature, including the original Fama–French three-factor and five-factor models, is developed and tested in this setting (Fama & French, 1993, 2015). Restricting the analysis to a single environment also improves comparability across firms and reduces the risk of conflating adjustment frictions with cross-country differences in accounting standards, market structure, investor composition, regulation, and macro-financial conditions. However, this also means that the results may differ in markets with different financial systems or corporate investment behavior.

In sum, the analysis is conducted within a clearly defined empirical setting, focusing on U.S. listed equities over the 1982–2024 period. These choices align the analysis with the same asset universe used by Fama and French (2015) in their five-factor paper, while extending the sample period. The results should therefore be interpreted within this empirical context and should not be assumed to generalize mechanically to other markets, asset classes, or sample periods.

1.3 Philosophy of Science

This paper adopts a scientific realist ontology, supported by a positivist epistemology, to guide the knowledge-development process. This philosophical position is appropriate because the research question relies on theoretical constructs that are not directly observable, while the empirical analysis is based on observable return data, accounting

variables, and statistical tests.

Scientific realism, as an ontological position, holds that true statements about the world and how it functions may exist beyond the level of observation, and that scientific progress consists in moving closer to approximately true representations of underlying mechanisms (Chalmers, 2013b). In this respect, the thesis is grounded in theoretical constructs that operate beyond the observable level. In particular, the analysis draws on q-theory of investment and consumption-based asset-pricing theory, which together provide the theoretical intuition for linking firm behavior to expected returns. These theories rely on formal mathematical structures to describe how firms behave, how value is determined, how adjustment frictions affect investment dynamics, and how expected returns may arise across states of the world. While such constructs are not directly observable in the data, the thesis treats them as meaningful representations of underlying economic mechanisms. Through a deductive approach, these theories are taken as the conceptual point of departure from which hypotheses are derived and subjected to empirical scrutiny (Ghauri et al., 2020).

Positivism shapes the epistemological perspective and thereby the process through which knowledge is developed. It assumes that knowledge can be advanced through systematic observation, measurement, and empirical testing of the external world (Easterby-Smith et al., 2018; Saunders et al., 2009). More specifically, the thesis adopts a falsificationist stance in which theoretically motivated hypotheses are confronted with observable data and evaluated through statistical analysis. From this perspective, empirical testing does not prove a theory to be true in any final sense, but instead develops knowledge by exposing theoretical predictions to tests that can be rejected statistically (Chalmers, 2013a). This is particularly relevant in the present paper, where both q-theory of investment and consumption-based asset-pricing models rely on economic assumptions about firms, investors, and markets, but must ultimately be evaluated by their empirical implications. Through factor construction, test portfolios, and tests of risk premia, the thesis therefore examines whether the predicted relation between adjustment frictions and expected returns is consistent with the data.

Ultimately, this philosophical position leads to a deductive and quantitatively oriented research design. The thesis starts from formal economic theory, derives hypotheses about adjustment frictions and expected returns, and evaluates these hypotheses using empirical asset-pricing methods. In this way, the philosophy of science clarifies how the research question, theoretical foundation, and empirical methodology are connected.

1.4 Research Structure

Before presenting the analysis in detail, it is useful to outline the structure of the paper. The roadmap is illustrated in Figure 1. The theoretical link between adjustment frictions and expected returns is first developed and then positioned within the existing literature before being translated into testable hypotheses. These hypotheses are evaluated through the research design, which relies on standard empirical asset-pricing tests, factor construction, and portfolio sorts. Finally, the empirical findings are presented and

interpreted before the paper turns to its limitations and avenues for future research.

The paper begins by establishing the theoretical background in Section 2. This section introduces q-theory of investment, consumption-based asset pricing, and the Fama–French Five-Factor Model. The purpose is first to develop the core intuition behind q-theory, with particular emphasis on the adjustment cost parameter, χ , which guides the speed of adjustment in firms’ responses to exogenous shocks. The section then turns to consumption-based asset-pricing theory to clarify how risk premia arise and why they depend on the relation between asset payoffs and good and bad states of the economy. This provides the conceptual link between firm-level adjustment dynamics and expected returns, although the paper does not derive a full general-equilibrium model capable of delivering a definitive prediction. With these two theoretical perspectives in place, the section finally compares their implications to the mechanisms embedded in the Fama–French Five-Factor Model. This comparison motivates the central aim of the paper, namely to examine whether adjustment frictions provide additional insight into the economic content of the five-factor model.

The paper then turns to the existing literature in Section 3. The review first examines the empirical investment literature on q-theory of investment, where average q is generally related to investment but often explains investment behavior less strongly than the theory would suggest. It then moves to production-based asset pricing, where q-theoretic mechanisms have proven more successful in explaining expected returns, particularly in the cross-section. Finally, the review considers the development of empirical factor models, beginning with the limitations of the CAPM and continuing through the Fama–French three-factor model, the broader factor-zoo debate, and then finally the five-factor model. Although the five-factor model stands as one of the leading benchmarks in empirical asset pricing, its theoretical foundation remains only partly established. Together, this section positions the paper at the intersection of q-theory, production-based asset pricing, and empirical factor models.

Building on both the theoretical background and the literature review, Section 4 develops the hypotheses that guide the empirical analysis. The first hypothesis examines whether a tradable SMQ (slow-minus-quick) factor, constructed from differences in estimated adjustment costs, earns a positive risk premium. The second hypothesis asks whether adding this adjustment cost factor to the Fama–French Five-Factor Model improves empirical asset-pricing performance. The hypotheses therefore address two related but distinct questions: first, whether adjustment frictions are reflected in the cross-section of expected returns, and second, whether they provide incremental explanatory power beyond that captured by the five-factor model.

The empirical implementation is presented in Section 5, which outlines the research design. This section describes the data sources, sample construction, firm characteristics, and estimation of the adjustment cost parameter, χ . Particular attention is given to the rolling-window estimation procedure, since χ is not directly observed and must be estimated from the investment–q relation. The section then explains how the Fama–French factors are replicated, how the SMQ (slow-minus-quick) factor is constructed, and how

the χ -sorted decile portfolios, 5×5 portfolios, and $2 \times 4 \times 4$ portfolios are formed. Finally, it introduces the econometric methods used to evaluate the hypotheses, including mean excess returns, time-series regressions, t -statistics, GRS tests, average absolute alphas, and average adjusted R^2 .

With the research design in place, Section 6 turns to the empirical results. The analysis begins with the one-dimensional evidence from χ -sorted decile portfolios and the SMQ factor, which provides the first assessment of whether adjustment costs are associated with expected returns. It then examines the broader factor structure through performance statistics and factor spanning regressions. The section subsequently turns to multidimensional portfolio sorts, where adjustment costs are examined jointly with size, profitability, book-to-market, and investment. This makes it possible to assess whether χ becomes more informative when firms are compared within similar characteristic groups. The section concludes with robustness checks based on alternative χ estimates at the firm, industry, and major-group levels.

Following the empirical analysis, Section 7 discusses the main limitations of the paper. These limitations relate to the absence of a fully unified general-equilibrium model, the assumptions underlying the estimation of χ , measurement error in average q , the choice of estimation window, and the level of aggregation used to estimate adjustment costs. The section also addresses practical implementation concerns, including transaction costs, short-selling constraints, small-firm liquidity, and the instability of mean-variance optimization. Finally, it discusses the external validity, especially given the focus on U.S. listed equities and the given sample period for empirical asset pricing tests.

These limitations naturally point to several avenues for future research, outlined in Section 8. This section highlights the need for a formal theoretical model linking adjustment costs to the stochastic discount factor, alternative methods for estimating the adjustment cost parameter, χ , the role of intangible capital, and extensions to other empirical settings. These future avenues for research are important for further strengthening and validating the arguments presented herein.

Finally, Section 9 concludes the paper by returning to the research question and summarizing the main findings. The final conclusion emphasizes that adjustment frictions do not appear to generate a clear unconditional risk premium, but that they contain information not fully captured by the five-factor model. In particular, the evidence is strongest among small firms, where the ability to convert growth opportunities into realized operating profitability is likely to be most important. Based on these findings, the paper argues that adjustment costs provide the preliminary empirical support for the view that the five-factor model may partly reflect economic mechanisms associated with q -theory of investment, while also motivating future research on this link.

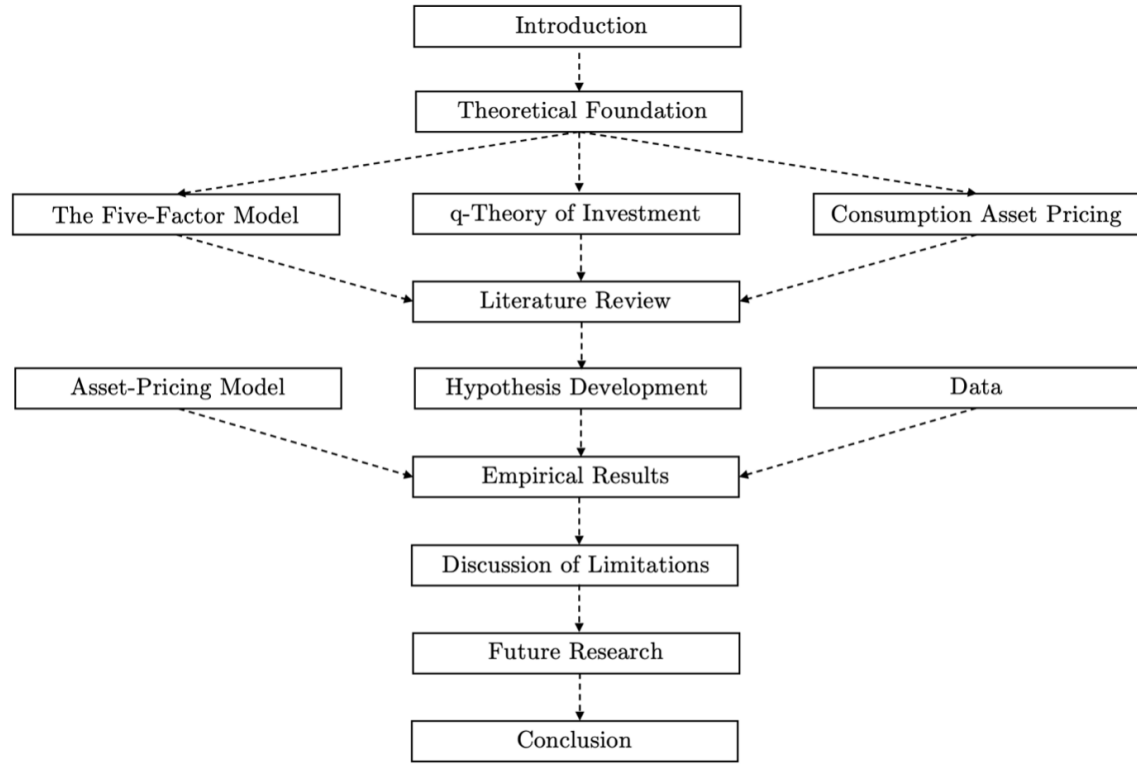


Figure 1: Diagram of the structure of the research paper
Source: Authors' own creation

2 Theoretical Background

The starting point of this paper arises from the possibility that the Fama–French Five-Factor Model may be interpreted through the lens of the q-theory of investment. The purpose of this section is therefore to outline the theoretical considerations that motivate this perspective. It begins from the production side, where firm investment is related to Tobin’s q and differences in firms’ responses to exogenous shocks are embedded in adjustment costs. The discussion then turns to the consumption side, where risk premia arise from the extent to which an asset’s payoffs co-move with good and bad states of the economy. Finally, these implications are compared to the five-factor model, whose factors may appear broadly consistent with the dynamics of q-theory. If so, heterogeneity in firms’ adjustment costs may help clarify their economic content and provide additional insight into the five-factor model.

2.1 q-Theory of Investment

The q-theory of investment begins from the idea that firms invest when the value of capital rises relative to its acquisition cost. This idea received its most influential empirical expression by Nobel Laureate James Tobin (1969), who first introduced the notion of average q , now known as Tobin’s q . That contribution helped launch a sequence of literature that united average q and convex adjustment costs within one common theoretical framework. This section outlines those foundations and highlights the central role of

adjustment costs in shaping firms' investment responses to exogenous economic shocks. To keep things transparent, our theory section follows the discrete-time setup of Obstfeld and Rogoff (1996a).

2.1.1 Value-Maximizing Representative Firm

The starting point of q-theory of investment is the behavior of the firm. On the production side of the economy, the firm is modeled as a representative firm maximizing the present value of all its future dividends distributed to shareholders:

$$V_t = \sum_{s=t}^{\infty} \left(\frac{1}{1+r} \right)^{s-t} [A_s F(K_s, L_s) - C(I_s, K_s) - I_s - L_s w_s], \quad \text{s.t.} \quad K_{s+1} = K_s + I_s \quad (1)$$

where V_t denotes firm value, r is the required rate of return, A_s captures firm productivity, $F(\cdot)$ is the production function, $C(\cdot)$ denotes the adjustment cost function, I_s denotes investment expenditure, K_s is the capital stock of the firm and L_s is total labor with w_s denoting the real wage. These components highlight three central elements to the q-theory of investment, namely the firm's productive capacity, the frictions associated with capital adjustment, and the dynamic link between current investment and future capital.

First, firm value ultimately depends on the ability to generate profits. This is captured by the production function, $F(\cdot)$, which describes the technological relationship between the inputs used in production and the quantity of goods or services the firm can produce. In this sense, it reflects the firm's productive capacity and thereby its potential to generate revenue. In economics, the production function is typically assumed to be increasing and concave in both capital and labor, reflecting that additional inputs raise output, but at a diminishing rate. Put simply, the first unit of capital or labor tends to contribute more to output than a later one.

Second, a defining feature of the q-theoretic literature is the presence of adjustment costs, represented by $C(\cdot)$. These costs reflect the frictions associated with changing the capital stock and need not be purely monetary, but may also arise through more indirect burdens, such as organizational strain, installation delays, or disruptions to ongoing operations. Expanding production may therefore be costly not only because new capital must be acquired, but also because the adjustment process itself absorbs resources. While adjustment costs may take different functional forms, the central assumption in q-theory is that they are convex.

Finally, because investment affects production only through its effect on the capital stock, it is necessary to consider the dynamic evolution of capital. This is captured by the law of motion for capital, $K_{s+1} = K_s + I_s$, which links current investment decisions to next period's capital stock. This constraint is intuitive: a firm that invests today does not benefit immediately from a higher capital stock, but only after some delay. For example, building a new production facility may take months before it becomes operational. More fundamentally, this intertemporal link is what makes the investment problem inherently

dynamic. However, to fully appreciate its implications, it is necessary to examine the exact nature of the adjustment cost function.

2.1.1.1 Adjustment Costs of the Firm

A central assumption in the neoclassical q-theory of investment is that firms face convex adjustment costs when adjusting their capital stock. Following this strand of literature, the adjustment costs are explicitly modeled as

$$C(I_s, K_s) = \chi \frac{I_s^2}{2K_s}, \quad \chi > 0 \quad (2)$$

where I_s is investment, K_s is the capital stock, and χ represents the adjustment cost parameter. Economically, χ occupies a central place because it governs the *speed* of capital adjustment: a high χ implies high adjustment costs, causing firms to adjust slowly over time, whereas a low χ implies lower adjustment costs and allows firms to adjust more quickly. The restriction $\chi > 0$ ensures that adjustment costs remain positive, since negative adjustment costs would imply that firms earn profits simply by investing. At the same time, despite its theoretical importance, the adjustment cost parameter remains inherently *unobservable*. This is precisely why χ is not only a central parameter in q-theory, but also one of its most widely discussed.

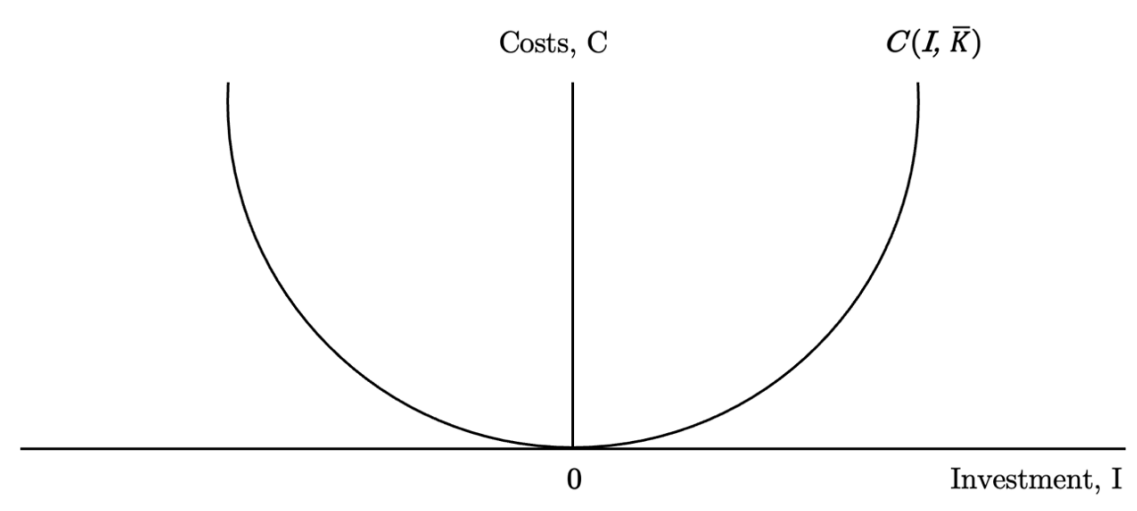


Figure 2: Adjustment costs as a function of gross investment when $K = \bar{K}$.

Source: Authors' own creation, inspired by Groth (2017).

Furthermore, the quadratic adjustment cost function captures an important feature of investment dynamics. As illustrated in Figure 2, holding the capital stock, K_s , fixed, adjustment costs rise with the absolute size of investment. Thus, both investment and disinvestment become more costly the further the firm moves away from its existing capital stock. The intuition is simple: small changes can usually be implemented with limited disruption, whereas large or rapid changes tend to involve disproportionately high installation, coordination, or restructuring. In this sense, adjustment costs penalize sharp changes in capital and therefore give firms an incentive to adjust gradually over

time rather than through large one-off changes.

Thus, the notion of convex adjustment costs has two important implications. First, the speed of capital adjustment is governed by χ , although this parameter is inherently unobservable. Second, larger changes in the capital stock imply higher adjustment costs, which gives firms an incentive to adjust gradually rather than through sharp one-off investments.

2.1.2 Constrained Optimization Problem

At the core of q-theory lies the firm's intertemporal investment behavior, governed by the *shadow price of capital*. To build intuition for this mechanism, it is useful to briefly introduce the firm's optimization problem and its interpretation. The firm maximizes its present value subject to the capital accumulation constraint, yielding the following

$$\mathcal{L} = \max_{\{L_s, I_s, K_{s+1}\}} \sum_{s=t}^{\infty} \left(\frac{1}{1+r} \right)^{s-t} [A_s F(K_s, L_s) - C(I_s, K_s) - I_s - L_s w_s + q_s (K_s + I_s - K_{s+1})] \quad (3)$$

where L_s , I_s and K_{s+1} denote the firm's choice variables in the dynamic optimization problem, while q_s is the Lagrange multiplier on the capital accumulation constraint, which formally denotes the shadow price of capital, also known as marginal q .

In the dynamic setup, labor L_t and investment I_t are current-period control variables, since they are chosen by the firm in period t . By contrast, the capital stock K_t is the state variable inherited from past decisions and therefore taken as given at the time of choice. Current decisions then determine next period's capital stock, K_{t+1} , through the capital accumulation constraint. This is what makes the problem dynamic, since the firm's current position reflects past decisions, while its current choices shape the constraints and opportunities it will face in the future.

Most importantly, the shadow price of capital, q , represents the value of one additional unit of capital. It is referred to as a shadow price because it is not directly observed in markets, but instead emerges from the firm's underlying optimization problem. Although abstract, a high shadow price indicates that additional capital is especially valuable to the firm, whereas a low shadow price implies the opposite. As such, q captures the marginal benefit of an additional unit of capital and thereby provides a signal about the firm's investment opportunities.

The optimization problem, therefore, clarifies the core logic of q-theory of investment: because firm value is maximized subject to the law of motion for capital, the firm cannot adjust its capital stock instantaneously. This dynamic constraint gives rise to the shadow price of capital, q , serving as a signal of how valuable additional capital is to the firm and, in turn, its incentive to invest.

2.1.2.1 Investment– q Relationship

This section turns to the investment– q relationship, which is central to q -theory, particularly in studies that aim to estimate the adjustment cost parameter, χ . From the firm’s optimality condition with respect to investment, I_s , the shadow price of capital can be written in general form, even before imposing a specific adjustment cost function

$$q_s = 1 + C_I(I_s, K_s), \quad (4)$$

where q_s denotes the shadow price of capital and $C_I(I_s, K_s)$ represents the marginal adjustment costs. Once the quadratic adjustment cost specification introduced earlier is imposed, this condition can be written more explicitly as

$$\frac{I_s}{K_s} = \frac{1}{\chi}(q_s - 1), \quad (5)$$

where I_s/K_s is the investment rate, measuring investment relative to the current capital stock. These expressions have three important implications. First, they provide the basic intuition for the firm’s investment decision. Second, under convex adjustment costs, they make the relationship between investment rate and q more explicit. Third, they make it possible, at least in principle, to estimate the adjustment cost parameter.

To begin, equation (4) shows that the shadow price of capital equals the marginal cost of acquiring an additional unit of capital: the purchase price of capital, normalized to one, plus marginal adjustment costs. Put differently, the firm invests up to the point where the marginal benefit of capital, measured by q_s , equals its marginal cost. This is important because it implies that, all else equal, a higher marginal cost must be matched by a higher marginal benefit.

This relationship becomes more informative once the quadratic adjustment cost specification is imposed. Equation (5) shows that the investment rate, I_s/K_s , is linearly related to q , with slope $1/\chi$. This makes the role of χ more transparent: a lower χ implies lower marginal adjustment costs and therefore a stronger investment response to a given change in q , whereas a higher χ implies higher marginal adjustment costs and thus a more sluggish response. This inference is imperative because it shows that χ captures the extent to which adjustment frictions limit the firm’s ability to respond to investment opportunities signaled by q . This is exactly why χ captures the *speed* of adjustment, since it governs how strongly investment responds to a given change in q .

Lastly, because the relationship is linear, χ is identified by the inverse slope coefficient under the convex adjustment cost function, as illustrated in Figure 3. Since adjustment costs are inherently unobservable, this linear form is important, especially to this thesis, because it makes indirect estimation feasible in practice using standard linear regression methods. More detail on the exact estimation procedure is provided in the research design. Nevertheless, to take this relationship to the data, both the investment rate and q must be measurable, which leads to the *Hayashi theorem* as the cornerstone in unifying the q -theory of investment.

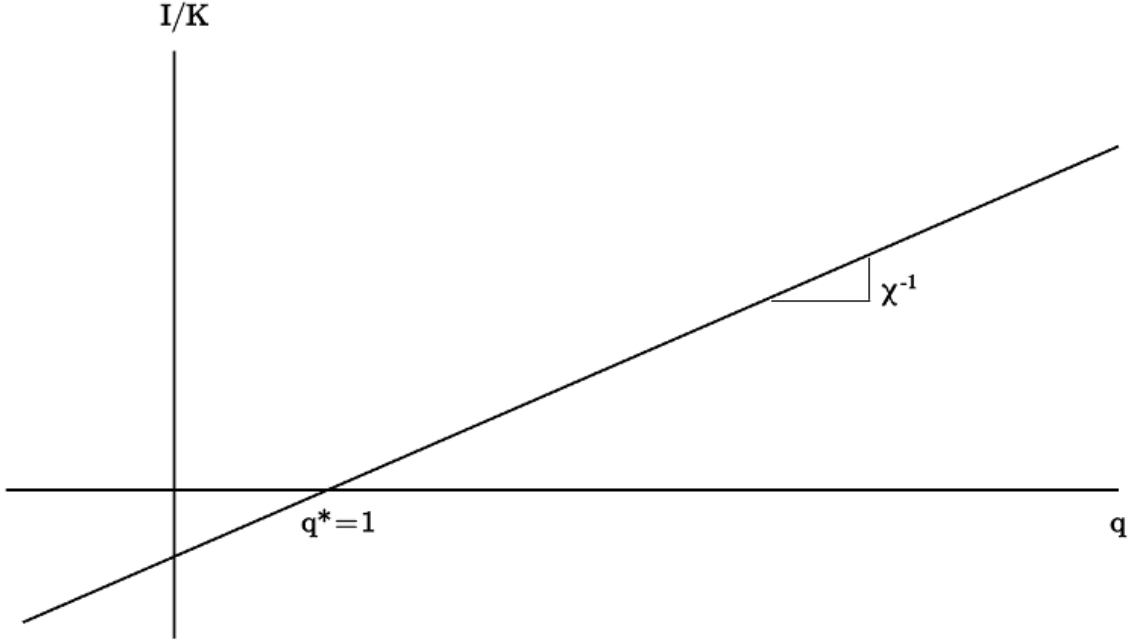


Figure 3: Linear investment– q relationship under quadratic adjustment costs.

Source: Authors' own creation, inspired by Groth (2017).

2.1.2.2 Marginal q and Average q

Thus far, the theory has focused on marginal q , which is central but not directly observable. This differs from Tobin's q , or average q , defined as the ratio of firm value to the capital stock, and is therefore empirically observable. Under a set of specific assumptions, however, Hayashi (1982) showed that marginal q and average q are equivalent. In that case, the firm's optimality condition with respect to K_{t+1} implies that

$$q_t K_{t+1} = \sum_{s=t}^{\infty} \left(\frac{1}{1+r} \right)^{s-t} \left[A_s F(K_s, L_s) - \chi \frac{I_s^2}{2K_s} - I_s - L_s w_s \right] = V_t, \quad (6)$$

which more explicitly can be rewritten as average q

$$q_t = \frac{V_t}{K_{t+1}}, \quad (7)$$

where V_t denotes the firm value and K_{t+1} denotes the capital stock. The intuition is closely related to the corporate finance distinction between growth and value firms: when average q is high, the firm's market value is higher than the replacement value of the capital stock, providing a positive investment signal to the firm (Groth, 2017). Conversely, when average q is low, the incentive to expand is weaker. Most importantly, this implies that the investment rate should be linearly related to average q , thereby making the investment– q relationship tractable for empirical analysis.

However, the equivalence between marginal q and average q rests on a set of restrictive assumptions. Specifically, the Hayashi theorem requires that (1) the production function

exhibits constant returns to scale, (2) the adjustment cost function is linearly homogeneous in investment and capital, and (3) the firm acts as a price taker in all markets. These assumptions are important because they ensure that firm value scales proportionally with the capital stock, allowing the average value of installed capital to coincide with the marginal value of an additional unit of capital (Obstfeld and Rogoff, 1996a). Although these conditions may appear restrictive, they provide the crucial empirical link between theory and measurement, a point revisited in the literature review.

In sum, under the Hayashi theorem, marginal q equals average q . Intuitively, this matters because average q reflects how financial markets value the firm's installed capital through firm value. Empirically, it is central because it makes it possible to estimate the unobservable adjustment cost parameter, χ . However, to understand the role of χ , it is necessary to examine how strongly investment responds to q .

2.1.2.3 Steady State Conditions and The Saddle Path Systems

The firm's dynamic investment behavior follows directly from its constrained optimization problem. In particular, the first-order conditions with respect to investment, I_s , and next period's capital stock, K_{s+1} , imply that the change in capital is given by

$$\Delta K_s = K_{s+1} - K_s = I_s = \left(\frac{q_s - 1}{\chi} \right) K_s, \quad (8)$$

while the change in the shadow price of capital, q , is given by

$$\Delta q_s = q_{s+1} - q_s = r q_s - A F_K(K_{s+1}, L_{s+1}) - \frac{\chi}{2} \left(\frac{I_{s+1}}{K_{s+1}} \right)^2. \quad (9)$$

Together, these two equations characterize the joint evolution of K and q , and thereby the firm's investment dynamics over time. To understand where this system converges, the steady state becomes the natural point of reference. Formally, the steady state is defined by $\Delta K = 0$ and $\Delta q = 0$, that is, a point at which neither the capital stock nor the shadow price of capital changes any further. These conditions can be represented by two curves. The $\Delta K = 0$ curve is given by

$$\bar{q} = 1 \quad (10)$$

which is horizontal and fixed, while the $\Delta q = 0$ curve is given by

$$q_s = \frac{1}{r} \left[A F_K(K_{s+1}, L_{s+1}) + \frac{\chi}{2} \left(\frac{I_{s+1}}{K_{s+1}} \right)^2 \right] \quad (11)$$

which is downward sloping due to a diminishing marginal product of capital. Their intersection determines the steady-state equilibrium, $(\bar{K}, \bar{q})$, where both the capital stock and the shadow price of capital remain constant. At this point, the return generated by an additional unit of capital exactly matches the return required by investors, such that $A_s F_K(K_s, L_s) = r$. Around this equilibrium, the system is characterized by a unique

saddle-path system, representing the adjustment path along which the firm converges toward its long-run steady state.

The system of equations, together with the saddle path system and steady state equilibrium, is illustrated in Figure 4. The key point is that the firm adjusts along the saddle path toward the steady state. Along this path, q determines whether the firm expands or contracts its capital stock through its investment. When $q > 1$, the marginal benefit of investment exceeds its marginal cost, giving the firm an incentive to invest and gradually increase its capital stock. When $q < 1$, the marginal benefit is below the marginal cost, so the firm disinvests and the capital stock declines. The speed of this adjustment is governed by the adjustment cost parameter, χ , which is precisely why it plays such a central role: a low χ implies that capital can be adjusted quickly, whereas a high χ makes adjustment slower. This process continues until the steady state is reached, where $q = 1$ and the firm has no incentive to adjust its capital stock any further. To better appreciate the role of χ , the analysis now turns to a setting in which the economy, and hence firms, are subject to exogenous shocks.

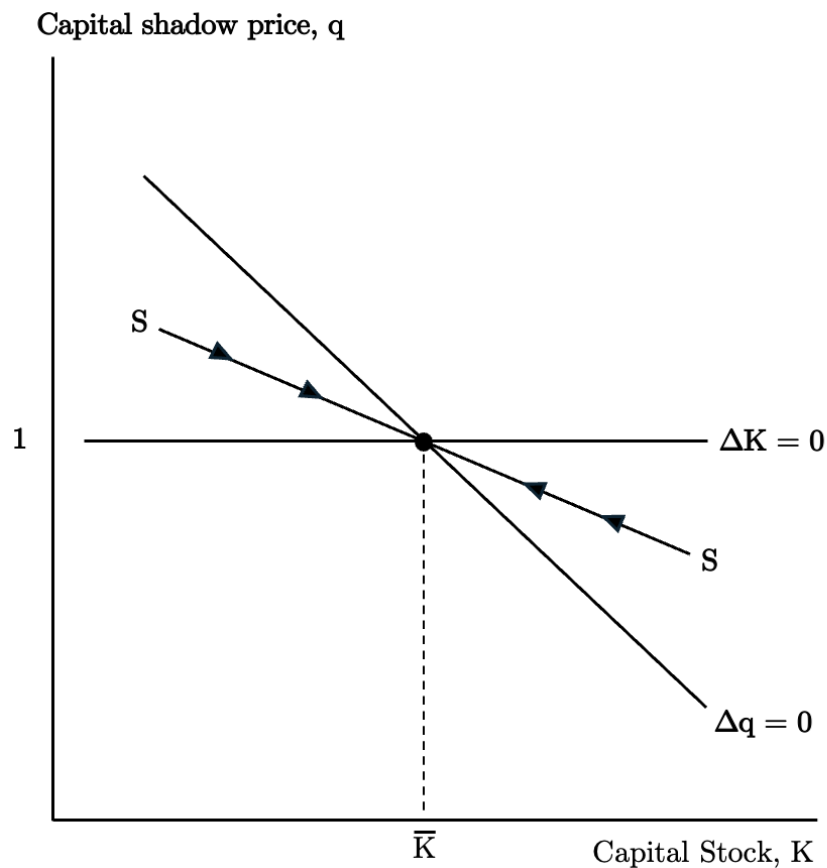


Figure 4: Saddle path system and dynamics of q investment model.
Source: Authors' own creation, inspired by Obstfeld and Rogoff (1996).

2.1.2.4 The Role of Permanent Productivity Shocks

A natural next step is to move beyond the steady state in isolation and examine how firms adjust when economic conditions change. This is precisely where adjustment frictions become most informative, because once firms are displaced from their initial optimum, differences in χ determine how quickly they are able to move toward a new one. For clarity, the following analysis abstracts from other sources of firm heterogeneity and assumes that firms differ only in their adjustment-cost parameter.

This mechanism can first be seen in Figure 5, which illustrates the case of a permanent increase in aggregate productivity. Suppose that firms initially start in the long-run steady-state equilibrium at point A. From this starting point, a permanent positive productivity shock shifts the $\Delta q = 0$ curve upward and thereby gives rise to a new steady state equilibrium at point C, associated with a higher capital stock. Since the capital stock is predetermined in the short run, firms cannot jump directly from A to C. Instead, adjustment begins through an immediate rise in the shadow price of capital, which moves the firm vertically from A to B. At point B, $q > 1$, so the marginal benefit of capital exceeds its marginal cost. The firm therefore increases investment and gradually expands its capital stock as it moves along the new saddle path from B to C, until the new steady state is reached.

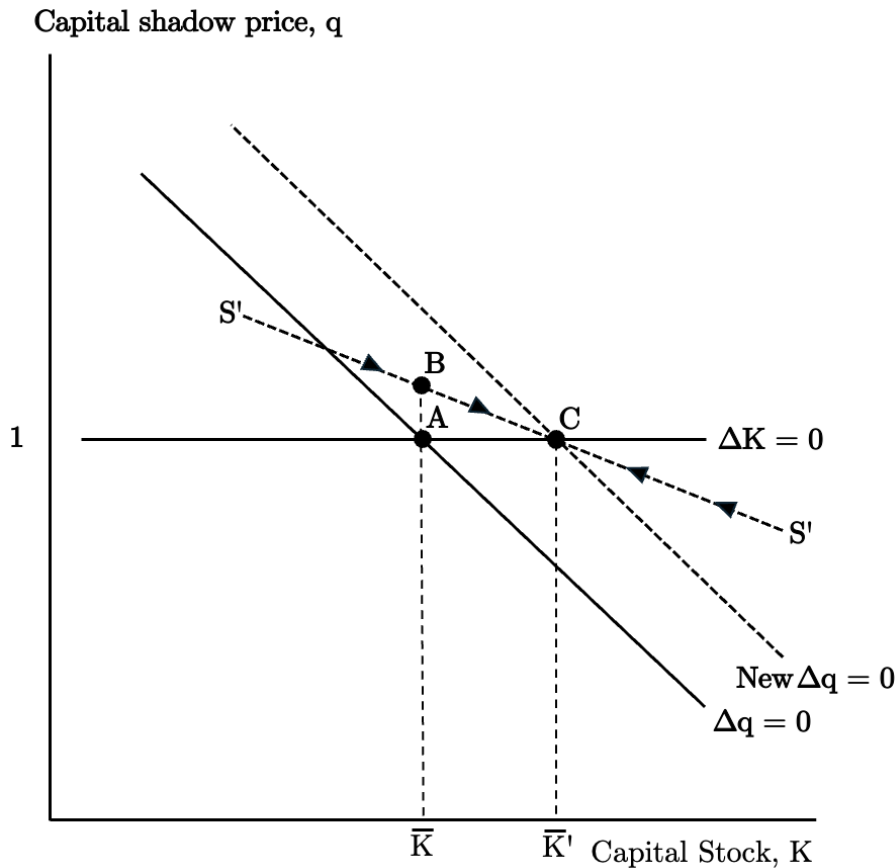


Figure 5: An unexpected permanent positive productivity shock.
 Source: Authors' own creation, inspired by Obstfeld and Rogoff (1996).

The same logic applies in reverse when productivity falls, as shown in Figure 6. In this case, a permanent decline in aggregate productivity shifts the $\Delta q = 0$ curve downward and implies a new steady state at point C with a lower capital stock. Once again, the capital stock cannot adjust instantaneously, so the firm cannot move directly from A to C. Instead, the shadow price of capital falls immediately and moves the firm vertically from A to B. At point B, $q < 1$, meaning that the marginal cost of capital now exceeds its marginal benefit. The firm therefore disinvests and gradually contracts its capital stock, precisely to avoid remaining burdened with capital that has become too costly relative to the cash flows that it generates. Over time, it hence moves along the new saddle path from B to C until the new steady state is reached.

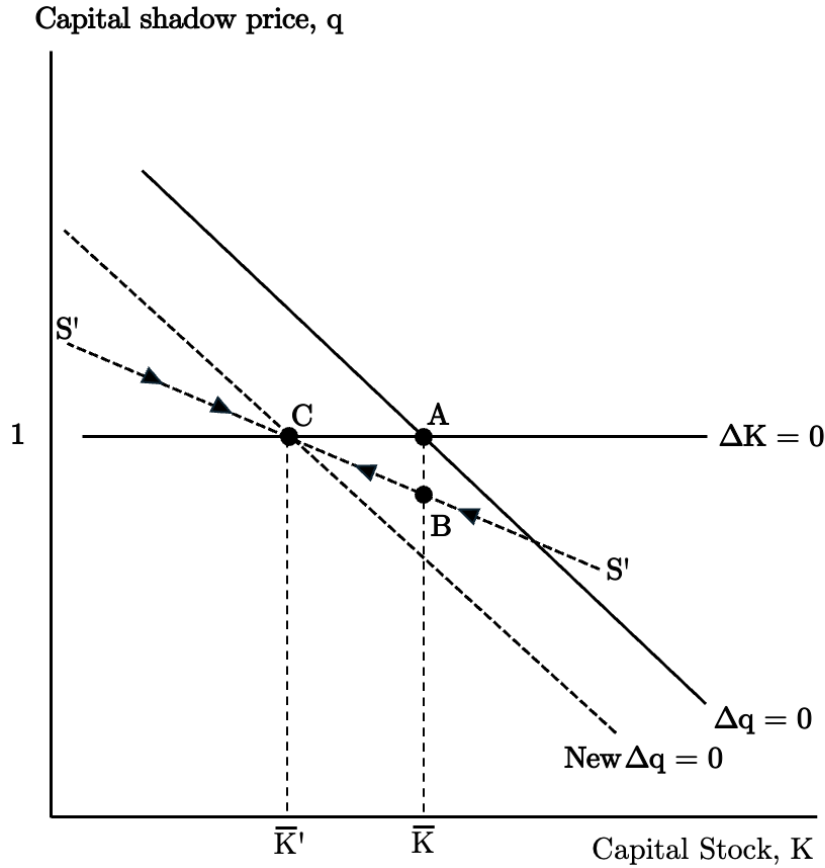


Figure 6: An unexpected permanent negative productivity shock.
Source: Authors' own creation, inspired by Obstfeld and Rogoff (1996).

Taken together, the two figures highlight the central role of the adjustment cost parameter, χ , for firm performance in the short and medium run. By governing the speed of adjustment, it determines how quickly firms can realign their capital stock with new economic conditions. Firms with a low χ are able to adjust more rapidly, whereas firms with a high χ respond more slowly. As a result, firms facing greater adjustment frictions are at a disadvantage regardless of the direction of the shock. Following a positive shock, they expand too slowly relative to the new optimum, whereas following a negative shock, they contract too slowly. In both cases, they remain away from the new steady state for longer. In any case, they lose the race. This is economically intuitive since, once

economic conditions change, firms seek to reoptimize their capital stock, but costly adjustment prevents an immediate response. Adjustment frictions therefore matter because they determine how quickly firms can adapt to a new economic environment.

2.1.3 Concluding Remarks

The purpose of this first part has been to establish the production-side intuition embedded in the q -theory of investment. At its core lies a systematic relationship between investment and marginal q : when the marginal value of an additional unit of capital exceeds its marginal cost, firms are incentivized to invest, and vice versa. Under the Hayashi theorem, marginal q coincides with observable average q , that is, firm value relative to the replacement cost of capital, making this relationship empirically tractable. Nevertheless, firms do not adjust their capital stock instantaneously, because adjustment costs introduce frictions that make investment gradual. As a result, the response to a given q -signal depends on how costly it is to change capital, making heterogeneity in adjustment costs central to understanding why firms may respond differently to the same aggregate shock, both in strength and persistence. However, the dynamics developed here remain partial-equilibrium in nature, as they show how firm value and investment respond to exogenous shocks but do not by themselves explain why risk premia arise. To establish the intuition behind this, we turn to the fundamentals embedded in the consumption side of the economy and the broader logic of asset pricing.

2.2 Consumption-Based Asset Pricing

An investor must decide how much to consume, how much to save, and which portfolio of assets to hold. These decisions are fundamental to asset pricing theory and to the determination of *risk premia*. For this reason, analyzing only the production side of the economy is not sufficient to fully understand the underlying sources of risk. In what follows, we therefore introduce the consumption side of the economy by examining the fundamental pricing equation and the economic intuition behind risk premia. The objective is to provide the theoretical intuition linking the q -theory of investment to the role of consumption risk. The setup follows the standard asset pricing treatment in Cochrane (2005a) and Obstfeld and Rogoff (1996b).

2.2.1 Utility Maximizing Representative Consumer

To fully appreciate the notion of asset prices and how expected returns are determined, it is necessary to consider investor incentives, which arise from the demand side of the economy. In asset pricing models, this role is represented by a utility-maximizing representative consumer who chooses how much to consume and how much to invest in financial assets

$$U_t = E_t \left[\sum_{s=t}^{\infty} \beta^{s-t} u(C_s) \right], \quad \text{s.t.} \quad C_s = L_s w_s + x_s (V_s + d_s) - x_{s+1} V_s, \quad (12)$$

where $u(C_s)$ is the period utility function defined over consumption, β is the consumer's subjective discount factor capturing impatience, w_s denotes the wage, L_s is labor supply, x_s represents the number of shares held in the firm, V_s is the ex-dividend value of the firm, and d_s is the dividend payment. Intuitively, the consumer receives labor income, dividends, and the value of held shares, while purchasing new shares for the next period. E_s denotes the conditional expectation operator, emphasizing that future variables are inherently uncertain and depend on the consumer's expectations given the information available at time s .

The utility function, $u(C_s)$, captures the fundamental preference for consumption itself. Throughout, it is assumed to be increasing, reflecting the preference for more consumption, and concave, reflecting diminishing marginal utility. You may enjoy a second cup of coffee, but it is usually less satisfying than the first. This implies that an additional unit of consumption is more valuable when consumption is low than when it is high. Investors, therefore, prefer smoother consumption paths across time and across states of nature. However, because future consumption C_{s+1} is uncertain, investors face risk regarding their future wealth and thus consumption opportunities.

In smoothing consumption over time under uncertainty, investors must balance their decisions dynamically. This is where the budget constraint becomes especially important, as it highlights the intertemporal trade-off between current consumption and the opportunity to transfer resources into future wealth and consumption. In the setup presented herein, shares in the firm are the vehicle through which investors transfer resources across time, since a share purchased today provides a claim to future dividends and firm value. The value of the representative firm therefore becomes central to the investor's consumption-saving decision and, ultimately, to the determination of asset prices.

2.2.2 Fundamental Pricing Equation

An investor who maximizes lifetime utility over consumption must decide how much to consume today and how much to save by purchasing shares in firms. In doing so, the investor chooses this period's consumption, C_s , and next period's share holdings, x_{s+1} . Solving this optimization problem yields the standard Euler equation governing optimal consumption and portfolio choice,

$$V_s = E_s [m_{s+1} (V_{s+1} + d_{s+1})] \quad (13)$$

where m_{s+1} is the stochastic discount factor (SDF), and $V_{s+1} + d_{s+1}$ is the firm's next-period payoff, consisting of its future value and dividend. This gives rise to the fundamental pricing equation: the value of the firm today equals the expected future payoff discounted by the SDF. Dividing both sides by V_s , the same condition can be written in terms of the firm's gross return,

$$1 = E_s [m_{s+1} R_{s+1}] \quad (14)$$

where the SDF and the gross return are defined as

$$m_{s+1} \equiv \beta \frac{u'(C_{s+1})}{u'(C_s)}, \quad R_{s+1} \equiv \frac{V_{s+1} + d_{s+1}}{V_s}. \quad (15)$$

From this setup, it becomes clear that the stochastic discount factor (SDF) is given by the investor's intertemporal marginal rate of substitution (MRS), while the gross return is the standard one-period return used in asset pricing. More specifically, the SDF depends on the investor's impatience, captured by β , as well as the ratio between the marginal utility of future consumption, $u'(C_{s+1})$, and the marginal utility of current consumption, $u'(C_s)$. In this respect, the ratio of these marginal utilities reflects how willing the investor is to give up one unit of consumption today in exchange for one unit of consumption tomorrow. At a deeper level, the SDF and the gross return reveal the inherent link between consumption choices and firm value, which lies at the core of this paper. On one hand, payoffs are discounted into present firm value according to the investor's willingness to exchange consumption through time, whereas on the other hand, the future payoff is shaped by the firm's ability to maximize future value and distribute dividends. In asset pricing, both sides must therefore be considered jointly.

Though abstract by construction, the underlying intuition is straightforward. Consider a stock whose next-period payoff is given by its future firm value and dividend. If the investor expects tomorrow to be a period in which consumption is relatively low, then future consumption becomes especially valuable, and the investor places greater value on the future payoff from the stock. Because the stock helps transfer wealth across time to a period when it matters more, the investor is willing to pay more for it today. For a given payoff, this raises firm value today and implies a lower gross return. The same logic works in reverse when tomorrow is expected to be a period of relatively high consumption. In that case, the future payoff is valued less, the investor is willing to pay less for the stock today, and the implied gross return is higher. This is the core intuition of asset pricing as firm value reflects both the firm's ability to generate future payoffs and the investor's willingness to discount those payoffs through time. At the intersection of these two forces lies the notion of expected returns and risk premia, namely the extent to which investors must be compensated for holding assets that are risky from the perspective of their consumption needs.

2.2.3 Consumption Risk and Asset Risk Premia

Most important in asset pricing theory is the notion of *risk premia*: the idea that investors require compensation for bearing systematic risks. Because investors buy and sell assets based on how payoffs align with their consumption needs across states, the pricing of financial assets ultimately reflects consumption risk. In particular, assets that expose investors to higher consumption risk must offer higher expected returns as compensation, and vice versa.

From the fundamental pricing equation, it is possible to derive an expression for

expected excess returns which is the risk premium associated with the asset

$$E_s[R_{s+1}] - R_f = -R_f \text{Cov}_s(m_{s+1}, R_{s+1}), \quad (16)$$

where R_f denotes the risk-free rate and $\text{Cov}_s(m_{s+1}, R_{s+1})$ represents the covariance between the SDF and the asset's gross return in the next period. Since the SDF is high in states in which future consumption is especially valuable relative to current consumption, this covariance captures whether the asset pays off in 'good' and 'bad' states of the economy. Assets that covary positively with the SDF provide payoffs in states in which future consumption is especially valuable. Because these assets effectively serve as a hedge against consumption risk, the investor is willing to accept a lower expected return. By contrast, assets that covary negatively with the SDF deliver low returns precisely in states where investors need them the most. Such assets worsen the investor's consumption risk and must therefore offer a positive risk premium, meaning that the investor only chooses to hold them if she expects a higher return as compensation.

To fully appreciate the interpretation, a practical example is useful. Imagine that an investor owns shares in a car manufacturer. In normal economic times, households purchase cars, the firm earns profits, and the stock delivers relatively solid returns. During a recession, however, households reduce spending as income and consumption fall. Fewer cars are sold, profits drop, and the stock return becomes low. Now consider the investor's position in that situation. Exactly during recessions, future consumption becomes a crucial concern. Because the asset performs badly precisely in those states in which future payoffs are most valuable to the investor, it worsens the investor's exposure to consumption risk. To be willing to hold such an asset, the investor must therefore be compensated through a higher risk premium.

In general, equity returns covary negatively with the SDF because firms tend to perform poorly in recessions, when future consumption is especially valuable relative to current consumption, and perform well in expansions, when future consumption is less of a concern. This negative covariance underlies the nature of the equity risk premium. Hence, the relevant question becomes not whether equity returns covary negatively with the SDF, but which firm characteristics are linked to a stronger negative covariance. Establishing this connection is difficult, but it is precisely required to make the q-theory of investment meaningful in an asset pricing context.

2.2.4 Risk Premia under q-Theory of Investment

Under the q-theory of investment, firm value reflects the present value of dividends distributed to shareholders. Because firm value is directly tied to the payoff of the asset, state-dependent movements in firm value provide a natural point of contact between the q-theory of investment and the consumption-based logic of risk premia. A useful starting point is, therefore, to consider how firm characteristics shape the response of firm value across economic states.

From this perspective, it is helpful to return to the saddle-path system implied by

the q-theory of investment and its dynamic responses to economic shocks. Although the model is not a complete general-equilibrium asset-pricing model, it still provides an economically useful link between aggregate conditions and changes in firm value. A positive productivity shock may be interpreted as a 'good' state of the economy: firms produce more efficiently, output and profits rise, aggregate income and consumption increase, and firm value rises. Conversely, a negative productivity shock may be interpreted as a 'bad' state of the economy: firms produce less efficiently, output and profits fall, aggregate income and consumption decline, and firm value falls. In this sense, the q-theory of investment offers an intuitive way to think about how firm value co-moves with aggregate economic states, and thus with the broad forces underlying the equity risk premium. However, when the focus shifts to what firm characteristics may be associated with differences in risk premia, the interpretation becomes somewhat ambiguous.

While many firm characteristics could in principle be associated with differences in risk premia, the speed of adjustment offers a particularly natural characteristic to examine within the q-theory of investment. The reason is that it governs the extent to which firms respond to aggregate economic shocks, and thus how movements in firm value persist across economic states. Firms with lower speeds of adjustment respond more slowly to economic shocks, so the effects of both positive and negative disturbances remain in firm value for longer, as firm value adjusts only gradually toward its new optimal level. This suggests that heterogeneity in the speed of adjustment may matter for firms' exposure to aggregate states and therefore may be relevant for asset pricing, since state-dependent variation in firm value is precisely where q-theory meets the consumption-based logic of risk premia. The implication, however, is not clear-cut. Firms with low speeds of adjustment appear to be more adversely affected in both good and bad states, which makes it difficult to determine whether they should be viewed as bearing greater systematic risk or simply as having lower valuations on average. Put differently, the present partial-equilibrium setup does not make it possible to determine with confidence how differences in the speed of adjustment map into the covariance between payoffs and the stochastic discount factor (SDF). Resolving that question would require a fully specified general-equilibrium asset-pricing model.

Despite these limited conclusions, an important implication remains: if q-theory of investment is informative about how firm value changes across aggregate economic states, and if risk premia depend on differences in payoffs across such states, then the speed of adjustment emerges as a particularly important characteristic to examine. It is precisely one of the core characteristics through which firms' exposure to aggregate economic shocks may differ. The central argument advanced here is, therefore, not that a complete theory has been established, but rather that it is both economically plausible and empirically relevant to examine whether differences in the speed of adjustment are associated with differences in risk premia.

2.2.5 Concluding Remarks

The purpose of this second part has been to establish the consumption-side logic underlying risk premia. At the most fundamental level, asset prices reflect investor preferences over time and across states of the economy. In this sense, the incentive to invest is tied to the opportunity to transfer current wealth into future consumption and thereby smooth consumption over time. However, because the future is inherently uncertain, investors face the risk of losses precisely when future consumption is mostly needed. This is the basic source of risk premia: assets whose payoffs co-move with ‘good’ and ‘bad’ states of the economy must offer compensation in the form of higher expected returns. The relevant question therefore becomes which types of firms have payoffs that are more strongly exposed to such economic states. From the q-theory of investment alone, that question cannot be answered with confidence, since a unified general-equilibrium model is not developed here. Nevertheless, to the extent that q-theory is informative about how firm value changes across economic states, the speed of adjustment remains an economically plausible and empirically relevant firm characteristic to examine.

2.3 Fama-French Five-Factor Model

Since the introduction of the three-factor model and its later extension to the five-factor specification, the Fama–French framework has remained central to academic debate. Its empirical performance is difficult to dismiss, yet its underlying economic intuition remains incomplete. In this thesis, we argue that the model shares important conceptual parallels with the q-theory of investment. Although this link is not fully pinned down in the existing literature, these parallels motivate closer empirical attention to the role of the speed of adjustment. While the proposed connection between the five-factor model and q-theory is developed in this thesis, the empirical specifications and factor-model testing framework build on Cochrane (2005b), Fama and French (2015), and Munk (2024d).

2.3.1 The Model

To appreciate the model and its possible relation to q-theory, it is useful first to present its specification. Formally, the Fama-French Five-Factor model can be written as

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,M}(R_{M,t} - R_{f,t}) + \beta_{i,SMB}SMB_t + \beta_{i,HML}HML_t + \beta_{i,RMW}RMW_t + \beta_{i,CMA}CMA_t + \epsilon_t \quad (17)$$

where $R_{i,t} - R_{f,t}$ denotes the excess return on asset i , $R_{M,t} - R_{f,t}$ the excess market return, SMB_t the size factor (small minus big), HML_t the value factor (high minus low), RMW_t the profitability factor (robust minus weak), and CMA_t the investment factor (conservative minus aggressive). The coefficients β measure the factor exposure of asset i , α_i denotes the abnormal return left unexplained by the model, and $\epsilon_{i,t}$ is the error term, representing the asset-specific return uncertainty.

Because the model is intended to explain expected returns, the five-factor model is

naturally interpreted as an empirical asset pricing model of risk premia. Its purpose is, in this regard, to account for differences in risk premia across firms: commonly referred to as the *cross-section*. In this setting, alpha is of central theoretical importance, since a correctly specified asset-pricing model should fully capture the systematic determinants of expected returns and hence leave no abnormal return unexplained. A nonzero alpha consequently indicates that the model remains incomplete, in the sense that some relevant dimension of return variation is not absorbed by the included factors. On the contrary, an alpha indistinguishable from zero suggests that the model explains the cross-section successfully (Cochrane, 2005b).

This is precisely where most of the influence of the five-factor model originates. Although the model is not without weaknesses, it has proven empirically difficult to dismiss, because it often leaves relatively small and statistically weak alphas compared with many competing specifications. Yet this empirical success does not by itself settle the deeper theoretical question of why the included factors should be linked to risk premia. As argued in the previous sections, such a link must ultimately rest on the relation between firm characteristics and different states of the economy. It is from this perspective that the comparison to the q-theory of investment becomes especially interesting.

2.3.2 A q-Theoretical Representation of the Five Factors

In the q-theory of investment, firm value reflects the discounted value of future operating profits, which depend on the firm's investment path and valuation signals captured by average q . These signals, in turn, depend on both aggregate economic conditions and the firm's position within the saddle-path system. This creates a natural link to the Fama–French Five-Factor Model, where profitability, investment, and value relate to the firm's adjustment dynamics, while market and size may capture broader economic states and differences in adjustment positions. Thus, while the five-factor model is not a structural model of q-theory, its factors may still reflect its economic mechanisms.

Operating profitability is a natural starting point because firm value ultimately depends on the firm's ability to generate future profits. This is not specific to the q-theory of investment, but follows from standard valuation logic: firm value depends on its ability to generate future cash flows that can be distributed to investors. From an asset-pricing perspective, this matters because future operating profits shape the firm's payoff, and if these payoffs vary systematically across good and bad states of the economy, profitability becomes relevant for understanding risk premia. However, the limitation of the profitability factor is that it measures current profitability rather than the full path of future profits. This is where the q-theory of investment adds a dynamic element. Future operating profits are not simply a static firm characteristic: they depend on valuation signals and the firm's optimal investment path. Profitability is therefore central to firm value, but it should be interpreted in relation to investment opportunities, where the q-theory provides a particularly useful perspective.

The investment and value factors are closely connected to q-theory through the investment– q relationship. Under the Hayashi theorem, marginal q equals average q , so

investment decisions can be linked to an observable valuation signal. When average q is high, firms have stronger incentives to invest, whereas when average q is low, investment incentives are weaker. Since book-to-market may be interpreted as an imperfect inverse proxy for average q , growth firms resemble high- q firms, while value firms resemble low- q firms. In this sense, the value factor reflects variation in q , while the investment factor captures variation in firms' investment rates, making them empirical counterparts to the investment- q relationship. This may help explain why Fama and French (2015) find a high positive correlation between value and investment and show that value becomes redundant once profitability and investment are included. Viewed through the lens of q -theory, this is economically intuitive: if book-to-market partly captures the inverse of average q , and investment responds to q , then the investment factor already absorbs part of the variation associated with the value factor. The close empirical relation between value and investment, therefore, provides one of the clearest links between the five-factor model and the q -theory of investment.

The market and size factors are less directly tied to the q -theory of investment, but they may still help characterize the environment in which firms adjust. The market factor captures broad aggregate conditions and can therefore be interpreted as a proxy for good and bad states of the economy. Since aggregate shocks affect firms' investment opportunities, operating profits, and firm values, the market factor provides useful information about the state-dependent environment surrounding the firm. Size may also be informative because small and large firms are likely to differ in maturity and their distance from their long-run steady state. For this reason, smaller firms may be located earlier in the saddle-path system, while larger firms may have more established operations and be closer to their long-run optimum or even beyond. As such, although market and size are not direct q -theoretic variables, they may still help describe the aggregate states and firm-level positions that shape investment dynamics.

Overall, the Fama-French Five-Factor Model contains several elements that appear related to the q -theory of investment. Profitability, investment, value, market, and size may all capture parts of the broader link between firm value, capital accumulation, valuation signals, aggregate states, and firms' positions in the saddle-path system. This interpretation should, however, remain cautious, since the present setup cannot derive exact sign predictions for risk premia. Even so, these connections motivate a closer empirical examination of the underlying economic link between the five-factor model and the q -theory of investment.

2.3.3 The Role of Adjustment Costs

The previous subsection argued that the Fama-French Five-Factor model may capture dynamics that are broadly consistent with the q -theory of investment. This interpretation is also present in Fama and French (2015), who compare their model to an alternative q -factor specification without directly presenting it as a q -theoretical expression. If the connection between the five-factor model and the q -theory of investment is indeed economically relevant, then the speed of adjustment appears to be a central missing element.

This dimension is central because adjustment costs determine how efficiently firms translate valuation signals, represented by q , into actual investment and thereby maximize their firm value. Firms with low adjustment costs can respond more quickly to both good and bad shocks to the economy and therefore respond more efficiently to changing economic conditions. By contrast, firms with high adjustment costs face greater frictions and adjust more sluggishly, which makes them especially vulnerable. These differences are relevant for asset pricing because they affect how future payoffs evolve across good and bad states of the economy. Hence, adjustment costs may represent an additional dimension of heterogeneity not fully reflected in the existing five-factor structure.

From this perspective, the adjustment-cost parameter, χ , may capture an additional source of economically relevant variation. Although the Fama–French factors already reflect differences in investment behavior, profitability, and valuation, they may not fully absorb the adjustment-friction dimension itself. If this dimension matters for returns, it may therefore show up as residual return variation in the form of nonzero alphas. This motivates testing whether an adjustment-cost-related characteristic adds incremental information beyond the Fama–French Five-Factor model and may improve the model by augmenting it with an adjustment-cost factor.

2.3.4 Concluding Remarks

The preceding discussion has examined the Fama–French Five-Factor Model through the lens of the q -theory of investment to develop an economic interpretation of its empirical success. Several dimensions appear consistent with mechanisms emphasized by the theory. Operating profitability provides a natural starting point, as firm value depends on the ability to generate future cash flows, although profitability itself measures current profitability rather than the full path of future profits. This is where q -theory adds a dynamic element through valuation signals and investment, which help determine the stream of future cash flows. In addition to these theoretical connections, the investment– q relationship may also be reflected indirectly in the model: under the Hayashi theorem, average q corresponds to marginal q , while book-to-market may serve as an imperfect inverse proxy for q . Since investment responds to q , the close relation between the value and investment factors appears fully consistent with the predictions by the investment– q relationship. Furthermore, while market and size are less directly tied to the q -theory of investment, they may still capture aggregate states and firm-level positions that shape the overall investment dynamics of the firm. If these parallels are economically linked, adjustment frictions become particularly relevant, since they determine how firms respond to economic shocks and represent a central dimension not directly captured by the existing five-factor structure.

3 Literature Review

Building on the preceding theoretical discussion, this section reviews the literature most relevant to q -theory, production-based asset pricing, and empirical factor models. The discussion begins with the investment literature, where a central question is whether average q is sufficient to explain firms' investment decisions. It then turns to the asset-pricing literature, focusing on production-based asset pricing, where the same economic logic, particularly the role of adjustment costs, becomes central to explaining expected returns. Finally, the section reviews empirical factor models and argues that the strong performance of newer specifications, especially the Fama–French Five-Factor Model, may partly reflect mechanisms closely related to the q -theory of investment.

3.1 q -Theory of Investment

A central idea in the q -theory of investment is that firms should invest when the value of installed capital exceeds its replacement cost. In empirical work, this idea is usually captured by average q , which has become a central benchmark for studying investment. Yet the evidence is less convincing than the theory would suggest. Although average q is generally related to investment, its explanatory power is often weak and sensitive to measurement, internal cash flow, and specification issues. Nevertheless, the literature points to a common conclusion: while q -theory has empirical struggles, its theoretical foundations remain highly influential.

3.1.1 Baseline Evidence: q Matters, but Weakly

Since Hayashi (1982) famously demonstrated that the unobservable marginal q equals observable average q under certain conditions – thereby making q -theory easier to test empirically – a substantial literature has examined the linear investment- q relationship. The broad conclusion is that q is positively related to investment, but that its explanatory power is weaker than implied by the q -theory of investment.

Building on the Hayashian theorem, while the contribution was still circulating as a working paper, Summers (1981) and Blanchard and Wyplosz (1981) conducted some of the first aggregate macro-level tests of the investment- q relationship using average q . Their findings provide some support for the view that q helps explain investment, though the relationship is positive but weak and specification-sensitive. Summers (1981), using annual U.S. data on nonfinancial firms from 1932 to 1978, finds that average q performs weakly in aggregate investment regressions, with coefficients ranging from -0.106 to 0.039 in the full sample and -0.010 to 0.042 in the postwar sample, with generally limited significance. This contrasts with his tax-adjusted q measure, which performs better. Equally, Blanchard and Wyplosz (1981), using quarterly U.S. data for 1953 to 1978, find positive effects and report an implied elasticity of about 0.43 , meaning a 1% increase in q is associated with a 0.43% increase in investment. Taken together, these papers offered early aggregate support for q -theory, while indicating that the empirical relation was

specification-dependent and not decisive, raising the first concerns that average q may be an imperfect proxy for marginal q .

In response to these early findings, Abel and Blanchard (1986) construct 12 alternative series for the unobservable marginal q based on the expected present value of future marginal profits and compare them with the conventional average q . Using aggregate U.S. manufacturing data from 1948 to 1979, they find that q is generally significantly related to investment but has limited explanatory power and leaves much of investment unexplained, with implied elasticities of investment with respect to q only around 0.1 to 0.3. Importantly, they conclude that these results are very similar to those obtained using average q , implying that average q is not a poor proxy for the theoretically relevant marginal q .

In sum, the empirical evidence on the investment– q relationship under convex adjustment costs suggests that q is positively related to investment, but less strongly than predicted by the q -theory of investment. This weakness is not limited to average q but appears to extend to alternative specifications of marginal q as well. The literature, therefore, begins from a central tension: q matters, but not nearly as strongly as the theory would suggest. This has motivated subsequent work on why the q -theory of investment fails to fully account for observed investment behavior.

3.1.2 Financial Constraints and the Cash Flow Debate

A large debate following the initially weak empirical performance of q -theory concerns whether investment is positively related to internal cash flow, which could challenge the prediction that q fully captures investment opportunities and internal finance should not matter. While early work points to financial constraints, subsequent research shows that this interpretation remains ambiguous and up for debate.

A central contribution in this regard is Fazzari, Hubbard, and Petersen (1988), who test whether firms' investment is determined solely by average q or whether internal finance also matters. The key idea is that if q -theory holds, firms should base investment decisions only on opportunities captured by q , while internal finance should not matter, since firms can raise external funds. Using data on U.S. manufacturing firms from 1970 to 1984, they estimate investment regressions with q and internal cash flow as explanatory variables, and compare results across firms classified by dividend payout behavior. Their main finding is that investment remains strongly sensitive to internal cash flow even after controlling for q , with much stronger sensitivity for low-dividend firms. Specifically, they find that the cash-flow coefficient for low-dividend firms is 0.670 versus 0.254 for high-dividend firms over 1970-75, and 0.461 versus 0.230 over the full 1970-84 period. Interpreting low-dividend firms as more financially constrained, they conclude that q alone does not fully explain investment behavior.

Building on this result, Gilchrist and Himmelberg (1995) revisit the same question but focus on the measurement of q . They argue that average q is a noisy proxy for marginal q , which can mechanically generate a spurious role for cash flow. Using a panel of 428 U.S.

manufacturing firms over 1979–1989, they construct a ‘fundamental q ’ measure and show that the cash-flow coefficient declines sharply and often becomes insignificant for firms with good access to capital markets. However, for more constrained firms, investment remains sensitive to cash flow, with differences relative to unconstrained firms statistically significant in 5 out of 8 specifications at the 5% level, though not uniformly across specifications. Together, these findings suggest that the excess sensitivity documented by Fazzari, Hubbard, and Petersen (1988) may partly reflect measurement error in q , while also leaving room for financial frictions.

At the same time, a more fundamental challenge to this interpretation is provided by Kaplan and Zingales (1997), who question whether higher investment–cash flow sensitivity signals tighter financing constraints. They also argue that dividend payout is not a reliable proxy for such constraints. Empirically, they revisit the same firms studied by Fazzari, Hubbard, and Petersen (1988), but replace dividend payout ratios with a more detailed firm-by-firm assessment of financial constraints. Re-estimating the investment regressions, they find that the cash-flow coefficient is in fact highest for the least constrained firms (around 0.70) and lower for more constrained firms (about 0.25 to 0.34). Effectively, this implies that investment–cash flow sensitivity cannot be taken as evidence of financing constraints and does not constitute a clear rejection of the q -theory of investment.

In light of the broader debate, the role of cash flow remains contested, with no definitive conclusion as to whether it reflects financing constraints or measurement problems in q . As such, q -theory was not abandoned: rather, its weak empirical performance became an unresolved puzzle. Accordingly, subsequent research has sought to reconcile theory and evidence by refining empirical specifications and improving measures of q .

3.1.3 Attempts to Improve the Empirical Performance of q -theory

Building on this motivation, the literature has pursued several approaches, ranging from alternative proxies for q to more advanced econometric methods addressing measurement error in average q . Despite these developments, a commonly used benchmark remains the physical-capital measure of q , as no clear consensus has emerged among the proposed solutions.

An example is Bond and Cummins (2001), who argue that the conventional stock-market-based measure of average q may be a noisy proxy for the fundamentals relevant for investment, since market valuations can deviate persistently from fundamental value. As a result, they construct a forward-looking alternative measure of q based on discounted analysts’ earnings forecasts. Using panel data on 1,066 U.S. firms over 1982–1999, they find that the coefficient on conventional q is small, around 0.017 to 0.022, whereas the coefficient rises to about 0.097 to 0.110 under their alternative measure. Consistent with this, they also find that financing constraints play no significant role in explaining investment behavior. More broadly, the literature has proposed a range of alternative q proxies, from simpler accounting-based approximations, such as Chung and Pruitt’s (1994) “simple q ,” to expectations-based measures, such as the “real q ” of Cummins,

Hassett, and Oliner (2006). Taken together, these approaches illustrate how multiple competing definitions of q have gradually emerged in the literature.

In contrast to these alternative proxies, Erickson and Whited (2000, 2006) focus on the econometric consequences of treating observed average q as a noisy measure of the true, unobservable marginal q . They show that correcting for measurement error can substantially improve the empirical fit of q -theory, with the R^2 increasing from about 0.22 under OLS to roughly 0.38–0.44 and cash flow becoming statistically insignificant. However, the importance of these econometric corrections remains debated. Almeida, Campello, and Galvao (2010) argue that the Erickson-Whited approach is sensitive to specification and produces unstable estimates, instead finding more robust results using instrumental-variable methods, with q coefficients of around 0.045–0.063 and cash-flow coefficients that remain positive at around 0.043–0.046. In response, Erickson and Whited (2012) revisit this comparison, aiming to defend their econometric approach. Taken together, this back-and-forth highlights the absence of a clear consensus on the most reliable econometric remedy for measurement error in q , beyond the traditional linear investment- q relationship.

Finally, at the frontier of this literature, Peters and Taylor (2017) propose a new measure of q that incorporates intangible capital and evaluate its performance relative to both the standard q proxy and the econometric literature on measurement error. Specifically, they compare the conventional physical-capital measure of q , defined as firm value net of current assets divided by physical capital, with their 'total q ', which adds intangible capital to the denominator. Using panel data on U.S. public firms from 1975 to 2011 and firm- and year-fixed-effects regressions of the investment- q relationship, they find that the within-firm R^2 increases from 0.139 to 0.209 for physical investment, from 0.266 to 0.279 for intangible investment, and from 0.250 to 0.327 for total investment. Nevertheless, despite these improvements, they emphasize that 'total q ' is by no means a perfect proxy, since intangible capital remains difficult to measure and q itself is still subject to measurement error. Their contribution may therefore be seen less as a rejection of the standard benchmark than as an attempt to extend and improve it.

Overall, a wide range of q proxies and econometric approaches have been developed, but no consensus has emerged to definitively resolve the empirical weaknesses of q -theory, leaving the physical-capital measure of q as a commonly used benchmark in practice. This weak empirical performance in the investment literature has motivated a shift in focus, where the underlying economic mechanisms of q -theory are examined in other contexts, particularly in asset pricing, where its empirical performance appears more promising.

3.2 Production-Based Asset Pricing

Although q -theory of investment remains empirically challenging in the investment literature, it has become increasingly influential in asset pricing. Its central appeal lies in the link between firms' investment decisions and expected returns through adjustment costs. In particular, when investment is costly to adjust, firms differ in how strongly they re-

spond to investment opportunities, which in turn generates variation in expected returns. Put differently, even if the relationship between investment and q is not one-for-one, the underlying economic mechanism of q -theory remains highly relevant for understanding expected returns in equity markets.

3.2.1 q -Theory in an Asset Pricing Context

The foundational contributions of Cochrane (1991, 1996) famously established the connection between investment, adjustment costs, and expected returns, thereby laying the foundation for production-based asset pricing. At the core of these contributions are the economic mechanisms of q -theory of investment, which subsequently inspired a broad strand of asset pricing research.

In his seminal paper, Cochrane (1991) argues that production-based asset pricing and q -theory are, at a theoretical level, two ways of expressing the same underlying economic mechanism, noting that “in theory, the two specifications are the same” (p. 218). The central intuition is that expected returns are governed by the firm’s marginal rate of transformation (MRT), which plays a role analogous to the marginal rate of substitution (MRS) in standard consumption-based asset pricing. Firms therefore invest until the return to physical investment equals the return required in capital markets. Using U.S. data from 1947 to 1987, he shows that the fitted forecasts of stock returns and investment returns are very similar, with correlations of 0.875 quarterly and 0.938 annually. He also argues that although q -theory fits poorly in much of the investment literature, it appears more convincing in an asset-pricing setting, which relies on higher-frequency variation and time-varying risk premia. Additionally, this formulation delivers more plausible estimates of adjustment costs: the implied fraction of investment lost is about 7%, while the implied fraction of output lost is only around 1%, helping to address the common criticism that the q -theory of investment requires implausibly large frictions (Cochrane, 1991).

By the same line of logic, Cochrane (1996) extends the production-side intuition from a time-series setting to the cross-section of returns and places greater emphasis on the convex adjustment-cost function. Specifically, he argues that “some adjustment cost is necessary to produce investment return variation anything like that observed in market returns” (p. 574). Using 186 quarterly observations, he constructs an investment factor model that includes both residential and non-residential aggregate investment. The empirical results are encouraging, though they still leave scope for further exploration. Based on the root mean square pricing error (RMSE), the model performs roughly in line with the CAPM for raw returns, with RMSEs of 0.11% and 0.09%, respectively. However, when both returns and factors are conditioned on lagged term spreads and dividend-price ratios to account for time-varying risk premia, the investment model performs substantially better, yielding an RMSE of 0.19% compared with 0.49% for the CAPM. The paper thus broadens the theory by showing that, once the framework moves beyond constant risk premia and allows for time-varying expected returns, it can “rehabilitate the q -theory view of investment” (p. 617).

As such, the initial work by Cochrane (1991, 1996) suggests that production-based

asset pricing models are not only closely related to the q-theoretic economic mechanism of firm behavior, but are more tractable when expressed in terms of asset prices, where investment, shaped by adjustment frictions, links firm decisions to expected returns.

3.2.2 Further Developments of Production-Based Asset Pricing

Subsequent empirical work builds on the production-based asset pricing model by taking its core mechanism to the data. The main implication is that the model performs reasonably well in explaining the cross-section of average returns, both relative to standard benchmark models and for conventional portfolio-sorts. Hence, these findings further support the empirical relevance of q-theory in an asset pricing context.

An early attempt to quantify this mechanism appears in Liu, Whited, and Zhang (2009), where stock returns are equal to levered investment returns under constant returns to scale. Using GMM on portfolios sorted by earnings surprises, book-to-market, and capital investment, the model performs well when asked only to match average returns. For earnings-surprise deciles, the mean absolute pricing error (MAE) is 0.7% per year, compared with 5.7% for the CAPM, 4.0% for the Fama–French three-factor model, and 3.6% for the standard consumption CAPM. Furthermore, across high-minus-low portfolios, pricing errors are similarly small: 0.4% for earnings surprises, 1.2% for book-to-market, and 0.5% for investment, versus errors ranging from about 6% to 18% in the competing models. However, once the model is required to match return variances as well, the implied expected-return errors become much larger. The key takeaway is therefore that the q-theoretical mechanism can be powerful in explaining average returns, relative to conventional benchmark models in the asset pricing literature.

A complementary extension is provided by Belo (2010), who embeds the production-side logic in an explicit stochastic discount factor (SDF) framework at the aggregate level. Rather than focusing on investment returns, the paper recovers the SDF directly from the MRT implied by firms’ production decisions, thereby establishing a formal equivalence between the MRT and the SDF. In a two-sector setting with durable and nondurable goods, the model links cross-sectional variation in expected returns to assets’ covariance with this production-based SDF. Empirically, using six Fama–French portfolios sorted on size and book-to-market, the model explains a substantial share of the cross-sectional variation in average returns, delivering an R^2 of 95.1% and a low mean absolute pricing error of 1.15% per annum. Moreover, the estimated MRT varies from about -0.24 in good states to 0.49 in bad states, implying that assets with more negative covariance with the SDF earn higher expected returns, exactly as expected. These findings are important because they show that production-based asset pricing can be embedded in a fully risk-based SDF representation, while also suggesting that such models do a relatively good job of capturing the risk-return trade-off across several portfolio sorts.

In this light, the empirical asset pricing literature suggests that production-based models compare fairly well with several alternative models. Above all, it suggests that the approach is most compelling not as a purely theoretical construction, but as an empirical lens on the cross-section of expected returns.

3.2.3 The Role of Adjustment Costs and the Cross Section

More recent work places even greater emphasis on the role of adjustment costs, a point already emphasized in the foundational contributions of Cochrane (1991, 1996). At the same time, the literature increasingly suggests that the main empirical strength of the production-based asset-pricing model lies more specifically in explaining cross-sectional return differences rather than time-series variation in returns.

Within this line of research, Li and Zhang (2010) move closer to a direct test of q-theory by asking whether the negative relation between investment and expected returns is stronger among firms facing higher adjustment costs. They proxy adjustment costs using observable financing-constraint measures such as firm size, payout ratios, and bond ratings. Using data from 1963–2008, they sort firms along these dimensions and estimate Fama–MacBeth regressions for several investment-related anomalies. The evidence is best described as qualified. For the investment-to-assets anomaly, the estimated slope is -0.85 in the small-asset tercile versus -0.33 in the large-asset tercile, -0.93 in the low-payout tercile versus -0.39 in the high-payout tercile, and -0.86 for firms without bond ratings versus -0.47 for firms with ratings. Yet this pattern is concentrated mainly in investment-to-assets and, more tentatively, asset growth, and it weakens in robustness tests. The paper is therefore informative without being conclusive: it offers only limited and indirect support for a friction-based interpretation. However, in our view, their measures are at best indirect and potentially poor proxies for true adjustment costs, as they appear more closely related to the financing constraints referenced in the cash flow debate, highlighting the broader difficulty of cleanly identifying adjustment frictions in the data.

A complementary perspective is offered by Belo, Deng, and Salomao (2024), who clarify where production-based asset pricing appears to be most empirically successful. Their results show that standard neoclassical investment models that rely on convex adjustment costs fit the cross-section of expected returns reasonably well, but perform poorly when confronted with the time-series aspect of returns. When the model is estimated using only cross-sectional moments, it attains a cross-sectional R^2 of 71% and pricing errors of roughly 1.4% per year, whereas the time-series R^2 falls to -93% . Moreover, once estimation begins to place weight on the model's time-series implications, a sharp trade-off emerges: gains in time-series fit come at the expense of markedly worse cross-sectional performance, and even an estimation designed to maximize time-series fit raises the time-series R^2 only to -4% . The broader implication is that production-based models appear far better suited to explaining differences in average returns across assets than to tracking realized return movements over time.

Overall, the literature points to two related conclusions. First, adjustment costs remain central to any credible production-based account of expected returns, yet they remain difficult to measure convincingly. Second, q-theory appears to be more empirically successful in the cross-section than in the time series. Together, these points suggest that the most promising next step is to study adjustment frictions more directly in relation to cross-sectional return differences.

3.3 Factor Models

While q-theory of investment has become increasingly influential in production-based asset pricing, factor models have gained broader prominence because of their strong empirical performance. Yet this literature remains contested due to persistent anomalies and its looser theoretical foundations. At the same time, the mechanisms underlying the Fama–French Five-Factor Model, which often serves as the leading benchmark, appear closely related to forces emphasized in q-theory. In this light, we argue that direct estimates of adjustment costs may provide a natural extension and help sharpen the interpretation of the Fama–French factors.

3.3.1 The Failure of CAPM

The Capital Asset Pricing Model (CAPM) provides a foundational framework for linking expected returns to systematic risk and was developed by Treynor (1961), Sharpe (1964), Lintner (1965), and Mossin (1966). It predicts that an asset’s expected return is determined solely by its sensitivity to the market portfolio: no firm characteristic should matter once market beta is controlled for. While theoretically appealing, early empirical tests began to question whether this single-factor structure adequately explains the cross-section of stock returns.

Some of the earliest influential tests already pointed to this problem. In portfolio-based tests, Black, Jensen, and Scholes (1972) show that high-beta portfolios have negative alphas and low-beta portfolios have positive alphas, with estimated intercepts ranging from about -0.19 to $+0.20$. This implies a flatter security market line than the CAPM predicts. Similarly, Fama and MacBeth (1973) find that the estimated market risk premium is positive on average, but weak and unstable over time. Using their two-pass regression approach, they report that from 1935 to 1940 the estimated premium is about 1.1% per month, but with a standard deviation of 11.6%, yielding a t -statistic of only 0.79. Rather than confirming the model, these results hence suggest that market beta is a weaker and less robust determinant of expected returns than the CAPM predicts. Even so, the exact implications for the CAPM remained debated, especially in light of what became known as ‘Roll’s critique’. Specifically, Roll (1977) argues that the CAPM cannot be tested cleanly because the true market portfolio is unobservable, casting doubt on how empirical tests of the model should be interpreted. Hence, these early contributions call into question the empirical performance of CAPM without fully rejecting it.

Later evidence made these shortcomings harder to ignore. Using the Fama–MacBeth regression approach, Fama and French (1992) examine a broader universe of NYSE, AMEX, and NASDAQ stocks from 1963 to 1990 and test whether market beta is sufficient to explain the cross-section of average returns once firm characteristics such as size and book-to-market are considered. They find that the estimated slope on market beta is only about 0.15% per month, with a standard error of 0.46, suggesting little relation between beta and average returns. At the same time, while market beta has little explanatory power, they document notable variation in average returns across both size and book-to-

market portfolios. Average returns fall from roughly 1.64% to 0.90% per month across size portfolios, while they rise from about 0.30% to 1.83% per month across book-to-market portfolios. In empirical terms, the *size* and *value* effects came to the forefront, not only as important empirical regularities, but also in the broader debate over the CAPM's explanatory power.

Taken together, the early empirical evidence does not imply that the CAPM is useless, but it did spark a broader academic debate. Market beta may matter in some settings, yet market risk alone does not appear sufficient to explain the full cross-section of expected returns. Instead, characteristics such as size and book-to-market seem to capture variation that beta leaves unexplained, thereby paving the way for the rise of multifactor asset pricing models.

3.3.2 Fama–French Three-Factor Model

Following the empirical weaknesses of the CAPM, Fama and French (1993) propose their three-factor model, augmenting the market factor with size and value factors to better explain the cross-section of expected returns. Although the model proves empirically successful, its economic interpretation remains contested.

In their pioneering paper, Fama and French (1993) develop their three-factor model by augmenting the market factor with a size factor (SMB) and a value factor (HML), both constructed as long-short portfolios based on market capitalization and book-to-market. To evaluate the model, they form 5×5 test portfolios sorted on size and book-to-market and estimate time-series regressions using monthly U.S. stock returns from 1963 to 1991. They find that average excess returns across the test portfolios range from 0.32% to 1.05% per month, generally declining with size and rising with book-to-market, and that the three-factor model sharply improves model fit: 21 of 25 portfolios have R^2 values above 0.90, compared with only 2 under the CAPM alone. They therefore conclude that the three-factor model captures much more of the cross-sectional variation in stock returns than a market-beta-only specification.

Yet this empirical success did not settle the underlying economic interpretation. Shortly after, Lakonishok, Shleifer, and Vishny (1994) challenge the risk-based view, arguing that value strategies outperform because investors systematically overprice growth stocks and underprice value stocks. In this respect, they report that value stocks earn average annual returns of 19.8%, compared with 9.3% for growth stocks. By contrast, Fama and French (1995) argue for a more rational-pricing interpretation by relating the size and value effects to long-run earnings fundamentals. Among small stocks, they show that high-value portfolios earn average monthly returns of 1.04% versus 0.49% for low-value portfolios, while among big stocks the corresponding returns are 0.72% and 0.38%. Accordingly, they interpret size and book-to-market as reflecting differences in firms' long-run profitability. Taken together, this debate suggests that although the empirical performance of the factors is well documented, their deeper economic interpretation remains open.

This line of research also prompted a reconsideration of whether the CAPM had been rejected too quickly. Concretely, Kothari, Shanken, and Sloan (1995) argue that its weak performance may partly reflect methodological choices. Using U.S. data from 1927 to 1990, they find that the market risk premium becomes economically large when betas are estimated from annual rather than monthly returns: 8.9% to 11.7% with the equally weighted index and 6.2% to 8.9% with the value-weighted index. They further suggest that the book-to-market effect is weaker because of survivorship bias in Compustat, with returns for small Compustat firms being 9 to 10 %-points higher than for small firms in the CRSP–Compustat sample and the estimated book-to-market effect about 40% lower among the 500 largest Compustat firms. In direct response, Fama and French (1996) challenge this interpretation, showing that annual and monthly market beta estimates are almost perfectly correlated, from 0.98 to 1.00, that returns in beta-decile portfolios rise only from 1.10% to 1.39%, and that size effects remain strong, with the smallest size decile earning 2.18% per month versus 0.87% for the largest. Finally, Fama and French (2004) later place this debate in a broader perspective through a comprehensive literature review and conclude that the central empirical problem persists. Specifically, the beta-return relation is too flat, while size and price-ratio variables explain return variation missed by market beta. Nevertheless, the persistence of the CAPM debate highlights a key point: despite weak empirical performance, it remains a theoretical cornerstone, whereas the Fama–French three-factor model, though empirically powerful, is theoretically more contested.

In sum, the Fama–French three-factor model marks a significant advance in empirical asset pricing by more effectively explaining the cross-section of expected returns. Yet despite its explanatory power, it remains debated because it still lacks a common theoretical grounding. Just as importantly, its empirical success helped inspire a growing number of additional factors constructed primarily from observed return patterns.

3.3.3 Factor Zoo, Anomalies and q-Theory of Investment

The empirical success of the Fama–French three-factor model helped launch a broader wave of proposed factors beyond market, size, and value. Since many of these factors were largely empirical constructs, scholars increasingly called for a stronger theoretical foundation, a concern that became known as the ‘*factor zoo*’. While several avenues have since been explored, the q-theory of investment stands out as one promising direction.

This broadening of the factor literature is reflected in a series of influential studies that identify return patterns beyond the original market, size, and value factors. Among the most cited are Jegadeesh and Titman (1993), Titman, Wei, and Xie (2004), and Novy-Marx (2013). To begin, Jegadeesh and Titman (1993) document the *momentum* anomaly by showing that strategies that buy past winners and sell past losers generate significant positive abnormal returns over 3 to 12 month horizons, with the most profitable strategy earning 1.49% per month using NYSE and AMEX stocks between 1965 and 1989. Beyond momentum, Titman, Wei, and Xie (2004) identify *investment* as another important anomaly, showing that low-investment firms outperform high-investment firms

by about 0.168% per month, even after controlling for standard factors. Along similar lines, Novy-Marx (2013) shows that *gross profitability* predicts returns, with profitable-minus-unprofitable firms earning 0.31% per month and a Fama–French three factor alpha of 0.52% per month. Together, these studies moved the literature beyond the original three-factor structure and revealed new dimensions of expected returns that appeared empirically important.

As this literature expanded, a central question became whether the growing number of return predictors reflected distinct sources of priced risk or overlapping empirical manifestations of similar underlying patterns. In this spirit, Cochrane (2011) coined the term ‘factor zoo’, arguing that much of the literature reflected a largely empirical search for patterns rather than a theory-driven understanding of asset pricing. By the same token, more recent evidence by Jensen, Kelly, and Pedersen (2023) suggests that this proliferation does not imply an ever-growing set of independent anomalies. Using a global data set of 153 factors across 93 countries, they show that these factors can be grouped into 13 overarching themes, indicating that many proposed factors may represent related implementations of a smaller set of common return patterns. Their findings thus suggest that the factor zoo may largely reflect overlapping empirical manifestations of broader themes, which only strengthens the need for more grounded theory.

One possible avenue in this respect is the q-theory of investment, which may provide a useful lens through which to interpret several anomalies in the factor literature. Hou, Xue, and Zhang (2015) develop an empirical q-factor model consisting of the market factor, a size factor, an investment factor, and a profitability factor. Based on this model, they report that 38 of their 74 primary anomalies are statistically insignificant in the broad cross-section in the period between 1972 and 2012. Among the remaining 35 significant anomalies², the q-factor model reduces the average absolute alpha to 0.20% per month, compared with 0.33% in the Carhart model and 0.55% in the Fama–French three-factor model. The broader lesson is therefore not that the anomaly literature is uninformative, but that its proliferation has strengthened the case for more grounded theory, with q-theory offering one promising candidate in this respect.

Thus, as the asset pricing literature moved beyond the CAPM toward broader multifactor structures, the expansion of the factor zoo reinforced a central point: empirical success alone is not enough, and progress requires a stronger theory that can relate these anomalies to common sources of risk and to an underlying economic mechanism.

3.3.4 Fama–French Five–Factor Model and q-Theory

At the forefront of the asset pricing literature stands the Fama–French Five-Factor model, which extends the market, size, and value factors with profitability and investment factors. Like the earlier three-factor specification, it has become highly influential because of its strong empirical performance and is now widely used as a benchmark in both academia

²The numbers do not sum exactly, since $74 - 38 \neq 35$, seemingly reflecting differences in how anomalies are classified and included across tests in Hou, Xue, and Zhang (2015).

and practice. Interestingly, we believe that the economic logic of the model bears a close resemblance to q-theory of investment.

As a direct extension of their earlier work, Fama and French (2015) propose a five-factor specification comprising the market factor, size (SMB), value (HML), profitability (RMW), and investment (CMA). Using time-series regressions on U.S. stock returns between 1963 and 2013, they preserve the basic structure of their earlier tests but expand the empirical design to include both 5×5 and $2 \times 4 \times 4$ test portfolio sorts. Their central result is straightforward: the five-factor model improves materially on both the three-factor model and the CAPM, delivering lower GRS statistics and smaller average absolute pricing errors across all six portfolio sets. For example, the average absolute pricing error falls from 0.102% to 0.094% for the 25 Size-B/M portfolios and from 0.182% to 0.103% for the 32 Size-OP-Inv portfolios, while the corresponding GRS statistics decline from 3.62 to 2.84 and from 4.38 to 2.92. As a result, the five-factor model came to occupy a central position as the new benchmark in empirical asset pricing.

Beyond its initial success, Fama and French (2016, 2017) show that the five-factor model remains empirically strong, particularly as the profitability and investment factors help absorb variation left unexplained by the three-factor model. In the U.S. market, Fama and French (2016) show that adding RMW and CMA reduces monthly average absolute pricing errors for several anomaly portfolios relative to their previous model, from 0.106% to 0.072% for Size- β portfolios, from 0.136% to 0.098% for Size-NI portfolios, from 0.217% to 0.131% for Size-Var portfolios, and from 0.222% to 0.120% for Size-RVar portfolios. Internationally, Fama and French (2017) show that when factors are constructed regionally, the five-factor model generally improves especially along the profitability and investment dimensions. For example, the average absolute alpha in the 32 Size-OP-Inv portfolios falls from 0.21% to 0.13% in North America, from 0.25% to 0.13% in Europe, and from 0.36% to 0.21% in Asia Pacific compared to their previous model. Taken together, the evidence suggests that profitability and investment capture key dimensions of the cross-section that previous models left unexplained.

The strength of the profitability and investment factors is particularly interesting because, at a broad level, they may appear economically consistent with the logic of q-theory. In fact, Fama and French (2015) themselves note that their specification is closest to the q-factor model by Hou, Xue, and Zhang (2015), which likewise emphasizes profitability and investment but omits HML. This is notable because the evidence by Fama and French (2015) themselves highlights that HML becomes largely redundant once profitability and investment are included: in their factor spanning tests, the HML intercept is small, at 0.04% per month, and statistically insignificant at conventional significance levels, while the correlation between HML and CMA is 0.70. Moreover, a four-factor model excluding HML performs almost identically to the full five-factor model: for the 25 Size-B/M portfolios, the GRS statistic is 2.84 under both specifications, for the 25 Size-OP portfolios it is likewise identical at 1.87, and for the 32 Size-OP-Inv portfolios it is 2.92 in both cases, with average absolute pricing errors that are virtually identical across specifications. In our view, these results are interesting because, to the extent that the value factor proxies for the inverse of average q, the redundancy of HML

and its close connection to CMA are roughly consistent with the well-known investment-q relationship. This remains an ambitious interpretation, but the empirical patterns seem sufficiently intuitive to raise the possibility that the Fama–French five-factor model may, at least in part, reflect some of the mechanism emphasized by q-theory.

In line with the rising popularity of the Fama–French Five-Factor model, a similar pattern seems to reappear: scholars and practitioners are equipped with an augmented and empirically stronger model of the cross-section that is once again subject to academic debate. However, this time, while not explicitly emphasized by Fama and French, the evidence appears much more closely related to something we have seen before, namely the q-theory of investment. While this remains an ambitious interpretation, we believe that it at least warrants further exploration.

3.4 Partial Conclusion

The investment and asset pricing literatures may be seen as two sides of the same coin: q-theory remains influential because of its theoretical coherence despite persistent empirical challenges, whereas the Fama–French Five-Factor model remains central because of its empirical strength even with a less explicit theoretical foundation. As we will argue, however, the literature may point to a deeper connection, namely that the five-factor model may reflect mechanisms closely related to q-theory of investment.

In the investment literature, the broad lesson is not that q-theory fails entirely, but that its empirical implementation remains imperfect. Although average q is generally related to investment, its explanatory power is often weak and sensitive to specification, measurement error, and financing frictions. The literature therefore leaves a puzzle rather than a clear rejection: q-theory remains theoretically influential, but its empirical performance is less convincing than its elegance would suggest. In this light, only the conventional measures of average q and the standard linear investment-q relationship seem to remain a common benchmark among scholars.

In parallel, a related line of research has gained prominence in asset pricing, namely production-based asset pricing. This literature builds directly on the q-theory of investment and suggests that its explanatory power strengthens once risk premia are allowed to vary over time, while performing relatively well in the cross-section. However, one issue remains unresolved. Although investment appears linked to expected returns, the precise role of adjustment costs remains insufficiently understood, despite being a central economic force. Effectively, this leaves a clear need for more structural work on adjustment-cost heterogeneity in the cross-section.

If we turn from production-based asset pricing to mainstream factor models, the Fama–French Five-Factor model stands at the forefront in both academia and practice. Yet because it lacks a fully explicit theoretical foundation, it has not fully replaced the CAPM. Against this backdrop, we argue that the five-factor model may be more closely related to the q-theory of investment than is usually acknowledged. In particular, the prominent roles of the investment and profitability factors appear intuitively connected

to the q-theory of investment. Similarly, the redundancy of the value factor, together with its high correlation to the investment factor, appears broadly consistent with the investment-q relationship, since book-to-market can be interpreted as an inverse valuation signal related to average q.

This interpretation is admittedly ambitious, but we believe the literature provides a sufficient basis to pursue it further. From this perspective, the natural next step is to bring adjustment costs explicitly into the empirical analysis.

4 Hypothesis Development

The theoretical background and the empirical literature point to a common mechanism, but also to an unresolved gap: the q-theory of investment is empirically strong in an asset pricing setting, yet to fully understand its economic relevance, further research is needed on the role of adjustment costs, and more particularly the speed of adjustment.

Theoretically, the q-theory of investment implies that firms respond to aggregate shocks through investment, and that this response is governed by adjustment frictions. The adjustment-cost parameter χ determines how quickly firms move toward their steady-state capital stock. Heterogeneity in χ may therefore affect how persistently shocks influence future operating profits and firm value. Since risk premia depend on how returns co-move with good and bad states of the economy, such heterogeneity may matter for expected returns. However, because this paper treats the production and consumption sides separately, it remains ambiguous whether χ should be associated with a clear risk premium. The claim is therefore more modest. If q-theory helps explain how firm value responds to aggregate shocks, and if risk premia reflect state-dependent payoffs, then adjustment-cost heterogeneity may plausibly be reflected in the cross-section.

The literature supports this intuition. Although the q-theory of investment has faced empirical challenges in investment settings, it has performed more convincingly in asset pricing. More generally, the literature suggests that q-theory performs particularly well in the cross-section and that adjustment costs are central to this result. At the same time, multi-factor models remain widely used because of their empirical strength, despite their weaker theoretical foundation. The Fama–French Five-Factor Model is especially relevant here as its performance is difficult to ignore, while its profitability, investment, and value factors appear sufficiently close to q-theory and the broader production-based literature. Moreover, Fama and French themselves note that their model is closest to the alternative q-factor model, although they do not make a stronger theoretical claim.

In light of these considerations, we argue that it is both theoretically plausible and empirically interesting to consider the Fama–French Five-Factor Model in the context of the adjustment-cost parameter. By doing so, we aim to clarify the role of adjustment-cost heterogeneity in production-based asset pricing and to provide deeper theoretical insight into the model.

4.1 Adjustment Frictions and Risk Premia

Firms with higher adjustment costs converge more slowly toward their steady state after economic shocks, which makes the effects on firm value more persistent. If expected returns reflect how firm payoffs covary with aggregate economic conditions, heterogeneity in the speed of adjustment, χ , may therefore generate systematic differences in risk premia. The central question, however, concerns the sign of the risk premium associated with χ , for which the theory developed in this paper does not yield a clear prediction. To guide the expected direction, we therefore turn to the asset-pricing literature, which generally finds that firms with lower investment rates earn higher expected returns. From the investment– q relationship, investment is primarily determined by the q -signal and moderated by χ . Holding q constant, a higher χ therefore implies a lower investment rate. If, all else equal, higher adjustment costs are the reason firms invest less, then high- χ firms may plausibly carry higher risk premia. We therefore label high- χ firms as ‘slow’ and low- χ firms as ‘quick’, reflecting their speed of adjustment, and construct a slow-minus-quick factor, SMQ, consistent with the Fama–French Five-Factor setup.

Hypothesis 1. *A tradable factor constructed from heterogeneity in adjustment costs, as a slow-minus-quick χ spread, earns a positive risk premium.*

Since this hypothesis is broad, it cannot be evaluated through a single statistical test and instead requires a holistic assessment. In this respect, a χ -related risk premium may appear both in unconditional excess returns and in alphas relative to empirical asset-pricing models, with the strongest evidence arising when both are present. However, special emphasis is placed on performance relative to the Fama–French Five-Factor model, as this specification appears closely related to the dynamic mechanisms underlying the q -theory of investment. Accordingly, the empirical analysis evaluates both the unconditional performance of the constructed adjustment-cost factor, SMQ, and its alphas relative to the benchmark models, with the five-factor specification serving as the central reference point.

4.2 Beyond the Fama-French Five-Factor Model

If the Fama–French Five-Factor model already captures part of the q -theoretic mechanism through its existing factor structure, then augmenting it with the SMQ factor may further improve its explanatory power. The argument is that the existing Fama–French factors may capture aspects of the economic mechanisms underlying the q -theory of investment, but they do not directly measure the speed of adjustment, which determines how quickly firms respond to changing economic conditions. Since SMQ is constructed from χ , it targets this adjustment-cost dimension more directly and may therefore capture cross-sectional return variation that remains unexplained by the five-factor model.

Hypothesis 2. *Augmenting the Fama–French Five-Factor model with an adjustment-cost factor improves empirical asset-pricing performance in the cross-section.*

Evaluating this hypothesis requires a holistic comparison of the Fama–French Five-Factor model and the augmented six-factor model across several sets of test portfolios. If SMQ captures an adjustment-friction dimension that is not fully spanned by the existing five factors, then adding SMQ should reduce the remaining pricing errors. However, such an evaluation is not clear-cut, since it cannot rely only on the number of statistically significant alphas. It must also consider the signs, magnitudes, and cross-sectional structure of the pricing errors across test portfolios. Evidence in favor of the hypothesis would therefore require a broad assessment and economically meaningful improvements in pricing performance, rather than isolated changes in individual portfolio alphas.

Overall, the two hypotheses follow a clear progression. The first asks whether adjustment frictions appear to be priced in the cross-section of returns, while the second asks whether they contain incremental explanatory power beyond the Fama–French Five-Factor model. Jointly, they examine whether heterogeneity in firms’ speed of adjustment helps explain expected returns, and thereby contributes to a clearer economic interpretation of the five-factor model through the lens of the q-theory of investment.

5 Research Design

This section outlines the research design of the study. It first introduces the data sources, sample construction, and firm characteristics underlying the analysis, with explicit attention devoted to the estimation of the adjustment-cost parameter. It then describes the construction of the tradable factors used in the empirical factor models and the test portfolios used to evaluate model performance. Finally, it presents the econometric methods used to assess unconditional excess returns and evaluate the asset-pricing performance of the benchmark models considered in the study.

5.1 Data Sources and Sample Construction

This study uses data from the WRDS CRSP–Compustat merged database, which combines monthly stock returns, the risk-free rate, and annual accounting information. The full sample spans from July 1962 to December 2024 and consists of monthly observations. It is from this full data source that we construct the sorting variables, tradable factors, and test portfolios.

The purpose of this dataset is twofold. First, it enables a full replication of the Fama–French factors, ensuring comparability and consistency with the methodology of Fama and French (1993, 2015). Second, it is used to estimate the adjustment-cost parameter and construct the associated adjustment-cost factor. As such, the sample is constructed using a purposive sampling approach aligned with this objective.

A relevant aspect of the empirical design is the distinction between the baseline universe and variable-specific universes. The baseline universe consists of firms drawn from NYSE, AMEX, and NASDAQ in the merged CRSP–Compustat database and defines the set of firms that may enter the empirical analysis. However, the effective set of firms used

in any given test depends on the availability of the variables required for that specific analysis. This distinction is important because portfolio sorts, factor construction, and estimation procedures rely on different subsets of firms, even though all analyses originate from the same underlying dataset.

5.1.1 Baseline Universe

The analysis is restricted to U.S. firms listed on the major stock exchanges, namely the NYSE, AMEX, and NASDAQ. To ensure comparability with the empirical asset-pricing literature, the sample is further restricted to common stocks, thereby excluding preferred shares, REITs, closed-end funds, and other non-operating securities. In addition, firms in financial industries, corresponding to SIC codes 6000–6999, are excluded following standard practice (Fama & French, 1992). To our understanding, these securities and firms are excluded because they are subject to distinct cash-flow structures, regulatory environments, balance-sheet characteristics, and valuation principles, making key variables such as book equity, debt, leverage, and investment not directly comparable.

5.1.2 Variable-Specific Asset Universes

While all analyses originate from the baseline universe described above, the effective set of firms used in any given estimation, portfolio sort, or factor construction depends on the availability of the specifically required variables. Each empirical exercise, therefore, relies on a variable-specific asset universe, which reflects the general sample restrictions together with the data requirements for the variables involved. For example, when constructing portfolios based on operating profitability, we require firms to have revenue and COGS for their construction. Similarly, the construction of other variables relies on different data inputs, which define separate sets of eligible firms. As a consequence, the cross-section of firms varies across portfolio sorts and factor constructions based on the local asset universe of each variable. These differences arise mechanically from data availability rather than from changes in the underlying common sampling approach.

5.2 Firm Characteristics and q-Theory Variables

This subsection defines the firm-level characteristics used throughout the empirical analysis. The variables serve two related purposes. First, the standard characteristics form the basis for Fama–French-style portfolio formation and replicated factor construction. Second, the q-theory-related variables are used to estimate the adjustment-cost parameter that forms the basis of the paper’s main empirical contribution. To ensure transparency, Appendix A.1 reports summary statistics for the firm characteristics and portfolio counts, while Appendix A.5 compares our replicated factors and portfolio returns with the corresponding data from the Kenneth French Data Library.

5.2.1 Standard Characteristics

The construction of the standard firm characteristics follows the methodology of Fama and French (1993, 2015). Since the original papers do not disclose all implementation details, and since the underlying construction code is not publicly available, the variables are reconstructed as closely as possible based on the available methodological descriptions. The characteristics are constructed using lagged accounting information in order to respect the timing conventions standard in empirical asset pricing.

5.2.1.1 Market Equity

The construction begins with market equity. For each firm i and month t , market equity is defined as

$$ME_{i,t} = |prc_{i,t}| \cdot shrout_{i,t}. \quad (18)$$

where prc denotes the stock price and $shrout$ denotes the number of shares outstanding. Throughout the paper, this characteristic plays a central role in both portfolio formation and return aggregation, since it determines firm size classifications and forms the basis for the value weights used to compute portfolio returns.

5.2.1.2 Book-to-Market

Book-to-market captures valuation signals by the market. Formally, it is defined using book equity and market equity from fiscal year $t - 1$:

$$BM_{i,t} = \frac{BE_{i,t-1}}{ME_{i,Dec(t-1)}}. \quad (19)$$

where $BE_{i,t-1}$ is book equity and $ME_{i,Dec(t-1)}$ is market equity in December of year $t - 1$. December market equity is used to follow the same dating convention as book equity. Book equity measures the accounting value of equity attributable to common shareholders and is defined as

$$BE_{i,t-1} = seq_{i,t-1} + txditc_{i,t-1} - PS_{i,t-1}. \quad (20)$$

where seq denotes total shareholders' equity, $txditc$ denotes deferred taxes, and PS denotes preferred stock. In the implementation, missing values of $txditc$ are set to zero, while observed values are included as reported. Preferred stock is subtracted to obtain book equity attributable to common shareholders and is defined hierarchically using redemption value ($pstkrv$), liquidation value ($pstkl$), or par value ($pstk$), when available. Throughout, firms are required to have positive book equity.

5.2.1.3 Operating Profitability

Operating profitability captures firms' operating efficiency and is defined as

$$OP_{i,t} = \frac{revt_{i,t-1} - cogs_{i,t-1} - xsga_{i,t-1} - xint_{i,t-1}}{BE_{i,t-1}}. \quad (21)$$

where *revt* denotes revenues, *cogs* denotes cost of goods sold, *xsga* denotes selling, general, and administrative expenses, and *xint* denotes interest expense. This is also one of the dimensions in which Fama and French (2015) are not fully transparent about the exact implementation, so the measure is reconstructed to the best of our ability. In particular, operating profitability is computed only for firms with positive book equity, *revt* and *cogs* must be observed; and missing values of *xsga* and *xint* are set to zero. This treatment is consistent with common empirical practice to avoid excluding firms that do not disclose these items (Hou et al., 2018). Importantly, we find that this implementation materially improves the quality of the replication.

5.2.1.4 Investment

Investment captures firms' growth in productive capacity and is measured as the growth rate of total assets

$$Inv_{i,t} = \frac{at_{i,t-1} - at_{i,t-2}}{at_{i,t-2}}. \quad (22)$$

so that investment reflects growth from $t - 2$ to $t - 1$, observable at the June t formation date. Again, the lag in the accounting variables is used to avoid look-ahead bias. In this respect, the use of $t - 2$ in the denominator ensures that the measure captures realized asset growth over a completed fiscal year based entirely on information available at the time of portfolio formation.

5.2.2 q-Theory-Related Variables

In addition to the standard firm characteristics from Fama and French (1992, 2015), used for portfolio formation, the empirical analysis requires a set of q-theory-related variables used for estimating our adjustment-cost parameter.

5.2.2.1 Tobin's q

Following the most common literature benchmark, Tobin's q is constructed as the ratio of firm market value to physical capital

$$q_{i,t} = \frac{(prcc_f_{i,t-1} \cdot csho_{i,t-1}) + (dltt_{i,t-1} + dlc_{i,t-1}) - act_{i,t-1}}{ppeg_{i,t-1}}, \quad (23)$$

where firm market value is defined as the market value of equity ($prcc_f \cdot csho$) plus total debt ($dltt + dlc$) minus current assets (act), and physical capital is proxied by the book value of property, plant, and equipment ($ppeg$). In our empirical implementation, the measure is computed only when firm market value is observed and $ppeg > 5$ million USD, thereby excluding firms with very low levels of physical capital, as is standard in the literature (Erickson & Whited, 2012; Gala et al., 2022; Peters & Taylor, 2017).

5.2.2.2 Investment Rate

The gross investment rate is defined as physical investment scaled by the existing stock of physical capital

$$\frac{I_{i,t}}{K_{i,t-1}} = \frac{capx_{i,t-1}}{ppeg_{i,t-2}}, \quad (24)$$

where investment is measured by capital expenditures ($capx$), and physical capital is proxied by the lagged book value of property, plant, and equipment ($ppeg_{t-1}$). Again, firms with $ppeg_t < 5$ million USD are excluded to avoid distortions from very small or thinly capitalized firms. Note also that this investment rate differs from the investment characteristic used by Fama and French (2015). The reason is that, in this part of the analysis, we aim to remain as consistent as possible with the q-theory literature on variable measurement, since this is crucial for the estimation of the adjustment-cost parameter.

5.3 Estimation of the Adjustment-Cost Parameter

The central contribution of this paper is to bring the adjustment-cost parameter into the Fama–French setting. Because this parameter is inherently unobservable and conceptually abstract, its estimation procedure requires due attention and deserves its own subsection for full transparency on its estimation approach.

5.3.1 Empirical Estimation of the Adjustment-Cost Parameter

In the theory section, it was shown that under a quadratic adjustment-cost function and the Hayashi assumptions, the investment–q relationship implies a linear relation between the investment rate and Tobin’s q. This provides a basis for estimating the adjustment-cost parameter using linear regression methods.

Following Peters and Taylor (2017), the investment–q relationship is estimated using an ordinary least squares (OLS) fixed-effects panel regression

$$\frac{I_{i,t}}{K_{i,t-1}} = \delta_i + \tau_t + \gamma q_{i,t-1} + \eta_{i,t}, \quad (25)$$

where $I_{i,t}/K_{i,t-1}$ is the investment rate, $q_{i,t-1}$ is lagged Tobin’s q, δ_i is the firm fixed effect, τ_t is the year fixed effect, γ is the slope coefficient, and $\eta_{i,t}$ is the error term. This specification is central because, under the theoretical investment–q relationship, the adjustment-cost parameter is given by the inverse of the slope coefficient. It can therefore be expressed as

$$\chi = \frac{1}{\gamma}, \quad (26)$$

where χ denotes the adjustment-cost parameter. This inversion underlies the structural estimation used throughout the paper and yields the empirical estimates of χ . Because χ is used as a sorting variable, it is estimated at each portfolio-formation date using only information available at that time, consistent with the Fama–French timing conventions.

The resulting estimates constitute the key characteristic in the empirical analysis and are used to construct both the adjustment-cost factor and the χ -sorted test portfolios.

5.3.2 Sample Restrictions in Estimating the Adjustment-Cost Parameter

Before proceeding to the remaining procedural specifications, it is important to note that the sample used to estimate the adjustment-cost parameter is further restricted in two ways. First, following standard practice, financial firms, corresponding to SIC codes 6000–6999, utilities, corresponding to SIC codes 4900–4999, and government entities, corresponding to SIC codes 9000 and above, are excluded (e.g., Belo et al., 2022; Peters & Taylor, 2017). These industries are typically excluded because regulatory constraints, public mandates, accounting structures, and sector-specific institutional features may distort the standard relation between investment and Tobin’s q . Second, both Tobin’s q and the investment rate are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme values, consistent with Peters and Taylor (2017). This adjustment is particularly important because the slope coefficient, γ , may be sensitive to outliers, which can produce unstable estimates of the adjustment-cost parameter. The issue is especially relevant given that Tobin’s q has been reported by several scholars to be noisy (e.g., Erickson & Whited, 2000, 2006; Hall, 2004). Together, the sample restrictions and variable adjustments are applied to stabilize the adjustment-cost parameter estimates and to keep the empirical setup as close as possible to Peters and Taylor (2017).

5.3.3 Controlling for Heterogeneity and Endogeneity

To estimate the adjustment-cost parameter as reliably as possible, it is important that the slope coefficient from the OLS regression is not systematically distorted by econometric biases. This is not a trivial task and has been a longstanding concern in the literature. To remain as objective as possible, this paper therefore follows the methodological approach of Peters and Taylor (2017), but still aims to present some basic arguments.

One concern arises from *omitted variable bias*, meaning that a model excludes variables that are correlated with the independent variable and also affect the dependent variable (Stock & Watson, 2020a). While the investment– q relationship theoretically implies that the investment rate should be closely related to q , concerns about omitted variable bias may still arise in empirical applications, as highlighted in the literature review. One issue in this context is the role of fixed effects, namely the possibility that unobserved firm-specific heterogeneity and common time-specific conditions may bias the slope estimator when working with panel data. For example, consider a firm with persistently better management. Such a firm may have a higher q because it is more productive, while also investing more because it is better able to identify and execute profitable projects. If this firm-specific characteristic is not controlled for, the regression may attribute part of this higher investment to q , thereby overstating the effect of q on investment. Similarly, if a macroeconomic boom increases both q and investment across firms, the estimated relationship may partly reflect this common shock rather than the direct effect of q on

investment. Consequently, following Peters and Taylor (2017), we include both firm and year fixed effects in the estimation of the adjustment-cost parameter.

A second concern is the possibility of *simultaneity bias*, meaning that two variables are determined at the same time, making it difficult to establish which one causes the other (Stock & Watson, 2020b). This problem may arise in the investment- q relationship, where both the investment rate and q are determined simultaneously and are inherently interrelated. Because of this, scholars typically use lagged capital and lagged q in their regression specifications (see e.g. Fazzari et al., 1988; Hall, 2004). In line with this literature, as well as Peters and Taylor (2017), we therefore lag both capital stock and q in the OLS fixed-effects panel regressions to reduce simultaneity bias in the slope estimator.

5.3.4 Determination of Estimation Window

A key difference between our paper and Peters and Taylor (2017) is that the adjustment-cost parameter is not treated as a single full-sample estimate. Instead, the parameter is used as a firm characteristic for both factor and test portfolio construction. For that reason, it must be estimated on an ongoing basis, as a full-sample estimate would introduce look-ahead bias into the sorting procedure. Because the estimated adjustment-cost parameter depends directly on the estimation design, the choice of estimation window deserves explicit attention.

This paper uses a 20-year rolling window to estimate the adjustment-cost parameter. The estimation is based on annual regressions, since both the investment rate and Tobin's q are constructed from annual accounting data. Accordingly, for each year t , the parameter is estimated using information from the previous 20 years, from $t - 20$ to $t - 1$, implying a 20-year burn-in period before the first estimate becomes available. In our sample, this means that the first estimate is obtained in 1982, so adjustment-cost estimates are available from 1982 to 2024. The resulting estimates are then merged back into the full dataset, allowing firms to be sorted based on their associated adjustment-cost parameter each year.

The choice of estimation window reflects three considerations. First, the adjustment-cost parameter may itself vary over time. For example, a manufacturing firm in the 1950s that sought to expand into China would likely have adjusted much more slowly than a firm in the 2020s, when digital communication tools such as Microsoft Teams have become standard. We therefore argue that a rolling window is preferable to an expanding window, since it allows the adjustment-cost parameter to account for more recent observations. Second, building on the same line of argument, there is a trade-off in selecting the length of the window: a long window may distort the adjustment-cost parameter by including older, less relevant observations, whereas a short window may suffer from too few observations to produce sufficiently precise estimates. Because of this trade-off, we need to strike a balance. Lastly, there is a practical consideration regarding the availability of the adjustment-cost estimates. A longer burn-in period delays the point at which the first estimates become usable for portfolio formation. We therefore need

to choose a window period and burn-in length that still leaves a sufficiently long time series of adjustment-cost estimates, allowing us to construct factors and test portfolios that remain broadly comparable to the benchmark of Fama and French (2015).

Based on these considerations, we select a 20-year rolling window, as we believe this choice best balances the considerations outlined above. For transparency, Peters and Taylor (2017) work with a 36-year sample period from 1975 to 2011, while our estimation window spans roughly half that length. Equally important, this setup yields adjustment-cost estimates from 1982 to 2024, which still provides a period that is reasonably comparable to the 1963–2013 sample studied by Fama and French (2015). Taken together, we believe that the choice provides a reasonable compromise between time variation, estimation precision, and sample coverage.

5.3.5 Baseline Estimation and Robustness

Since our results are sensitive to the methodology used to estimate the adjustment-cost parameter, we consider both a baseline specification and alternative estimation approaches to assess robustness. In the following, we outline the main considerations underlying the baseline specification and the robustness checks.

Our baseline specification estimates the OLS fixed-effects panel regression after pooling firms at the industry level. Pooling firms at the industry level is widely used in the literature and, to our knowledge, represents one of the more established approaches (e.g., Belo et al., 2022; Gertner et al., 2002; Peters & Taylor, 2017). However, the literature does not appear to provide a single unified definition of how industries should be measured. As such, we define industries using 4-digit SIC codes, as we believe this classification provides a relatively granular industry definition while still allowing for a sufficient number of observations within each industry.

The baseline method is, nevertheless, not without limitations. In particular, the estimation involves a trade-off between firm-level noise and aggregation bias. More granular estimation may better capture heterogeneity in adjustment costs, but may also produce noisier estimates due to measurement error and limited time-series variation. Broader pooling, by contrast, may yield more stable estimates but obscure relevant heterogeneity within groups (Belo et al., 2022). To assess the robustness of our results to this trade-off, we estimate the adjustment-cost parameter at both the major-group level and the firm level. At the major-group level, we retain the OLS fixed-effects panel specification but define industries using 2-digit SIC codes. This provides a more aggregated, but potentially more stable, alternative to the 4-digit SIC benchmark. At the firm level, we estimate the slope separately for each firm using an OLS time-series regression. Since firms are no longer pooled, the model does not have a panel structure and therefore cannot include firm and year fixed effects. Furthermore, the number of observations underlying each estimate is also materially smaller. To reduce excessive noise from very short firm-level time series, we require at least five usable observations before estimating a firm-level adjustment-cost parameter. This restriction may introduce some survivorship bias, but the specification remains informative because it tests whether the results persist when

adjustment costs are estimated at the most granular level.

Overall, the baseline specification seeks to balance granularity and estimation stability by estimating adjustment costs at the industry level. However, given that the results presented in this paper are sensitive to the research design, we also conduct robustness checks to examine whether the findings persist under both broader industry aggregation and fully firm-level estimation.

5.4 Portfolio Formation

The portfolio construction serves two main purposes. First, it enables the construction of tradable long–short factors, including both the Fama–French five factors and our adjustment-cost factor. Second, it provides diversified test portfolios used to evaluate the performance of empirical asset-pricing models. Since both our factors and test portfolios aim to replicate the results in the Kenneth French Data Library (2026), the constructed factors and portfolios are directly comparable to established benchmarks in the literature. Again, for transparency, Appendix A.5 compares our replicated factors and portfolio returns with the corresponding data from the Kenneth French Data Library.

5.4.1 General Portfolio Formation Procedure

Before turning to the specific factor and test-portfolio constructions, this subsection outlines the common portfolio formation procedure used across most portfolio sorts. These shared elements include the timing convention, NYSE breakpoints, size splitting, and the monthly computation of value-weighted returns.

5.4.1.1 Portfolio Formation Timing

All portfolios are formed annually at the end of June in line with the timing conventions of Fama and French (1993, 2015). To be specific, firms are assigned to portfolios in June of year t using only information available at that date, and the resulting portfolios are held from July of year t to June of year $t + 1$. Consistent with the asset-pricing literature, size is measured by market equity at the June t formation date, whereas accounting-based characteristics are measured using fiscal-year $t - 1$ information, and fiscal-year $t - 2$ where required. For the adjustment-cost parameter, which is specific to our contribution, the estimate used in June of year t is based on regressions that use information available only up until year $t - 1$.

5.4.1.2 NYSE Breakpoints

The assignment of firms to portfolios is determined using NYSE-based breakpoints. For each sorting variable, the relevant breakpoints are computed from NYSE firms in June of year t , and these breakpoints are then applied to the full universe of NYSE, AMEX, and NASDAQ firms. This follows the standard portfolio construction methodology by Fama and French (1993, 2015). The use of NYSE breakpoints is important because the exchange composition of the sample is not neutral. As Fama and French (1993) note,

most AMEX and NASDAQ stocks are smaller than the NYSE median firm. These firms may be numerous in terms of the number of listed companies, but they account for a much smaller share of total market capitalization. Consequently, computing breakpoints from the full NYSE–AMEX–NASDAQ universe would make the thresholds more sensitive to the large number of relatively small firms outside the NYSE, even though these firms are less important in value-weighted terms. Using NYSE breakpoints hence helps prevent portfolio definitions from being dominated by microcaps and makes the resulting portfolios more comparable to the empirical asset-pricing literature.

5.4.1.3 Size Splits

Across all factors and test portfolios, except for the market portfolio and the χ -decile portfolios, size serves as a central sorting dimension. This follows standard portfolio construction methodology in empirical asset pricing, as used by Fama and French (1993, 2015). Size is important because characteristic-based return patterns may differ substantially across size groups. Fama and French (2008), for example, show that some anomalies are concentrated among microcaps, while others are more pervasive across both small- and big-cap stocks. Using size as a primary sorting dimension therefore makes it possible to assess whether return patterns associated with book-to-market, profitability, investment, or χ persist across size groups, rather than being driven by a particular segment of the market.

5.4.1.4 Monthly Rebalancing and Return Computation

Unlike the characteristic portfolio sorts, which are formed annually, portfolio returns are calculated on a monthly basis (Fama & French, 1993, 2015). Specifically, portfolio returns are computed as a value-weighted average

$$R_{p,t} = \frac{\sum_{i \in p} ME_{i,t-1} \cdot R_{i,t}}{\sum_{i \in p} ME_{i,t-1}}, \quad (27)$$

where $R_{p,t}$ denotes the return on portfolio p in month t , $R_{i,t}$ is the return on stock i in month t , and $ME_{i,t-1}$ is firm i 's market equity at the end of month $t - 1$. This formula calculates the portfolio return as a weighted average of individual stock returns, where larger firms receive greater relative weight. Value-weighting reflects the relative economic size of firms and ensures that portfolio returns are not driven disproportionately by small-cap stocks (Fama, 1998). Additionally, using $ME_{i,t-1}$ ensures that portfolio weights are predetermined at the beginning of the return period, while $R_{i,t}$ reflects the realized return during that period. Ultimately, the value-weighted return ensures that each firm's contribution to portfolio performance is proportional.

5.4.2 Factor Construction

This subsection outlines the construction of the traded factors used in the empirical analysis. The factor set consists of the Fama–French five factors together with an adjustment-cost factor constructed from the estimated adjustment-cost parameter, χ . Since these

factors form the core explanatory variables in the subsequent asset-pricing tests, it is important to describe their construction in a clear and transparent manner.

5.4.2.1 General Factor Construction Procedure

Factor construction follows the standard asset-pricing procedures of Fama and French (1993, 2015). First, the market factor is defined as the excess return on the market portfolio for the full U.S. equity universe. By contrast, the remaining factors are constructed using 2×3 portfolio sorts, with SMQ representing an analogous extension of the standard Fama–French procedure. Firms are independently sorted by size and by the relevant characteristic using NYSE breakpoints. The size sort classifies firms as small (S) and big (B) using the NYSE median, while the characteristic sort classifies firms into three groups using 30/40/30 NYSE breakpoints. The intersection of these sorts produces the underlying 2×3 portfolios used in factor construction. The size factor (SMB) is constructed as an average of the long–short portfolios between small and big firms across the underlying 2×3 sorts for value, profitability, and investment. Meanwhile, the remaining factors for value (HML), profitability (RMW), investment (CMA), and adjustment-cost (SMQ) are constructed as long–short portfolios using the extreme characteristic portfolios, while the middle 40% is excluded. Unlike the market factor, these long–short factors are already self-financing portfolio returns and therefore do not require a separate subtraction of the risk-free rate (Munk, 2024d).

5.4.2.2 The Fama–French Five Factors

We begin with the Fama–French five factors, which we seek to replicate. This provides the benchmark factor set for the empirical analysis and ensures that our constructed factors remain directly comparable to the standard asset-pricing factors. To our best ability, we aimed to follow the same approach as Fama and French (1993, 2015).

The Market Factor. The market factor is constructed as the excess return

$$MKT_t = R_{M,t} - R_{f,t}, \quad (28)$$

where $R_{M,t}$ denotes the value-weighted return on the full U.S. equity universe, corresponding to all firms in NYSE, AMEX, and NASDAQ, and $R_{f,t}$ denotes the monthly risk-free rate, proxied by the one-month U.S. Treasury bill. Unlike the remaining factors, the market factor is not formed from characteristic-sorted portfolios.

Value (High-minus-Low). For book-to-market, firms are classified as low (L), medium (M), or high (H). The value factor is defined as

$$HML_t = \frac{1}{2} (R_{S,H,t} + R_{B,H,t}) - \frac{1}{2} (R_{S,L,t} + R_{B,L,t}), \quad (29)$$

where $R_{S,H,t}$ and $R_{B,H,t}$ denote the returns on the small and big high book-to-market portfolios, while $R_{S,L,t}$ and $R_{B,L,t}$ denote the returns on the corresponding low book-to-market portfolios. The factor thus captures the return spread between value and growth stocks.

Profitability (Robust-minus-Weak). For operating profitability, firms are classified as weak (W), medium (M), or robust (R). The profitability factor is defined as

$$RMW_t = \frac{1}{2} (R_{S,R,t} + R_{B,R,t}) - \frac{1}{2} (R_{S,W,t} + R_{B,W,t}), \quad (30)$$

where $R_{S,R,t}$ and $R_{B,R,t}$ denote the returns on the small and big robust operating profitability portfolios, while $R_{S,W,t}$ and $R_{B,W,t}$ denote the returns on the corresponding weak operating profitability portfolios. The factor thus captures the return spread between firms with high and low operating profitability.

Investment (Conservative-minus-Aggressive). For investment, firms are classified as conservative (C), medium (M), or aggressive (A). The investment factor is defined as

$$CMA_t = \frac{1}{2} (R_{S,C,t} + R_{B,C,t}) - \frac{1}{2} (R_{S,A,t} + R_{B,A,t}), \quad (31)$$

where $R_{S,C,t}$ and $R_{B,C,t}$ denote the returns on the small and big conservative investment portfolios, while $R_{S,A,t}$ and $R_{B,A,t}$ denote the returns on the corresponding aggressive investment portfolios. The factor thus captures the return spread between firms with conservative and aggressive investment.

Size (Small-minus-Big). The size factor is constructed as the average of the size spreads obtained from the three underlying Fama–French characteristics

$$SMB_t = \frac{1}{3} (SMB_t^{HML} + SMB_t^{RMW} + SMB_t^{CMA}), \quad (32)$$

where SMB_t^{HML} , SMB_t^{RMW} , and SMB_t^{CMA} denote the size spreads constructed from the book-to-market, profitability, and investment sorts, respectively. These spreads are defined as

$$SMB_t^{HML} = \frac{1}{3} (R_{S,L,t} + R_{S,M,t} + R_{S,H,t}) - \frac{1}{3} (R_{B,L,t} + R_{B,M,t} + R_{B,H,t}), \quad (33)$$

$$SMB_t^{RMW} = \frac{1}{3} (R_{S,W,t} + R_{S,M,t} + R_{S,R,t}) - \frac{1}{3} (R_{B,W,t} + R_{B,M,t} + R_{B,R,t}), \quad (34)$$

$$SMB_t^{CMA} = \frac{1}{3} (R_{S,A,t} + R_{S,M,t} + R_{S,C,t}) - \frac{1}{3} (R_{B,A,t} + R_{B,M,t} + R_{B,C,t}). \quad (35)$$

In this paper, the adjustment-cost parameter χ is not incorporated into the SMB construction. Including a fourth χ -based size spread would alter the definition of SMB relative to the standard Fama–French benchmark and would therefore weaken direct comparison with the benchmark models. The analysis consequently retains the conventional SMB definition and evaluates the incremental contribution of SMQ separately.

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HML	$\frac{1}{2}(R_{S,H} + R_{B,H})$	−	$\frac{1}{2}(R_{S,L} + R_{B,L})$	→	HML	$= \frac{1}{2}(R_{S,H} + R_{B,H}) - \frac{1}{2}(R_{S,L} + R_{B,L})$
RMW	$\frac{1}{2}(R_{S,R} + R_{B,R})$	−	$\frac{1}{2}(R_{S,W} + R_{B,W})$	→	RMW	$= \frac{1}{2}(R_{S,R} + R_{B,R}) - \frac{1}{2}(R_{S,W} + R_{B,W})$
CMA	$\frac{1}{2}(R_{S,C} + R_{B,C})$	−	$\frac{1}{2}(R_{S,A} + R_{B,A})$	→	CMA	$= \frac{1}{2}(R_{S,C} + R_{B,C}) - \frac{1}{2}(R_{S,A} + R_{B,A})$
SMQ	$\frac{1}{2}(R_{S,S} + R_{B,S})$	−	$\frac{1}{2}(R_{S,Q} + R_{B,Q})$	→	SMQ	$= \frac{1}{2}(R_{S,S} + R_{B,S}) - \frac{1}{2}(R_{S,Q} + R_{B,Q})$

$$\begin{array}{lcl}
 \text{SMB}^{\text{HML}} & = & \frac{1}{3}(R_{S,L} + R_{S,M} + R_{S,H}) - \frac{1}{3}(R_{B,L} + R_{B,M} + R_{B,H}) \\
 \text{SMB}^{\text{RMW}} & = & \frac{1}{3}(R_{S,W} + R_{S,M} + R_{S,R}) - \frac{1}{3}(R_{B,W} + R_{B,M} + R_{B,R}) \\
 \text{SMB}^{\text{CMA}} & = & \frac{1}{3}(R_{S,A} + R_{S,M} + R_{S,C}) - \frac{1}{3}(R_{B,A} + R_{B,M} + R_{B,C})
 \end{array}
 \rightarrow \text{SMB} = \frac{1}{3} (\text{SMB}^{\text{HML}} + \text{SMB}^{\text{RMW}} + \text{SMB}^{\text{CMA}})$$

Figure 7: Factor construction using 2×3 sorts. HML, RMW, CMA, and SMQ are formed from the extreme characteristic portfolios within each size group, while SMB is constructed separately as the average size spread across the three conventional benchmark sorts.

5.4.2.3 The Adjustment-Cost Factor

The adjustment-cost factor represents the central empirical contribution of this paper and is constructed to capture return differences associated with cross-sectional variation in firms' speed of adjustment. The sorting variable is the estimated adjustment cost parameter, χ , obtained from the estimation procedure described earlier.

Adjustment Cost (Slow-minus-Quick). Firms are classified as slow (S), medium (M), or quick (Q), for which the adjustment-cost factor, denoted SMQ_t , is defined as

$$SMQ_t = \frac{1}{2} (R_{S,Slow,t} + R_{B,Slow,t}) - \frac{1}{2} (R_{S,Quick,t} + R_{B,Quick,t}), \quad (36)$$

where $R_{S,Slow,t}$ and $R_{B,Slow,t}$ are the returns on the small and big slow-adjusting firms, while $R_{S,Quick,t}$ and $R_{B,Quick,t}$ are the returns on the corresponding quick-adjusting firms. For clarity, slow corresponds to firms with high adjustment costs, χ , whereas quick corresponds to firms with low adjustment costs. As motivated in the hypothesis development, firms with higher adjustment costs face greater frictions in adjusting their capital stock and therefore respond more slowly to changes in investment opportunities. A positive realization of SMQ_t indicates that slow-adjusting firms outperform quick-adjusting firms, and vice versa. As hypothesized, we expect SMQ_t to earn a positive risk premium.

5.4.3 Test Portfolio Design

In addition to constructing traded factors, the empirical analysis requires a set of test portfolios to evaluate the performance of empirical asset-pricing models. The use of test portfolios serves two main purposes. First, it provides a structured way to generate cross-sectional variation in expected returns. Second, by aggregating individual securities into portfolios, it reduces idiosyncrasies and ensures that the resulting test portfolios are sufficiently diversified. This diversification is important because asset-pricing tests rely on systematic return variation rather than firm-specific shocks. As the construction follows the methodology of Fama and French (1993, 2015), the test portfolios are directly assumed to be reasonably diversified. To provide transparency, Appendix Tables A.2 and A.3 report the average number of firms in each portfolio.

5.4.3.1 The χ -Decile Portfolios

Inspired by Frazzini and Pedersen (2014), we consider decile portfolios sorted by χ . Firms in the lowest decile correspond to those with the lowest estimated adjustment costs, while firms in the highest decile correspond to those with the highest estimated adjustment costs. The purpose of these decile portfolios is to provide insight into the cross-sectional relation between adjustment costs and expected returns. In particular, they allow for an assessment of whether average returns vary monotonically across the adjustment-cost parameter, χ , which serves as a direct test of the economic mechanism implied by the q -theory of investment.

5.4.3.2 Two-Dimensional 5×5 Portfolios

Consistent with Fama and French (1993, 2015), we use 5×5 portfolios formed on size and a second characteristic as test portfolios to evaluate model performance. Besides the standard Fama–French firm characteristics, the second sorting variable also includes our adjustment-cost parameter, χ , as an analogous extension of the standard sorting procedure. For each characteristic, firms are independently assigned to one of five size portfolios and one of five characteristic portfolios using NYSE quintile breakpoints measured at portfolio formation. The resulting 25 portfolios are given by the intersections of these two classifications. These test portfolios allow us to examine whether the asset-pricing models can account for return variation across portfolios jointly sorted on size and the relevant firm characteristic.

5.4.3.3 Three-Dimensional $2 \times 4 \times 4$ Portfolios

To study model performance in a higher-dimensional setting, we also construct $2 \times 4 \times 4$ portfolios following the sorting logic of Fama and French (2015). Firms are first split into two size groups, small and big, using the NYSE median. Within each size group, firms are then independently assigned to quartile portfolios for each of the two additional characteristics, using NYSE breakpoints. In addition to the standard Fama–French characteristics, we also include specifications in which one of the sorting variables is the adjustment-cost parameter, χ , as an analogous extension of the standard procedure. The intersections of these classifications produce a total of 32 portfolios for each set of three sorting variables. Relative to the 5×5 independent sorts, the $2 \times 4 \times 4$ portfolios provide a more conditional test of model performance because the two non-size characteristics are sorted within small and big firms separately. This is useful when firm characteristics are correlated. For example, a 5×5 sort on size and book-to-market may still reflect profitability-related variation if book-to-market and profitability are correlated, whereas the $2 \times 4 \times 4$ portfolios make it possible to examine book-to-market-related return patterns within profitability groups, and vice versa. The procedure therefore provides a sharper description of return variation by separating the effects of each characteristic.

5.5 Empirical Asset-Pricing Methodology

This section outlines the empirical asset-pricing methodology used to evaluate the performance of both the constructed factors and the test portfolios. It introduces the statistical tools used to assess whether the factors earn significant risk premia and how well the empirical asset-pricing models explain the cross-section of returns, based on time-series regressions estimated over the period from July 1982 to December 2024. In this way, the section links the portfolio construction in the previous section to the formal asset-pricing tests that follow.

5.5.1 Unconditional Mean

The unconditional mean provides a first assessment of whether a factor or test portfolio earns a statistically significant excess return. Theoretically, this is a test of whether the asset carries an unconditional risk premium (Munk, 2024a). The sample mean excess return is defined as

$$\mu_i = \frac{1}{T} \sum_{t=1}^T (R_{i,t} - R_{f,t}), \quad (37)$$

where $R_{i,t}$ denotes the return on factor or portfolio i , and $R_{f,t}$ denotes the risk-free rate. The mean excess return μ_i thus measures whether the factor or test portfolio delivers, on average, a significant return beyond the risk-free rate. Note that the factors are by construction self-financing portfolios and therefore already represent excess returns. For this reason, the risk-free rate is not subtracted from the factors. To evaluate whether the average excess return differs from zero, the following hypothesis is tested

$$H_0 : \mu_i = 0 \quad \text{vs.} \quad H_1 : \mu_i \neq 0. \quad (38)$$

A rejection of the null indicates that the average excess return is statistically different from zero, implying the presence of a risk premium. A significantly positive mean excess return indicates a positive unconditional risk premium, while a significantly negative mean excess return indicates that the asset earns less than the risk-free rate on average.

5.5.2 Time-Series Regressions

To test the performance of empirical asset-pricing models, we perform time-series regressions estimated using OLS. The general regression specification is given by

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}F_{1,t} + \beta_{i,2}F_{2,t} + \cdots + \beta_{i,k}F_{k,t} + \varepsilon_{i,t}, \quad (39)$$

where $R_{i,t} - R_{f,t}$ represents the excess return on asset i in month t , and $F_{1,t}, F_{2,t}, \dots, F_{k,t}$ denote the set of traded risk factors included in the empirical asset-pricing model. The coefficients $\beta_{i,1}, \beta_{i,2}, \dots, \beta_{i,k}$ measure the exposure of asset i to each factor, also known as the factor loadings, while the intercept α_i captures the average abnormal excess return embedded in asset i not explained by factor risk. Under a correctly specified asset-pricing model, α_i should not be statistically different from zero. Lastly, the error term $\varepsilon_{i,t}$ represents the idiosyncratic component (Munk, 2024d).

5.5.2.1 Alpha

The primary object of interest is the intercept, α_i , which measures the abnormal return of asset i (Munk, 2024d). To evaluate whether this abnormal return is statistically distinguishable from zero, the following hypothesis is tested

$$H_0 : \alpha_i = 0 \quad \text{vs.} \quad H_1 : \alpha_i \neq 0. \quad (40)$$

Rejection of the null hypothesis indicates that the model fails to fully explain the

expected excess return of asset i , implying that part of the return is not fully explained by the included risk factors. A significantly positive alpha indicates that the asset earns an abnormal return above what is implied by the empirical asset-pricing model and thus may be interpreted as evidence of a positive risk premium not captured by the model. Conversely, a significantly negative alpha indicates that the asset earns an abnormally low excess return relative to the model (Munk, 2024d). The interpretation of α_i , however, depends on the testing context. In asset-pricing tests conducted on test portfolios, insignificant alphas are favored, as they indicate that the model successfully prices the cross-section. In factor spanning tests, by contrast, a significantly positive alpha is favored, as it implies that the candidate risk factor contains incremental pricing information beyond that captured by the existing factor set. The former type of test is applied to test portfolios, whereas the latter is applied to factors themselves.

5.5.2.2 Betas

The slope coefficients, $\beta_{i,k}$, measure the co-movement of the dependent variable with the underlying risk factors (Munk, 2024d). To assess whether a given factor has explanatory power, the following hypothesis is tested

$$H_0 : \beta_{i,k} = 0 \quad \text{vs.} \quad H_1 : \beta_{i,k} \neq 0. \quad (41)$$

Rejection of the null hypothesis indicates that factor k contributes to explaining the time-series variation in the excess return of asset i , conditional on the other factors included in the model. A significantly positive factor loading indicates positive exposure to the factor, meaning that the asset tends to co-move positively with that factor, whereas a significantly negative factor loading indicates the reverse. These loadings therefore capture the extent to which asset i responds to each risk factor or factor-mimicking portfolio included in the empirical asset-pricing model (Munk, 2024d). For example, if $\beta_{i,\text{MKT}} = 1.5$, a 1%-point increase in the market factor is associated with a 1.5%-point increase in the excess return of asset i , holding the other factors constant. This interpretation applies whether asset i is a test portfolio or a candidate factor portfolio.

5.5.2.3 Adjusted R^2

Model fit is evaluated using the adjusted R^2 , also denoted $\bar{R}^2$. The standard R^2 measures the fraction of the time-series variation in the dependent variable that is explained by the systematic risk exposures captured by the included factors (Munk, 2024d). However, because R^2 increases when additional regressors are added, an increase in R^2 does not necessarily imply an improvement in model fit. For this reason, we use $\bar{R}^2$, which accounts for the number of regressors included in the model and therefore provides a more conservative measure (Stock & Watson, 2020a). A higher $\bar{R}^2$ indicates that the model explains a larger share of fluctuations in the dependent variable over time, whereas a lower $\bar{R}^2$ indicates the reverse. However, $\bar{R}^2$ does not assess whether the model correctly prices assets, since it provides no information about whether estimated alphas are statistically or economically different from zero. For example, an adjusted $\bar{R}^2$ of 0.85 implies that 85% of the time-series variation in excess returns is explained by the model, while the re-

maining 15% is unexplained. Even so, the model may still produce large and statistically significant alphas, indicating pricing errors relative to the model. Hence, the adjusted $\bar{R}^2$ is best evaluated together with the alphas and factor loadings.

5.5.2.4 OLS Assumptions

The time-series regressions are estimated using OLS, which rests on a set of classical assumptions. Following Wooldridge (2013), these assumptions can be divided into three assumptions related to the unbiasedness and consistency of the OLS estimator, plus two additional assumptions related to the variance of the estimator. This distinction is useful because the assumptions serve different purposes. The first three assumptions concern whether OLS estimates the population parameters correctly, while the two additional assumptions are used to study the variance and efficiency of OLS specifically.

The first three assumptions are (1) linearity in parameters, (2) no perfect collinearity, and (3) zero conditional mean. Linearity in parameters means that excess returns are modeled as a linear function of the factor returns, where the intercept and factor loadings are the parameters to be estimated. No perfect collinearity means that none of the explanatory variables are a perfect linear combination of the others. Zero conditional mean means that the error term in each period has an expected value of zero conditional on the explanatory variables. These are the key assumptions for unbiasedness and consistency, since they require the model to be correctly specified. These assumptions provide the basis for interpreting the estimated alphas and factor loadings as reliable population regression coefficients. In the time-series setting, these assumptions are important for relying on appropriate large-sample regularity conditions.

The last two assumptions are (4) homoskedasticity and (5) no serial correlation. Homoskedasticity means that the conditional variance of the error term is constant over time, while no serial correlation means that errors from different time periods are uncorrelated conditional on the explanatory variables. These assumptions are not required for unbiasedness, but they are required for the usual OLS standard errors to be valid and for OLS to be efficient among linear unbiased estimators. If these assumptions are violated, the estimated alphas and factor loadings may still be unbiased under the first three assumptions, but the conventional standard errors, t -statistics, and conclusions about statistical significance may be unreliable.

These concerns are especially relevant in financial return data. Financial returns often display volatility clustering, making the homoskedasticity assumption unlikely to hold exactly (Ruppert & Matteson, 2015a). In addition, financial data often exhibit serial correlation in the residuals if the model does not fully capture the dynamic structure of returns or if shocks persist across periods (Ruppert & Matteson, 2015b). The empirical analysis therefore does not rely on conventional standard errors. Instead, as explained below, all reported t -statistics are based on Newey–West standard errors, which adjust the estimated standard errors for heteroskedasticity and autocorrelation in the residuals. This adjustment does not change the estimated alphas or factor loadings, but it changes the estimated uncertainty around them and therefore affects statistical inference.

5.5.3 Statistical Inference

Statistical inference is central to the empirical analysis because estimated excess returns, alphas, and factor loadings are only informative if they differ statistically from zero. This section presents the methods of statistical inference used throughout the paper and thereby provides the foundation for evaluating our hypotheses.

5.5.3.1 *t*-Statistic

For statistical inference on the estimated parameters, we use the *t*-statistic. For a generic parameter estimate $\hat{\theta}$, the corresponding *t*-statistic is given by

$$t = \frac{\hat{\theta} - \theta_0}{SE(\hat{\theta})}, \quad (42)$$

where $\hat{\theta}$ is the estimate obtained from the sample, θ_0 is the hypothesized value under the null hypothesis, and $SE(\hat{\theta})$ denotes the Newey–West standard error of the estimate, used to account for heteroskedasticity and autocorrelation in the errors. Throughout the empirical analysis, the parameter of interest $\hat{\theta}$ may represent the sample mean of excess returns, the alpha, or a beta from a pre-specified empirical asset-pricing model.

All tests are conducted as two-sided tests, such that both positive and negative deviations from zero are taken into account when assessing statistical significance. This is appropriate because the analysis is concerned with whether the estimated parameter differs from zero, rather than whether it differs in a particular direction. Accordingly, the null hypothesis is specified as $\theta_0 = 0$ throughout the paper. In this setting, a null value of zero corresponds to no average excess return, no abnormal return, or no factor exposure, depending on the parameter under consideration. Statistical significance is assessed using conventional critical values corresponding to the 10%, 5%, and 1% significance levels.

To address the issues of heteroskedasticity and serial correlation, statistical inference throughout the paper is based on Newey–West standard errors. In this respect, the lag length determines how many previous periods are taken into account when estimating the standard errors. Throughout the paper, a lag length of 12 months is applied, corresponding to one year of monthly data. This is a natural choice in a monthly setting because it encompasses one full seasonal cycle, ensuring that any calendar-related patterns in the residuals are accounted for, and aligns with the standard convention in the empirical asset-pricing literature (e.g., Ang et al., 2006; Green et al., 2017; Heston & Sadka, 2008).

5.5.3.2 Gibbons, Ross, and Shanken (GRS) Test

Given the breadth of our empirical results, we complement the individual parameter tests with the *GRS test statistic* by Gibbons, Ross, and Shanken (1989), which jointly evaluates the alphas across test portfolios. Since the GRS test evaluates the joint null hypothesis that all alphas associated with a pre-specified empirical asset-pricing model are equal to zero, the hypotheses are given by

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_N = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \alpha_i \neq 0. \quad (43)$$

Failure to reject H_0 implies that the model cannot be rejected as jointly pricing the given set of test portfolios, whereas rejection indicates that at least one portfolio exhibits a non-zero pricing error and hence that the model fails to fully explain the cross-section of expected returns (Cochrane, 2005b).

In our empirical analysis, the GRS test is applied within each portfolio set separately and not across all test portfolios jointly. Thus, similar to Fama and French (2015), each of the four 5×5 portfolio sets is evaluated individually, and each of the six $2 \times 4 \times 4$ portfolio sets is likewise evaluated individually. For example, we do not pool different portfolio families into one combined test, such as combining the Size $\times$ B/M and Size $\times$ OP portfolios into a single set of 50 test assets. This design keeps the comparison tied to specific portfolio families and avoids introducing additional dependence across our test-portfolio sets.

Furthermore, it is worth noting that the GRS test is typically rejected in empirical asset-pricing tests (Fama & French, 2015). For this reason, while failure to reject the null hypothesis is viewed as desirable, the GRS statistic is used primarily as an indicative tool. In this respect, a lower GRS statistic suggests that the model comes closer to failing to reject the null, and therefore points to relatively better model performance, whereas a higher GRS statistic suggests the reverse. In this sense, the GRS results in this paper are interpreted less as a binary pass-fail criterion and more as a structured way to compare the relative pricing performance of competing models across portfolio sets.

5.5.3.3 Additional Holistic Model Performance Metrics

Because we primarily use the GRS test statistic as an indicative tool, and because the GRS statistic is not without limitations, we supplement it with additional performance metrics aimed at providing a more holistic perspective. In this respect, we use the absolute average alpha and the average adjusted R^2 as complementary metrics.

In line with Fama and French (2015), the absolute average alpha is given by

$$A|\alpha| = \frac{1}{N} \sum_{i=1}^N |\hat{\alpha}_i|, \quad (44)$$

where N denotes the number of test portfolios in the given portfolio set, and $\hat{\alpha}_i$ is the estimated alpha for portfolio i . Since alpha is a measure of abnormal return, a higher absolute average alpha indicates a larger average pricing error for a given portfolio set, whereas a lower absolute average alpha indicates the reverse. The reason for using absolute values is that positive and negative alphas might otherwise offset one another. Hence, the absolute average alpha provides an informative summary of the overall magnitude of pricing errors across the portfolio set.

The average adjusted R^2 is given by

$$A(\bar{R}^2) = \frac{1}{N} \sum_{i=1}^N \bar{R}_i^2, \quad (45)$$

where N denotes the number of test portfolios in the given portfolio set, and $\bar{R}_i^2$ denotes the adjusted R^2 from the time-series regression for portfolio i . Because $\bar{R}_i^2$ measures the proportion of time-series variation in portfolio i explained by the included risk factors, a higher average adjusted R^2 is desirable, whereas a lower average $\bar{R}_i^2$ suggests the reverse. It therefore provides information about the overall time-series fit of a given model within a portfolio set. Since $\bar{R}_i^2$ is not constructed as a signed pricing-error measure, there is no reason to use absolute values here.

Together with the GRS test statistic, the purpose of these performance metrics is to provide a holistic perspective on the benchmark empirical asset-pricing models and to help guide the direction of the analysis, especially given the breadth of our results.

5.5.4 Mean-Variance Frontier

As a supplementary portfolio-based interpretation, we examine whether the adjustment-cost factor expands the mean-variance frontier. Since the Fama–French factors and the adjustment-cost factor, SMQ, are tradable excess-return portfolios, the tangency portfolio can be formed directly from the factors (Munk, 2024d). From this setup, the tangency portfolio weights are given by

$$w_F^{TP} = \frac{\Sigma_F^{-1} \mu_F}{\mathbf{1}^\top \Sigma_F^{-1} \mu_F}, \quad (46)$$

where, w_F^{TP} denotes the vector of tangency portfolio weights, μ_F denotes the vector of expected factor excess returns, Σ_F denotes the variance-covariance matrix of factor excess returns, and $\mathbf{1}$ is a vector of ones. The tangency portfolio is mean-variance efficient among all portfolios formed from the available factors. In an equilibrium multi-factor model with traded factor portfolios, this tangency portfolio achieves the largest attainable Sharpe ratio (Munk, 2024b, 2024d).

The Sharpe ratio, which the tangency portfolio maximizes, is defined as the excess return relative to the standard deviation,

$$SR_p = \frac{\mu_p}{\sigma_p}, \quad (47)$$

where μ_p denotes the expected excess return on portfolio p and σ_p its return standard deviation. This measure is central in empirical finance because it summarises the risk-return tradeoff in a single number (Munk, 2024a).

The portfolio-based intuition follows from the relation between factor spanning and mean-variance efficiency. In this setting, the relevant consideration is whether the candidate factor adds information beyond the existing factor opportunity set. An alpha of zero means that the mean excess return of the candidate factor is fully explained by its exposure to the benchmark factors. The factor is therefore spanned by the existing factor set and should not expand the mean-variance frontier. Conversely, if a tradable factor earns a nonzero alpha relative to the benchmark factors, it contains a return component that is not spanned by the existing factor portfolios. In mean-variance terms, this implies that the factor should expand the efficient frontier and increase the Sharpe ratio

of the tangency portfolio (Munk, 2024d). Hence, if the adjustment-cost factor, SMQ, is not fully spanned by the Fama–French five factors and earns a nonzero FF5 alpha, the corresponding portfolio implication is that adding it should improve the attainable mean-variance tradeoff.

5.6 Use of Statistical Software and Artificial Intelligence

This study relies on statistical software and programming tools to conduct the empirical analysis. Data processing, factor construction, portfolio construction, and econometric estimations are performed in *R*, using the *RStudio* integrated development environment. This setup supports systematic data processing, reproducible factor and portfolio construction, and consistent implementation of the econometric estimations.

Generative Artificial Intelligence (GenAI) has been used in a supportive manner throughout the research process. In particular, GenAI has been used for language refinement, grammatical checks, and assistance with coding in *R*, including debugging, troubleshooting, and suggestions for possible implementation approaches in parts of the empirical analysis. However, all empirical design choices, analytical judgments, interpretations of the results, and final conclusions remain the authors’ own. The authors have independently reviewed, understood, and verified the implementation of the empirical analysis. The use of GenAI tools complies with applicable CBS guidelines regarding academic integrity and the responsible use of GenAI in master’s theses.

6 Empirical Results

This section presents the empirical evidence on whether adjustment-cost heterogeneity matters for expected returns. The results suggest that the adjustment cost parameter, χ , does not generate a strong unconditional return premium on its own, as raw excess returns are not clearly monotonic and SMQ has limited standalone pricing power. The evidence becomes more informative once the role of adjustment cost is evaluated relative to the five-factor model and interacted with size, profitability, book-to-market, and investment. In this setting, χ captures a conditional dimension of return variation, especially among small firms where adjustment frictions are likely more economically relevant. Overall, the results are broadly consistent with a q-theoretic interpretation and point to adjustment frictions as a meaningful dimension in the cross-section of expected returns.

6.1 The SMQ Strategy and Adjustment Cost Decile Portfolios

We begin by examining the one-dimensional evidence on the role of adjustment cost. This setting is the most transparent, as it isolates the relationship between χ and excess returns without conditioning on other firm characteristics. The analysis covers the χ -sorted decile portfolios, the SMQ factor constructed from these portfolios, and the five-factor loadings that help interpret the economic content of χ -related return variation.

While the unconditional excess-return evidence is limited, the one-dimensional setting provides essential grounding for the richer multidimensional analysis that follows.

6.1.1 The Slow-Minus-Quick Factor and χ -Sorted Decile Portfolios

Panel A of Table 1 reports mean monthly excess returns, monthly standard deviations, annualized Sharpe ratios, and CAPM, FF3, and FF5 alphas for the ten χ -sorted decile portfolios and the slow-minus-quick factor, SMQ. The main result is that χ does not generate a strong unconditional return premium when examined in isolation, but that SMQ earns a positive abnormal return relative to the five-factor model.

Across the χ -sorted decile portfolios, mean monthly excess returns are positive and statistically significant, ranging from 0.62% to 0.96%. However, there is no clear monotonic relation between χ and excess returns. The highest- χ portfolio does not earn the highest excess return, and notably, the lowest- χ ($P1$) and highest- χ ($P10$) portfolios exhibit identical excess returns of 0.75%. The positive excess returns across all deciles should therefore primarily be interpreted as reflecting the broad realized equity premium over the sample period, rather than abnormal performance specific to any particular χ group. The same conclusion applies to volatilities, Sharpe ratios, and alphas. Monthly standard deviations range from 4.62% to 5.82%, with similar values at the extremes, including 4.90% for the lowest- χ portfolio ($P1$) and 4.68% for the highest- χ portfolio ($P10$). Annualized Sharpe ratios range from 0.37 to 0.64, again without a clear ordering in χ , and with similar values at the extremes, 0.53 for $P1$ and 0.55 for $P10$. Under the CAPM, none of the decile portfolios earn statistically significant alphas, with monthly alphas ranging from -0.24% to 0.16% . Under FF3, the decile alphas remain economically small and mostly insignificant, ranging from -0.17% to 0.16% , with only $P7$ marginally significant at the 10% level. The FF5 alphas are also non-monotonic, ranging from -0.24% to 0.20% . The lowest- χ portfolio earns a significantly negative FF5 alpha of -0.21% per month, while $P7$ earns a marginally significant positive alpha of 0.20% , but the highest- χ portfolios do not continue this pattern. Thus, the decile-level evidence does not support a clean standalone adjustment cost premium. The absence of monotonicity is central: sorting firms only on estimated adjustment costs does not, by itself, generate a systematic relation between χ and portfolio performance across excess returns, risk, Sharpe ratios, or alphas from various empirical asset-pricing models.

Table 1: Excess returns, alphas, and FF5 factor loadings for χ -sorted decile portfolios

US equities, July 1982–December 2024. At the end of each June, firms are sorted into decile portfolios based on the lagged SIC 4-digit industry-level χ estimate using NYSE breakpoints, where P1 denotes the lowest- χ portfolio and P10 the highest- χ portfolio. The SMQ (slow minus quick) factor is a self-financing factor constructed from 2×3 Size- χ sorts, similar to the Fama–French factor construction, and is long high- χ firms and short low- χ firms. Portfolios are formed in June of year t and held from July of year t to June of year $t + 1$. Panel A reports monthly excess returns, CAPM, FF3, and FF5 alphas, as well as monthly standard deviation and annualized Sharpe ratios. Panel B reports the FF5 factor loadings. Newey–West t -statistics based on 12 lags are reported in parentheses, where *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Portfolio	P1 (Quick firms)	P2	P3	P4	P5	P6	P7	P8	P9	P10 (Slow firms)	SMQ
<i>Panel A: Excess returns and benchmark alphas</i>											
Excess return	0.75*** (3.87)	0.87*** (4.31)	0.71*** (3.31)	0.62** (2.41)	0.83*** (3.66)	0.82*** (3.77)	0.96*** (3.68)	0.91*** (4.26)	0.79*** (4.09)	0.75*** (3.66)	0.09 (0.75)
CAPM alpha	-0.02 (-0.21)	0.16 (1.20)	-0.10 (-0.58)	-0.24 (-1.49)	-0.02 (-0.23)	0.04 (0.50)	0.06 (0.51)	0.08 (0.75)	0.09 (0.83)	0.01 (0.08)	0.07 (0.58)
FF3 alpha	-0.07 (-0.72)	0.08 (0.74)	-0.16 (-0.97)	-0.17 (-1.07)	0.01 (0.13)	0.05 (0.59)	0.16* (1.73)	0.15 (1.42)	0.07 (0.71)	-0.02 (-0.26)	0.20* (1.89)
FF5 alpha	-0.21** (-2.12)	-0.12 (-1.31)	-0.24 (-1.42)	-0.01 (-0.08)	0.01 (0.09)	-0.02 (-0.25)	0.20* (1.93)	0.13 (1.05)	-0.12 (-1.14)	-0.14 (-1.64)	0.32*** (3.19)
Std. Dev.	4.90	4.76	5.46	5.82	5.33	4.96	5.61	5.30	4.62	4.68	2.32
Sharpe ratio	0.53	0.64	0.45	0.37	0.54	0.58	0.59	0.59	0.59	0.55	0.13
<i>Panel B: FF5 factor loadings</i>											
$R_M - R_f$	1.02*** (37.34)	0.98*** (32.17)	1.06*** (17.04)	1.00*** (29.11)	1.05*** (42.73)	1.02*** (34.47)	1.07*** (40.41)	1.04*** (24.94)	0.94*** (27.97)	0.98*** (39.36)	-0.06** (-2.03)
SMB	0.18*** (4.68)	0.10* (1.80)	0.22*** (2.69)	0.11* (1.90)	0.11* (1.94)	-0.08 (-1.65)	0.07 (1.55)	-0.14*** (-3.00)	0.06 (0.97)	0.10*** (2.87)	-0.06 (-1.38)
HML	0.10 (1.35)	0.14** (2.20)	0.11 (0.74)	-0.02 (-0.22)	-0.10 (-1.39)	-0.07 (-0.73)	-0.29*** (-6.27)	-0.36*** (-4.54)	-0.13** (-2.26)	0.03 (0.57)	-0.44*** (-7.19)
RMW	0.26*** (4.55)	0.38*** (3.89)	0.08 (0.62)	-0.21* (-1.90)	0.00 (0.03)	0.17* (1.94)	-0.06 (-1.31)	-0.05 (-0.79)	0.29*** (2.60)	0.22*** (3.74)	-0.31*** (-3.70)
CMA	0.14 (1.36)	0.22* (1.82)	0.29* (1.69)	-0.37** (-2.38)	0.01 (0.11)	-0.00 (-0.02)	-0.07 (-0.81)	0.24* (1.93)	0.39*** (2.96)	0.14* (1.68)	0.09 (0.86)

Turning to the SMQ factor, the raw excess return evidence is consistent with the weak return spread across the decile portfolios. Specifically, SMQ earns a monthly excess return of only 0.09%, which is statistically insignificant, and its annualized Sharpe ratio is only 0.13. Hence, SMQ does not earn a strong unconditional premium. This is consistent with the theoretical ambiguity in linking q -theory to risk premia, where the partial-equilibrium setup does not yield a clear prediction for an unconditional adjustment-cost risk premium. However, the picture changes once we account for empirical asset pricing models. The SMQ alpha is small and insignificant under the CAPM, increases to 0.20% and becomes marginally significant under FF3, and rises further to 0.32% per month under FF5, where it is significant at the 1% level. This is the key result from Panel A. The adjustment cost parameter is not strongly priced in isolation, but SMQ contains return variation that is not fully spanned by the Fama–French Five-Factor model.

6.1.2 Adjustment-Cost Factor Loadings

Panel B of Table 1 reports FF5 factor loadings for the χ -sorted decile portfolios and the SMQ factor. These loadings allow us to decompose the one-dimensional χ excess returns and alphas, and offer early support for the q -theoretic interpretation.

Starting with the SMQ factor, the statistically significant loadings are related to MKT, HML, and RMW. Although the market loading is statistically significant, its magnitude is small at -0.06 ($t = -2.03$), suggesting only limited exposure to broad market movements. The more informative loadings are therefore HML and RMW. Specifically, SMQ loads negatively and significantly on HML and RMW, at -0.44 ($t = -7.19$) and -0.31 ($t = -3.70$), respectively. This suggests that the high- χ minus low- χ return spread is primarily related to the value and profitability dimensions of the five-factor model. The negative HML loading indicates that slow-adjusting firms behave more like growth firms relative to quick-adjusting firms, while the negative RMW loading indicates that they behave more like weak-profitability firms. From a q -theoretical perspective, higher adjustment costs may prevent these firms from responding quickly to growth opportunities, causing them to remain growth-like and weakly profitable for longer.

Turning from the SMQ factor to the χ -sorted decile portfolios, Figure 8 provides graphical evidence on the structure of the factor loadings through a simple bar chart. It shows that the market and size loadings do not provide much information about the χ dimension. Market betas are uniformly close to one and highly statistically significant across all decile portfolios, ranging from 0.94 ($t = 27.97$) to 1.07 ($t = 40.41$). This is expected, since the test assets are diversified equity portfolios with substantial exposure to common market risk. For SMB, the loadings are positive for most deciles, but they do not display a clear χ -related ordering, and statistical significance does not vary systematically across the χ distribution. Thus, neither the market nor size loadings appear to provide a systematic pattern across the χ -sorted portfolios.

The clearest decile-level pattern is instead found in HML. The value loadings broadly decline as χ increases, although not completely monotonically. The lowest- χ portfolios have positive HML loadings, around 0.10 ($t = 1.35$) and 0.14 ($t = 2.20$) for $P1$ and $P2$,

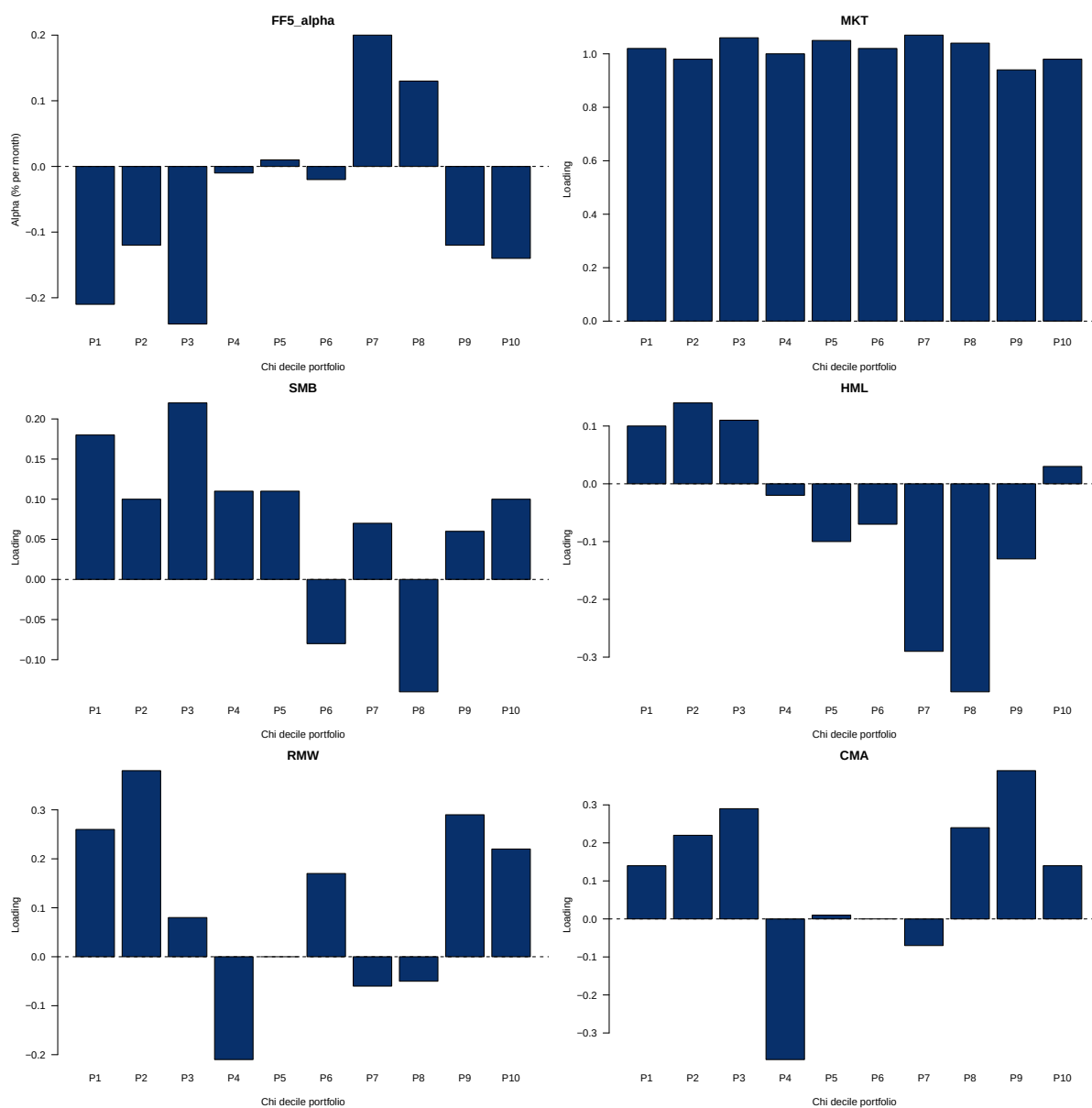


Figure 8: Bar chart of FF5 alphas and factor loadings for the χ -sorted decile portfolios.
Source: Authors' own creation.

while several higher- χ portfolios have negative HML loadings, reaching -0.29 ($t = -6.27$) and -0.36 ($t = -4.54$) for $P7$ and $P8$. This supports the interpretation that firms with higher adjustment costs are more growth-like. In q-theoretic terms, these firms may take longer to convert growth opportunities into realized capital, book value, and cash flows. As a result, the gap between market value and book value may persist for longer, making high- χ firms resemble growth firms. Nevertheless, this interpretation should be made cautiously, since it is most directly applicable when high market values reflect positive growth opportunities. Following negative shocks, the reverse mechanism may also apply, where slow-moving firms may take longer to reduce their capital stock and therefore remain more value-like for longer.

The RMW and CMA loadings are less ordered in the decile portfolios and display more pronounced non-linearities. RMW is positive and significant for both the lowest- and highest- χ portfolios: $P1$ and $P2$ load at 0.26 ($t = 4.55$) and 0.38 ($t = 3.89$), respectively, while $P9$ and $P10$ load at 0.29 ($t = 2.60$) and 0.22 ($t = 3.74$). This suggests that both quick- and slow-adjusting firms appear more robust-profitability-like in the one-dimensional sort, while the middle deciles are closer to neutral. A similar pattern applies to the CMA loadings, although it is important to emphasize that the significance is primarily in the χ range, where $P8$, $P9$ and $P10$ load positively at 0.24 ($t = 1.93$), 0.39 ($t = 2.96$), and 0.14 ($t = 1.68$) respectively. These nonlinearities are important rather than merely problematic. Although RMW and CMA are difficult to interpret cleanly in the one-dimensional sort, we later show that these exposures, particularly the profitability loading, become more informative in the two- and three-dimensional test portfolios.

6.1.3 Partial Conclusion

The one-dimensional evidence yields a coherent but partial picture. The adjustment-cost parameter does not generate a strong unconditional risk premium. Sharpe ratios, raw excess returns, and alphas show no clear monotonic ordering across χ -sorted decile portfolios. Meanwhile, the more important result comes from the SMQ factor. While SMQ likewise earns no unconditional risk premium, its FF5 alpha is positive and statistically significant, suggesting that the adjustment-cost dimension contains information not fully captured by the five-factor model. This supports the broader interpretation that χ may provide a meaningful dimension in addition to the FF5 structure. The factor loadings provide further insight into this finding: the SMQ loadings indicate that slow-adjusting firms appear more growth-like and weak-profitability-like, while quick-adjusting firms are more value-like and robust-profitability-like. Taken together, while the evidence does not support Hypothesis 1 in terms of a clear positive unconditional χ -related premium, the positive and statistically significant FF5 alpha for SMQ reveals a priced χ -related component and thus supports the hypothesis through a conditional risk premium relative to FF5. This is explored further in the multidimensional portfolio tests.

6.2 Factor Characteristics and Spanning Tests

Having established the one-dimensional properties of the SMQ factor and the χ -sorted decile portfolios, we now turn to a more systematic examination of the full set of factors. To assess how SMQ relates to the Fama–French factors and to understand the structure of the six-factor investment opportunity set, this section considers MKT, SMB, HML, RMW, CMA, and SMQ in terms of their summary statistics and factor spanning regressions. This analysis serves two purposes. First, it situates SMQ relative to the existing factors and sheds light on the nature of the incremental information it contains. Second, it provides a foundation for the efficient-frontier analysis, where we examine whether adding SMQ to the FF5 factor set expands the mean–variance opportunity set.

6.2.1 Performance Statistics

Panel A of Table 2 reports mean monthly excess returns, standard deviations, and annualized Sharpe ratios for the six factors over the sample period from July 1982 to December 2024. Across factors, the summary statistics appear broadly consistent with prior empirical evidence and the economic characteristics of the sample period.

The market factor shows a mean monthly excess return of 0.79% and a monthly standard deviation of 4.49%. The corresponding annualized Sharpe ratio of 0.61 indicates relatively strong risk-adjusted performance and is somewhat higher than broad long-sample benchmarks often discussed in the macro-finance literature, where market Sharpe ratios are commonly placed in the range of approximately 0.25–0.50 (Cochrane, 2017). This likely reflects the characteristics of the 1982–2024 period, which, despite several major downturns, was a favorable environment for U.S. equities (Ritter & Warr, 2002).

Similarly, RMW and CMA both earn positive and statistically significant mean excess returns, again with relatively strong Sharpe ratios. For RMW, the mean monthly return is 0.38% ($t = 3.24$), and the annualized Sharpe ratio is 0.58, placing it close to the market factor in risk-adjusted terms. CMA shows a similar, although more moderate, pattern, earning a mean monthly return of 0.21% ($t = 2.26$), with an annualized Sharpe ratio of 0.39. These results suggest that the profitability and investment effects are evident in factor returns over our sample period.

Table 2: Summary statistics and spanning regressions for factors

US equities, July 1982–December 2024. The factor set consists of the market excess return ($R_M - R_f$), SMB (small minus big), HML (high minus low), RMW (robust minus weak), CMA (conservative minus aggressive), and SMQ (slow minus quick), where SMQ is constructed from SIC 4-digit industry-level estimates of χ . At the end of each June, firms are sorted based on the lagged characteristic using NYSE breakpoints. Panel A reports mean monthly excess returns in percent, monthly standard deviations in percent, and annualized Sharpe ratios. Panel B reports time-series regressions, where the intercept is denoted by α , the slope coefficients are factor loadings, and the final row reports the adjusted R^2 . All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	$R_M - R_f$	SMB	HML	RMW	CMA	SMQ
<i>Panel A: Performance statistics</i>						
Mean excess return	0.79*** (3.90)	0.06 (0.45)	0.18 (1.04)	0.38*** (3.24)	0.21** (2.26)	0.09 (0.75)
Std. Dev.	4.49	3.01	3.06	2.27	1.90	2.32
Sharpe ratio	0.61	0.07	0.20	0.58	0.39	0.13
<i>Panel B: Factor spanning test</i>						
α	1.17*** (6.92)	0.24 (1.52)	-0.04 (-0.39)	0.56*** (5.26)	0.26*** (3.86)	0.32*** (3.19)
$R_M - R_f$		0.08** (1.98)	-0.01 (-0.27)	-0.10** (-2.58)	-0.07*** (-3.12)	-0.06** (-2.03)
SMB	0.20* (1.69)		0.05 (0.93)	-0.28*** (-5.05)	-0.02 (-0.59)	-0.06 (-1.38)
HML	-0.06 (-0.27)	0.10 (0.97)		0.15 (1.40)	0.46*** (11.00)	-0.44*** (-7.19)
RMW	-0.51*** (-2.81)	-0.60*** (-4.26)	0.15 (1.56)		-0.19*** (-3.28)	-0.31*** (-3.70)
CMA	-0.77*** (-2.87)	-0.09 (-0.62)	0.98*** (11.39)	-0.40*** (-3.66)		0.09 (0.86)
SMQ	-0.32** (-2.26)	-0.12 (-1.29)	-0.45*** (-8.31)	-0.31*** (-4.05)	0.04 (0.89)	
$\bar{R}^2$	0.18	0.21	0.65	0.37	0.57	0.40

By contrast, SMB, HML, and SMQ do not earn statistically significant mean excess returns. Hence, the size, value, and adjustment-cost effects are not strongly present in our sample from an unconditional perspective. For SMQ, this is expected, as it is consistent with the one-dimensional evidence documented in Table 1. For SMB and HML, the result is more interesting, as it differs from Fama and French (2015), who document positive mean excess returns for the size and value factors. In our sample, SMB earns only 0.06% per month ($t = 0.45$), with an annualized Sharpe ratio of 0.07, while HML earns 0.18% per month ($t = 1.04$), with a Sharpe ratio of 0.20. This is not surprising, however, as it is broadly consistent with the existing literature, which has documented the instability of the size premium and the particularly weak performance of the value premium in more recent decades (Fama & French, 2021; Schwert, 2002).

6.2.2 Factor Spanning Tests

Panel B of Table 2 reports factor-spanning regressions that provide additional insight into the behavior of SMQ and its relation to the broader Fama–French Five-Factor structure. In particular, apart from its significant positive FF5 alpha of 0.32% ($t = 3.19$), it seems to play a more central role in the relationship between value and profitability than previously suggested. Specifically, in the HML and RMW regressions, the SMQ loadings are -0.45 ($t = -8.31$) and -0.31 ($t = -4.05$), respectively, and the previously documented relationship between HML and RMW seems to disappear.

This pattern is informative for understanding the role of SMQ. In Fama and French (2015), HML and RMW exhibit significant positive mutual spanning loadings, whereas this relation appears to become neutral once SMQ is included. To verify that this change is not merely driven by differences in sample periods, we estimate FF5-only spanning regressions in Appendix Table A.4. The results confirm that the positive relation between HML and RMW remains present when SMQ is excluded. Specifically, HML and RMW continue to load significantly positively on each other in the FF5-only spanning regressions. The disappearance of this relationship in the six-factor specification is therefore consistent with the argument that the inclusion of SMQ absorbs part of the common variation between value and profitability.

This helps clarify the economic mechanism behind SMQ’s positive FF5 alpha. The key point is not only that SMQ earns a significant FF5 alpha, but also that its inclusion changes the conditional relation between HML and RMW. Without SMQ, value and profitability appear positively related. However, once SMQ is introduced, this relation becomes neutral, while SMQ loads negatively on both HML and RMW. This suggests that part of the usual value-profitability relation may reflect an underlying adjustment-speed component. From an economic perspective, quick-adjusting firms may respond more rapidly to valuation signals and therefore become more value-like, as faster adjustment brings market valuations closer to underlying fundamentals. By the same logic, rapid adjustment may allow firms to realize investment opportunities more quickly, leading to stronger profitability. By contrast, slow-moving firms may remain growth-like and weakly profitable for longer time horizons, as adjustment frictions delay their response to

valuation signals. In this sense, SMQ appears to capture a conditional dimension linking value and profitability within the FF5 factor space.

6.2.3 The Efficient Frontier

Table 3 reports the tangency-portfolio properties under the five-factor and six-factor specifications. Under FF5, the tangency portfolio earns a mean monthly excess return of 0.38% with a monthly standard deviation of 1.03%, yielding an annualized Sharpe ratio of 1.27. When SMQ is added, the mean monthly excess return falls to 0.31%, but the monthly standard deviation falls more sharply to 0.77%. As a result, the annualized Sharpe ratio increases to 1.41. For this reason, the improvement does not come from a higher expected return but instead from a stronger reduction in risk. While the six-factor tangency portfolio gives up some expected return, it also reduces volatility proportionally more. This shift in the risk-return trade-off and the resulting expansion of the efficient frontier are illustrated in Figure 9.

Table 3: Tangency-portfolio weights and portfolio summary across factor sets

US equities, July 1982–December 2024. This table reports TP (tangency-portfolio) weights for the factors included in the Fama-French five-factor model and the six-factor specification augmented with the SMQ (slow minus quick) factor. The factor set consists of the market excess return ($R_M - R_f$), SMB (small minus big), HML (high minus low), RMW (robust minus weak), CMA (conservative minus aggressive), and SMQ (slow minus quick), where SMQ is constructed from SIC 4-digit industry-level estimates of χ . The first six columns report the weights in each of the factors. The final three columns report the mean monthly excess return, monthly standard deviation, and annualized Sharpe ratio implied by the tangency portfolio under each of the factor specifications. Portfolio weights, mean excess returns, and standard deviations are reported in percent.

	$R_M - R_f$	SMB	HML	RMW	CMA	SMQ	Mean	Std. Dev.	Sharpe ratio
Five-Factor TP	0.18	0.08	-0.16	0.40	0.49	–	0.38	1.03	1.27
Six-Factor TP	0.13	0.06	-0.02	0.33	0.32	0.19	0.31	0.77	1.41

Mechanically, this is expected, since SMQ contributes primarily through its covariance structure rather than through a large unconditional risk premium. SMQ does not substantially raise the mean excess return of the tangency portfolio, but its negative loadings on HML and RMW make it useful for hedging parts of the value and profitability exposures in the full factor set. In variance terms, adding a factor with a low standalone mean excess return can still improve the tangency portfolio if its covariance terms reduce total portfolio risk sufficiently. This interpretation is reflected in the estimated tangency-portfolio weights. When SMQ is added, it receives a positive weight of 0.19, while the overall allocation across the original FF5 factors becomes less concentrated. The weight on MKT falls from 0.18 to 0.13, RMW from 0.40 to 0.33, and CMA from 0.49 to 0.32, while the short position in HML shrinks from -0.16 to -0.02 . The relatively high weight on SMQ is notable given its low mean return, indicating that its role in the optimal portfolio is driven more by its covariance with the other factors than by its standalone performance. Taken together, the diversification benefits of SMQ increase the Sharpe ratio of the factor set from 1.27 to 1.41.

The economic intuition is that adjustment frictions capture a dimension of firm heterogeneity that is not fully spanned by the standard FF5 factors. Firms with high adjustment costs are less able to change their capital stock or investment policy quickly when economic conditions change, while firms with low adjustment costs are more flexible and can respond more rapidly to shocks and investment opportunities. The SMQ portfolio, which is long slow-moving firms and short quick-moving firms, captures return variation associated with this difference in adjustment speed. Despite its low unconditional excess return and Sharpe ratio, its value lies in the fact that it covaries differently with the standard value and profitability factors. Ultimately, this makes SMQ useful as a hedging or diversifying factor within the broader factor set.

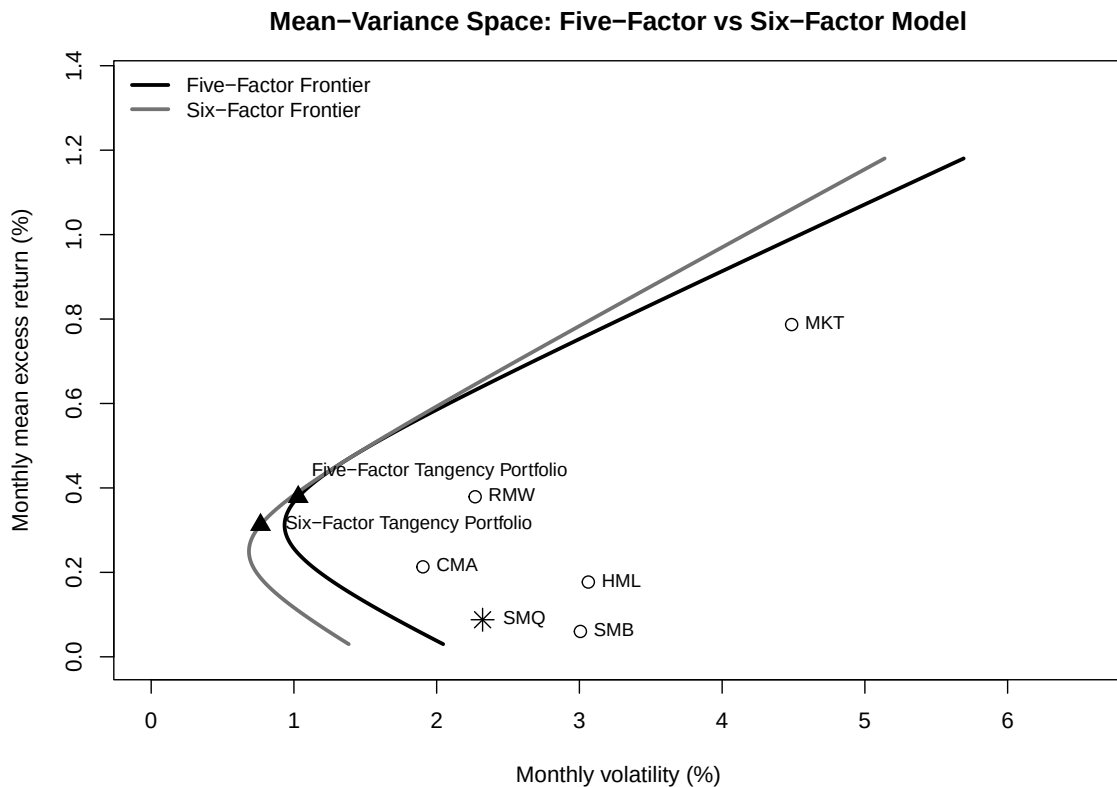


Figure 9: Comparison of Tangency Portfolios and Efficient Frontiers for the Fama-French Five-Factor model and the six factor model, which includes SMQ.

Source: Authors' own creation.

6.2.4 Partial Conclusion

The factor characteristics and spanning tests provide direct insight into the role of SMQ within the broader Fama-French Five-Factor structure. The spanning regressions show that SMQ is economically informative: it loads negatively on HML and RMW, and its inclusion neutralizes the value-profitability relationship. This suggests that adjustment-cost heterogeneity captures part of the structure embedded in the FF5 factors. Firms with high adjustment costs appear more growth-like and weak-profitability-like, consistent with the idea that adjustment frictions delay the conversion of investment opportunities into realized capital and earnings. The efficient-frontier results reinforce this

interpretation. Adding SMQ improves the mean–variance opportunity set by increasing the tangency-portfolio Sharpe ratio primarily through lower volatility rather than higher expected returns. SMQ therefore acts as a diversifying factor that hedges part of the risk embedded in value and profitability. Economically, this is intuitive because value and profitability are closely related to firms’ ability to respond to valuation signals and translate growth opportunities into realized capital and profits. If adjustment frictions slow this process, then SMQ may capture a related dimension that is not fully spanned by these standard factors. Its hedging value may therefore arise because it adds information about firms’ adjustment frictions that the FF5 factors do not fully capture.

6.3 Adjustment Costs in a Multidimensional Setting

The one-dimensional and factor-level evidence suggests that the adjustment cost parameter, χ , contains information not fully captured by FF5, and that its role is especially important in relation to the value and profitability dimensions. However, while the relation between χ and value appears relatively systematic, the profitability relation remains mostly non-linear. Turning to the multidimensional portfolio tests, these conditional patterns are examined more directly. By jointly sorting on size, χ , and operating profitability, book-to-market, or investment, we assess whether χ becomes more informative within Fama–French dimensions. The section first presents excess returns for the multidimensional portfolios, and then evaluates model performance across portfolio sorts from a holistic perspective, before focusing on the two-dimensional Size– χ sort and the three-dimensional χ sorts that form the core empirical evidence.

6.3.1 Mean Excess Returns of Test Portfolios

Tables 4 and 5 report mean monthly excess returns for the 25 and 32 portfolio families. These tables provide descriptive evidence on whether the standard Fama–French characteristics remain visible in our updated 1982–2024 sample, and whether the added χ dimension generates additional cross-sectional structure. Across the portfolio families, excess returns are generally positive, with most portfolios earning roughly 0.70–1.10% excess return per month. Overall, the conventional effects are present in the excess returns, though their strength varies across characteristics. Operating profitability and investment show the clearest and most consistent spreads, while the size and value effects are weaker and less uniform across the different portfolio sorts.

In the two-dimensional 5×5 sorts in Table 4, the conventional effects are broadly visible, though with varying strength. The size effect is essentially absent on average: the small-minus-big spread averages close to zero across B/M quintiles, with the sign alternating across columns. The value effect is more noticeable, with an average high-minus-low B/M spread of 0.29 %-points, though it is concentrated among smaller firms and turns slightly negative for big stocks. Operating profitability is the strongest effect in the table, with an average high-minus-low OP spread of 0.44 %-points, fairly consistent across size groups. The investment effect is also clearly present, with low-investment firms outperforming high-investment firms by 0.26 %-points per month on average, con-

sistent with the investment premium. The Size- χ portfolios show no systematic pattern across either dimension, with the high-minus-low spread averaging only 0.03 %-points and alternating in sign, consistent with the earlier decile evidence that adjustment costs do not map cleanly onto average excess returns.

Table 4: Mean monthly excess returns for 25 test portfolios

US equities, July 1982–December 2024. This table reports average monthly excess returns in percent, for the four sets of 25 test portfolios formed using NYSE breakpoints: Size-B/M, Size-OP, Size-Inv, and Size- χ . Each block reports a 5×5 sort, where rows denote size quintiles and columns denote characteristic quintiles.

	Size-B/M					Size-OP				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.23	0.85	0.81	1.03	1.15	0.52	0.93	1.13	0.94	1.02
2	0.64	0.91	0.98	1.10	0.86	0.63	0.75	0.94	1.03	1.07
3	0.69	0.95	0.90	0.94	0.98	0.59	0.78	0.85	0.95	1.07
4	0.87	0.89	0.86	0.90	0.96	0.60	0.83	0.97	0.93	0.99
Big	0.86	0.82	0.85	0.79	0.77	0.53	0.68	0.71	0.82	0.92

	Size-Inv					Size- χ				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.96	0.95	0.98	0.89	0.43	0.88	0.66	0.75	0.94	0.87
2	0.94	0.90	0.99	1.00	0.64	0.86	0.76	0.77	0.99	0.79
3	0.96	0.95	0.91	0.88	0.71	0.80	0.87	0.83	0.80	1.05
4	0.82	0.93	0.91	0.96	0.84	0.85	0.74	0.88	0.98	0.94
Big	1.03	0.83	0.83	0.83	0.79	0.81	0.65	0.84	0.99	0.72

The three-dimensional $2 \times 4 \times 4$ sorts in Table 5 provide a sharper picture. Operating profitability is the clearest dimension: the high-minus-low OP spread has the expected sign in 23 out of 24 comparisons across the three OP-sorted panels, with average spreads of 0.37 %-points in the Size-B/M-OP portfolios, 0.40 %-points in the Size-OP-Inv portfolios, and 0.24 %-points in the Size-OP- χ portfolios. The investment effect is also more visible in the three-dimensional portfolio sorts, with the low-minus-high investment spread having the expected sign in 20 out of 24 comparisons across the three investment-sorted panels, and average spreads of 0.31 %-points in the Size-OP-Inv portfolios and 0.22 %-points in the Size-Inv- χ portfolios. A notable exception is the small-firm, low-operating-profitability, high-investment portfolio, which earns a mean excess return of essentially zero, -0.002% per month, corresponding to the same corner of the characteristic space that FF5 is known to struggle with (Fama & French, 2015).

Table 5: Mean monthly excess returns for 32 test portfolios

US equities, July 1982–December 2024. This table reports mean monthly excess returns in percent for six sets of 32 test portfolios formed using NYSE breakpoints. Each portfolio family is based on a $2 \times 4 \times 4$ sort. Within each panel, the left block reports small firms and the right block reports big firms. Row labels denote quartiles of the first non-size characteristic in the panel title, from low (Q1) to high (Q4), while columns denote quartiles of the second non-size characteristic, also from low to high.

	Small				Big			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
<i>Panel A: Size-B/M-OP</i>								
Low B/M	0.20	0.66	0.81	0.93	0.66	0.92	0.88	0.86
Q2	0.58	0.87	0.97	1.10	0.60	0.76	0.84	0.98
Q3	0.79	0.93	1.19	1.19	0.77	0.81	0.88	0.97
High B/M	0.86	1.03	1.44	1.28	0.72	0.82	1.10	0.81
<i>Panel B: Size-B/M-Inv</i>								
Low B/M	0.71	0.99	0.93	0.55	0.99	0.84	0.90	0.83
Q2	0.88	0.89	0.96	0.95	0.95	0.83	0.78	0.73
Q3	1.08	0.99	1.09	0.94	0.85	0.89	0.78	0.84
High B/M	1.10	1.03	0.85	1.19	0.91	0.69	0.89	0.91
<i>Panel C: Size-OP-Inv</i>								
Low OP	0.72	0.86	0.59	-0.00	0.77	0.51	0.71	0.61
Q2	0.98	0.93	0.91	0.63	0.86	0.65	0.94	0.67
Q3	1.26	1.01	1.03	0.91	1.03	0.90	0.76	0.83
High OP	1.25	1.05	1.05	0.91	1.05	0.91	0.85	0.90
<i>Panel D: Size-OP-χ</i>								
Low OP	0.31	0.35	0.61	0.63	0.81	0.50	0.64	0.89
Q2	0.69	0.78	0.86	1.04	0.76	0.81	0.83	0.80
Q3	1.03	1.10	0.96	0.91	0.79	0.68	0.94	0.85
High OP	0.99	0.96	0.94	0.96	0.86	0.80	1.12	0.71
<i>Panel E: Size-B/M-χ</i>								
Low B/M	0.68	0.72	0.75	0.75	0.86	0.67	0.96	0.78
Q2	0.95	0.96	0.98	1.07	0.79	0.76	0.95	0.80
Q3	0.74	1.14	1.06	1.16	0.76	0.80	0.98	1.00
High B/M	1.17	0.94	1.06	0.98	0.97	0.55	0.82	1.00
<i>Panel F: Size-Inv-χ</i>								
Low Inv	0.97	0.84	0.93	1.01	0.86	0.72	1.14	0.95
Q2	0.84	0.96	0.97	1.19	0.74	0.71	0.91	0.75
Q3	0.95	0.98	0.96	1.00	0.84	0.70	0.94	0.77
High Inv	0.64	0.67	0.63	0.56	0.64	0.73	0.95	0.84

By contrast, value remains more uneven, with the high-minus-low book-to-market spread having the expected sign in only 16 out of 24 comparisons across the three B/M-sorted panels. The χ dimension is also not clearly ordered in raw excess returns, with the high-minus-low χ spread having the expected sign in only 17 out of 24 comparisons across the three χ -sorted panels. This is important for the interpretation of the paper: the adjustment-cost parameter, χ , does not appear to generate a clear systematic pattern in the raw excess returns, but, as we will show later, its role becomes more visible once these excess returns are evaluated relative to the FF5 model.

Overall, the mean excess returns suggest that the standard Fama–French dimensions are present in the data, albeit slightly less sharply than in the sample studied by Fama and French (2015). The χ dimension does not generate a clear pattern in the excess returns, which includes both the Size– χ portfolios and the three-dimensional sorts, although the expected sign structure remains in approximately two-thirds of the spreads. With the excess-return patterns in mind, this leads us to the subsequent portfolio-pricing tests, where we examine whether CAPM, FF3, FF5, and FF5 + SMQ can capture the cross-sectional variation among these test portfolios.

6.3.2 Model Performance

Table 6 reports overall pricing performance based on the GRS statistic, average absolute alpha, and average adjusted $\bar{R}^2$. Performance generally improves as the model expands from CAPM to FF3 to FF5, but the additional gain from adding SMQ is modest and uneven across portfolio families.

In Panel A, the two-dimensional 5×5 sorts follow a broadly familiar pattern. The GRS statistic declines meaningfully from FF3 to FF5, for instance, from 4.52 to 3.45 for Size–B/M and from 3.54 to 2.79 for Size–Inv, but the further step to FF5 + SMQ adds little, with average absolute alphas and average $\bar{R}^2$ virtually unchanged across all four portfolio families. The main exception is Size– χ , where CAPM produces the lowest GRS statistic, 1.41, and is the only model that fails to reject the joint null at the 5% significance level. This is noteworthy but should not be overstated. As already shown, the mean excess returns for the Size– χ portfolios cluster around 0.80–0.90%, with little cross-sectional variation to explain, making them relatively easy to capture by CAPM.

Panel B provides a similar story for the three-dimensional $2 \times 4 \times 4$ sorts. For the non- χ portfolio sorts, FF5 and FF5 + SMQ deliver nearly identical average absolute alphas and average $\bar{R}^2$, with average absolute alphas remaining at 0.09–0.11 across Size–B/M–OP, Size–B/M–Inv, and Size–OP–Inv. The more visible improvements from adding SMQ appear in the χ -related 32-portfolio families, but these gains are mainly concentrated in the GRS statistic and remain modest for average absolute alpha and average $\bar{R}^2$. The clearest improvement appears in Size–Inv– χ , where average absolute alpha falls from 0.17 to 0.12. Even here, however, $\bar{R}^2$ barely moves, and the GRS statistic remains statistically significant throughout.

Table 6: GRS Test Statistics, Absolute Average Alpha and Average Adjusted R^2 for 25- and 32-portfolio test portfolios.

US equities, July 1982–December 2024. The table reports summary statistics for CAPM (Capital Asset Pricing Model), FF3 (Fama-French Three-Factor Model), FF5 (Fama-French Five-Factor Model), and the six-factor model that augments FF5 with the SIC 4-digit industry-level slow-minus-quick factor (FF5 + SMQ). Panel A reports results of asset pricing tests for the individual 25-portfolio families, and Panel B reports results for the individual 32-portfolio families. For each set of portfolio regressions, the table reports the GRS test statistic, the average absolute alpha $A|\alpha_i|$, and the average adjusted R^2 , denoted $A(\bar{R}^2)$. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	CAPM			FF3			FF5			FF5 + SMQ		
	GRS	$A \alpha_i $	$A(\bar{R}^2)$	GRS	$A \alpha_i $	$A(\bar{R}^2)$	GRS	$A \alpha_i $	$A(\bar{R}^2)$	GRS	$A \alpha_i $	$A(\bar{R}^2)$
<i>Panel A: 25-portfolio test assets</i>												
Size-B/M	4.67***	0.17	0.72	4.52***	0.13	0.90	3.45***	0.10	0.91	3.21***	0.10	0.91
Size-OP	2.70***	0.18	0.76	2.76***	0.16	0.89	1.74**	0.06	0.91	1.70**	0.06	0.91
Size-Inv	3.62***	0.16	0.76	3.54***	0.11	0.91	2.79***	0.09	0.92	2.46***	0.08	0.92
Size- χ	1.41*	0.11	0.73	2.59***	0.13	0.86	2.98***	0.18	0.87	2.44***	0.13	0.88
<i>Panel B: 32-portfolio test assets</i>												
Size-B/M-OP	1.93***	0.22	0.67	2.13***	0.15	0.83	1.55**	0.11	0.85	1.65**	0.11	0.85
Size-B/M-Inv	2.76***	0.16	0.69	2.56***	0.12	0.85	1.86***	0.09	0.86	1.73***	0.09	0.86
Size-OP-Inv	3.29***	0.23	0.73	3.42***	0.18	0.85	1.93***	0.10	0.87	1.81***	0.10	0.87
Size-OP- χ	2.18***	0.19	0.69	3.03***	0.18	0.80	2.52***	0.17	0.82	2.10***	0.15	0.83
Size-B/M- χ	2.09***	0.15	0.67	2.12***	0.13	0.80	2.04***	0.13	0.81	1.67**	0.12	0.82
Size-Inv- χ	2.72***	0.16	0.69	2.75***	0.15	0.81	2.14***	0.17	0.83	1.72***	0.12	0.84

Overall, FF5 + SMQ struggles with the same test portfolios as FF5, and the incremental contribution of SMQ is not systematic across portfolio families. Although not reported directly in the main text, we confirm that the same conclusion holds at the more granular portfolio level for the non- χ -sorted portfolios, as shown in Appendix A.3. These results therefore provide limited support for Hypothesis 2, since augmenting FF5 with SMQ does not lead to consistent and economically meaningful improvements in pricing performance across the cross-section. Instead, the gains are largely confined to χ -sorted portfolios. Given this limited aggregate improvement, the subsequent analysis focuses more directly on the χ -sorted portfolios, allowing for a more granular assessment of where the FF5 alpha associated with SMQ is most pronounced.

6.3.3 The Two-Dimensional 5×5 Size- χ Sort

The two-dimensional Size- χ sort shows that the adjustment cost parameter, χ , is informative across the size distribution and appears closely connected to the value dimension, while the size-related pattern is mainly concentrated in the highest χ quintile and is more closely tied to operating profitability. Intuitively, adjustment costs matter when firms have future investment opportunities that must be converted into realized capital, cash flows, and operational profit. Among the slowest-moving firms, size and operating profitability become especially important because small, weak-profitability firms may be less able to improve operations quickly and may remain constrained for longer.

Appendix Table A.8 reports alphas for the 25 Size- χ portfolios under CAPM, FF5, and FF5 + SMQ. Consistent with the holistic model-performance results, CAPM produces no statistically significant alphas and performs well, although this should not be overstated given the limited return variation in the Size- χ portfolios. FF5 + SMQ mitigates several alpha estimates, especially among low- χ firms, suggesting that SMQ captures some χ -related variation left unexplained by FF5. However, this is partly mechanical since SMQ is constructed from χ . The most informative evidence therefore comes from the FF5 alpha patterns, where the Size- χ sort reveals clearer structure.

As presented in Panel A of Table 7, the FF5 alphas generally increase with χ from the lowest to the fourth χ quintile. In the lowest χ quintile, alphas are negative across all size groups and statistically significant for the three middle size groups, ranging from -0.29% per month ($t = -2.54$) to -0.17% ($t = -1.91$). In the fourth χ quintile, the pattern reverses: alphas are positive across all size groups and statistically significant for four out of five test portfolios, with estimates ranging from 0.24% ($t = 2.73$) to 0.32% ($t = 2.79$). This suggests that moving from low to high χ is associated with higher abnormal returns. However, the pattern weakens in the highest χ quintile, where the size effect becomes more dominant: small high- χ firms earn an FF5 alpha of 0.24% ($t = 2.08$), while big high- χ firms earn -0.18% ($t = -2.12$), producing a small-minus-big spread of 0.42% ($t = 3.07$). Thus, the two-dimensional sort shows that χ matters, but that the size effect becomes especially important among the slowest-adjusting firms relative to the FF5 model.

Panels B and C of Table 7 provide further insight and show that the χ dimension is

Table 7: FF5 alphas, HML loadings, and RMW loadings for Size- χ portfolios

US equities, July 1982–December 2024. This table reports the FF5 (Fama-French Five Factor model) alphas and selected factor loadings for the 25 Size- χ portfolios formed using NYSE breakpoints and based on SIC 4-digit industry-level estimates for χ . Panel A reports intercepts from time-series regressions, denoted by α , which is expressed in monthly percentage. Panels B and C report the corresponding HML (high minus low) and RMW (robust minus weak) factor loadings. The final column in Panels B and C reports the high-minus-low ($H - L$) χ coefficient from a direct time-series regression of the high-minus-low χ spread excess return within each size quintile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	χ					
	Low	2	3	4	High	H-L
<i>Panel A: FF5 alpha</i>						
Small	-0.13 (-0.74)	-0.26* (-1.67)	-0.10 (-1.02)	0.32*** (2.79)	0.24** (2.08)	0.37* (1.67)
2	-0.29** (-2.54)	-0.33** (-2.03)	-0.23*** (-2.65)	0.29*** (3.63)	-0.10 (-1.04)	0.19 (1.49)
3	-0.29*** (-2.65)	-0.16 (-1.05)	-0.16 (-1.43)	0.04 (0.58)	0.17* (1.88)	0.46*** (4.12)
4	-0.17* (-1.91)	-0.22 (-1.25)	-0.12 (-1.36)	0.30*** (3.57)	0.02 (0.20)	0.19* (1.77)
Big	-0.13 (-1.33)	-0.04 (-0.27)	0.05 (0.75)	0.24*** (2.73)	-0.18** (-2.12)	-0.05 (-0.45)
<i>Panel B: HML loading</i>						
Small	0.30*** (4.24)	0.31*** (3.94)	0.17** (2.02)	-0.36*** (-6.51)	-0.10 (-1.38)	-0.41*** (-6.08)
2	0.31*** (5.23)	0.21*** (3.06)	0.06 (0.69)	-0.32*** (-7.79)	-0.03 (-0.26)	-0.33*** (-3.63)
3	0.39*** (5.96)	0.18*** (2.80)	0.12 (1.14)	-0.27*** (-6.03)	-0.07 (-1.46)	-0.46*** (-7.44)
4	0.28*** (5.15)	0.18* (1.74)	0.01 (0.11)	-0.31*** (-6.42)	0.02 (0.20)	-0.26*** (-3.32)
Big	0.08 (1.38)	0.00 (0.04)	-0.10 (-1.61)	-0.38*** (-11.18)	-0.08* (-1.89)	-0.16* (-1.93)
<i>Panel C: RMW loading</i>						
Small	0.17* (1.88)	-0.08 (-0.89)	-0.15** (-2.26)	-0.56*** (-8.63)	-0.50*** (-6.40)	-0.67*** (-5.54)
2	0.43*** (5.05)	0.23*** (2.79)	0.06 (0.56)	-0.35*** (-5.53)	0.01 (0.10)	-0.42*** (-5.19)
3	0.38*** (4.76)	0.11 (1.58)	0.16 (1.49)	-0.19*** (-3.90)	0.08 (1.19)	-0.29*** (-3.38)
4	0.25*** (3.11)	-0.07 (-0.67)	0.16 (1.47)	-0.24*** (-4.96)	0.18* (1.86)	-0.07 (-0.85)
Big	0.28*** (4.38)	-0.17 (-1.56)	0.09 (1.49)	-0.01 (-0.30)	0.31*** (3.62)	0.03 (0.36)

closely related to HML exposure, while the size dimension within the highest χ quintile is more closely associated with RMW exposure. As χ increases, HML loadings generally decline, indicating that slow-adjusting firms exhibit more growth-like factor exposure. In addition, within the highest χ quintile, where the size effect appears more pronounced, RMW loadings tend to decline as firm size decreases. Overall, high- χ firms appear more growth-like and carry higher risk, while within the highest- χ group, this risk is concentrated among smaller firms that also appear more weak-profitability-like.

Economically, this provides a somewhat coherent interpretation of the Size- χ evidence. A lower speed of adjustment makes it harder for firms to adjust quickly in response to valuation signals and thereby narrow the wedge between book and market values, which helps explain why slow-adjusting firms exhibit more growth-like factor exposure and earn higher FF5 alphas. Among the slowest-adjusting firms, this adjustment friction appears to interact with size and profitability, as smaller firms earn higher FF5 alphas and also exhibit more weak-profitability-like factor exposure. This suggests that slow adjustment may be especially consequential for small firms. This is consistent with the intuition embedded in the q -theory of investment. Smaller firms may be further from their steady-state capital stock and therefore more dependent on the speed at which investment opportunities are translated into realized operating profitability. Large firms, by contrast, may be better able to absorb the risks associated with a lower speed of adjustment because they already have more installed capital, current cash flows, and production capacity. A low speed of adjustment therefore appears to amplify profitability-related differences across size groups, which helps explain why the size effect in FF5 alphas appears strongest among the slowest-adjusting firms.

6.3.4 Three-Dimensional $2 \times 4 \times 4$ χ -Sorts

Moving to the three-dimensional $2 \times 4 \times 4$ portfolio sorts provides more structure on the role of adjustment costs. Across all three χ -sorted portfolio families, a consistent pattern emerges: among small firms, FF5 alphas generally increase from low to high χ , along with decreasing HML and RMW factor loadings. Figure 10 makes this visually clear by plotting average FF5 alphas, HML loadings, and RMW loadings across χ quartiles. Evidently, FF5 alphas tend to start negative and increase with χ across small firms, while HML and RMW loadings seem to decrease. This suggests that quick-adjusting small firms are associated with lower risk than predicted by the FF5 model, while the declining loadings indicate that high- χ small firms are more growth-like and weak-profitability-like, once again confirming the previous findings in this paper. Together, we interpret these results as showing that adjustment costs are most informative among small firms, where the speed of converting investment opportunities into realized capital and earnings is likely to be most economically important.

6.3.4.1 Size-OP- χ Test Portfolios

Table 8 reports FF5 alphas and factor loadings for the 32 Size-OP- χ portfolios, while a full comparison across benchmark models is provided in Appendix Table A.12. The

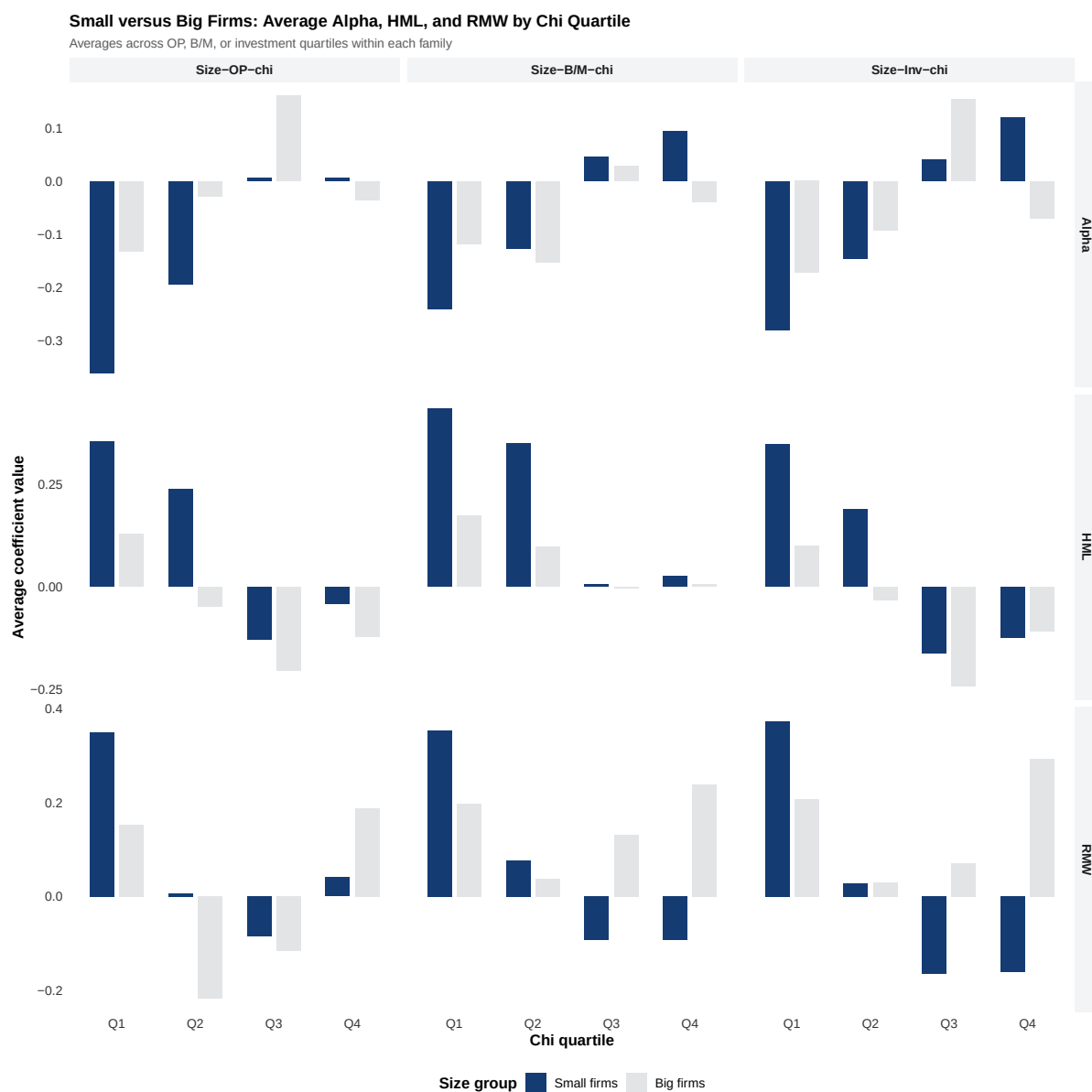


Figure 10: Bar chart of average Fama-French Five-Factor alphas, HML loadings, and RMW loadings across χ quartiles for small and big firms.

Source: Authors' own creation.

central finding is that adjustment costs matter most for small firms, which are typically earlier in their life cycle and face greater challenges in converting growth opportunities into operating profits.

Appendix Table A.12 reports model alphas under CAPM, FF5, and FF5 + SMQ. The broad conclusions are similar to those previously reported: adding SMQ to FF5 does not appear to provide substantially better performance relative to the benchmark models. Beyond these direct comparisons, the CAPM alphas are consistent with the standard profitability anomaly, lending credibility to the portfolio sorts. Since SMQ earns a positive and statistically significant alpha relative to the FF5 model, and since this is also where the adjustment-cost pattern is clearest, we focus on these alphas to examine the adjustment-cost effect more structurally.

Turning to Table 8, there is a clear pattern among FF5 alphas and the corresponding HML and RMW loadings. Starting with Panel A, the adjustment-cost effect is considerably clearer for small firms than for large firms. Among small firms, alphas generally increase from the lowest to the highest χ quartile within most profitability groups, whereas this pattern is less pronounced among large firms. This pattern for small firms is especially strong in the two lowest profitability groups, where the high-minus-low χ spreads are 0.79% ($t = 3.20$) and 0.60% ($t = 4.08$) per month for low-OP and OP 2 small firms, respectively. Within these alpha spreads, most of the significant alphas appear among the quickest-adjusting firms. To be specific, three out of four small-firm alphas are negative and statistically significant in the lowest χ quartile, with values of -0.68% ($t = -3.76$), -0.46% ($t = -4.24$), and -0.23% ($t = -2.01$) for low-OP, OP 2, and high-OP small firms, respectively. The evidence hence suggests that adjustment-cost heterogeneity is a more consequential source of cross-sectional return variation among small firms, where the ability to convert growth opportunities into realized capital, cash flows, and earnings is likely to be especially important.

The factor loadings in Panels B and C provide further insight into the behavior of the FF5 alphas. Across both small and large firms, HML loadings generally decline as χ increases, indicating that high- χ firms become more growth-like. RMW loadings, however, decline structurally only for small firms. This suggests that, among small firms where the adjustment-cost effect is strongest, slow-adjusting firms exhibit increasingly weak profitability. Economically, this is consistent with the idea that small firms are more dependent on adjusting capital efficiently to realize operating profits, whereas profitability among large firms is less sensitive to such frictions. Large firms typically already have substantial capital in production and can therefore rely more on existing operational capacity to sustain profitability, whereas small firms depend more on adjustment itself to build that capacity. Thus, for small firms, the speed of adjustment has more direct implications for operating profitability.

Table 8: FF5 alphas, HML loadings, and RMW loadings for Size-OP- χ portfolios

US equities, July 1982–December 2024. This table reports selected time-series regression results for the $2 \times 4 \times 4$ Size-OP- χ portfolios under the Fama–French Five-Factor model. At the end of each June, firms are assigned to two size groups using the NYSE median and then independently sorted into four operating profitability portfolios and four adjustment-cost χ portfolios using NYSE quartile breakpoints. Panels A–C report FF5 intercepts, HML loadings, and RMW loadings, respectively. The $H - L$ χ column reports the high-minus-low χ spread within each OP quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: FF5 alpha</i>										
Low OP	-0.68*** (-3.76)	-0.44** (-2.01)	0.07 (0.72)	0.11 (0.90)	0.79*** (3.20)	-0.01 (-0.05)	-0.26 (-1.35)	0.13 (0.97)	0.26* (1.86)	0.26 (1.27)
2	-0.46*** (-4.24)	-0.21 (-1.48)	0.02 (0.20)	0.14 (1.32)	0.60*** (4.08)	-0.24 (-1.37)	0.12 (0.64)	0.12 (1.18)	-0.13 (-1.06)	0.11 (0.61)
3	-0.07 (-0.55)	0.09 (0.75)	0.04 (0.42)	-0.08 (-0.98)	-0.01 (-0.07)	-0.11 (-0.90)	0.12 (0.73)	0.11 (0.90)	-0.04 (-0.37)	0.07 (0.39)
High OP	-0.23** (-2.01)	-0.21* (-1.89)	-0.10 (-1.16)	-0.14 (-1.60)	0.09 (0.69)	-0.17 (-1.48)	-0.10 (-0.84)	0.28*** (2.83)	-0.23* (-1.94)	-0.06 (-0.36)
<i>Panel B: HML loading</i>										
Low OP	0.28** (2.52)	0.17 (1.32)	-0.45*** (-9.28)	-0.45*** (-7.41)	-0.73*** (-5.73)	0.35** (2.32)	0.10 (0.80)	-0.07 (-0.74)	-0.15** (-2.10)	-0.50*** (-2.89)
2	0.46*** (6.12)	0.31*** (4.37)	-0.06 (-1.01)	0.01 (0.20)	-0.45*** (-5.66)	-0.08 (-0.97)	-0.21* (-1.82)	-0.14** (-2.07)	-0.02 (-0.26)	0.06 (0.46)
3	0.30*** (3.29)	0.19*** (3.03)	-0.04 (-0.95)	0.07 (0.91)	-0.23*** (-4.91)	0.21** (2.38)	0.18** (2.06)	-0.20*** (-3.38)	-0.17*** (-3.92)	-0.38*** (-3.91)
High OP	0.38*** (4.96)	0.28*** (4.99)	0.04 (0.69)	0.20** (2.25)	-0.18*** (-3.15)	0.03 (0.38)	-0.27*** (-4.04)	-0.41*** (-7.61)	-0.14** (-2.39)	-0.17 (-1.39)
<i>Panel C: RMW loading</i>										
Low OP	-0.21* (-1.94)	-0.84*** (-7.11)	-0.88*** (-17.70)	-0.85*** (-9.08)	-0.64*** (-3.78)	-0.35*** (-2.95)	-0.56*** (-3.81)	-0.70*** (-5.87)	-0.39*** (-4.09)	-0.05 (-0.35)
2	0.38*** (4.46)	0.06 (0.81)	-0.02 (-0.36)	0.13* (1.74)	-0.25*** (-3.12)	0.17** (2.09)	-0.35** (-2.38)	-0.22*** (-2.87)	0.28*** (3.64)	0.11 (0.89)
3	0.51*** (5.75)	0.24*** (3.91)	0.13** (2.58)	0.35*** (4.59)	-0.15** (-2.41)	0.20*** (3.86)	-0.25** (-2.56)	0.24*** (4.14)	0.40*** (4.24)	0.20* (1.66)
High OP	0.73*** (8.02)	0.57*** (8.46)	0.43*** (7.91)	0.52*** (7.25)	-0.20*** (-3.46)	0.59*** (7.54)	0.29*** (4.82)	0.22*** (4.20)	0.46*** (4.98)	-0.13 (-1.41)

6.3.4.2 Size-B/M- χ Test Portfolios

Table 9 reports FF5 alphas and factor loadings for the 32 Size-B/M- χ portfolios, while a full comparison across benchmark models is provided in Appendix Table A.13. The central finding mirrors the Size-OP- χ sorted results: adjustment costs matter most for small firms, as these firms are more dependent on effectively converting growth opportunities into operating profitability.

Appendix Table A.13 reports model alphas under CAPM, FF5, and FF5 + SMQ. The broad conclusions carry over: augmenting FF5 with SMQ provides only limited additional explanatory power, and the adjustment-cost effect is most visible in the FF5 alphas. A notable difference relative to the Size-OP- χ sorts is that CAPM produces only two weakly significant alphas across the entire Size-B/M- χ portfolio family, suggesting that CAPM performs relatively well among these portfolio sorts. However, this likely reflects the fact that B/M variation generates less extreme return dispersion than OP variation, leaving less mispricing for CAPM to miss, consistent with evidence that the value effect has weakened in recent periods. For these reasons, and consistent with our earlier approach, we concentrate on the FF5 alphas.

Turning to Table 9, the adjustment-cost effect is again considerably clearer for small firms than for large firms. This is most evident in the statistically significant high-minus-low χ spreads for small firms, at 0.48% ($t = 2.99$), 0.30% ($t = 2.70$), and 0.55% ($t = 2.63$) for low-B/M, B/M 2, and B/M 3, respectively. Again, the significant alphas underlying these spreads appear mainly in the lowest χ quartile, where two out of four FF5 alphas are negative and statistically significant: -0.40% ($t = -3.35$) for low-B/M firms and -0.38% ($t = -2.84$) for B/M 3. The pattern is less pronounced among large firms, reinforcing the conclusion that adjustment-cost heterogeneity is most consequential for small firms.

Turning to the factor loadings in Panels B and C, the HML and RMW patterns again support our earlier interpretation. HML loadings decline with χ for both small and large firms, consistent with slow-adjusting firms appearing more growth-like, likely because they are less able to invest optimally in response to growth opportunities, so expected growth is reflected in market valuations but not yet in book values, leaving a larger wedge between the two. The RMW structure is less systematic for large firms but seems to decline for small firms as χ increases: for example, at the lowest χ quartile, RMW loadings are positive and significant across most sorts, with values of 0.49 ($t = 5.31$), 0.44 ($t = 6.07$), 0.34 ($t = 4.62$), and 0.15 ($t = 1.56$), while at the highest χ quartile, two of the four loadings turn significantly negative at -0.32 ($t = -4.39$) and -0.24 ($t = -2.30$). This confirms the same story as before. Among small firms, adjustment costs are associated with profitability exposure because they determine whether growth opportunities translate into realized operating profit.

Table 9: FF5 alphas, HML loadings, RMW loadings, and CMA loadings for Size-B/M- χ portfolios

US equities, July 1982–December 2024. This table reports selected time-series regression results for the $2 \times 4 \times 4$ Size-B/M- χ portfolios under the Fama–French five-factor model. At the end of each June, firms are assigned to two size groups using the NYSE median and then independently sorted into four book-to-market portfolios and four adjustment-cost χ portfolios using NYSE quartile breakpoints. Panels A–D report FF5 intercepts, HML loadings, RMW loadings, and CMA loadings, respectively. The $H - L$ columns report high-minus-low χ spreads within each B/M quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: FF5 alpha</i>										
Low B/M	-0.40*** (-3.35)	-0.20* (-1.80)	0.08 (1.13)	0.08 (0.92)	0.48*** (2.99)	-0.10 (-0.86)	0.03 (0.25)	0.24*** (2.62)	-0.05 (-0.48)	0.05 (0.33)
2	-0.17 (-1.59)	-0.04 (-0.38)	0.07 (0.82)	0.13 (1.56)	0.30*** (2.70)	-0.10 (-0.85)	-0.12 (-0.83)	0.05 (0.42)	-0.10 (-0.93)	-0.01 (-0.05)
3	-0.38*** (-2.84)	0.14 (0.83)	0.02 (0.16)	0.17 (1.37)	0.55*** (2.63)	-0.21 (-1.41)	-0.08 (-0.50)	0.01 (0.05)	-0.03 (-0.26)	0.17 (0.96)
High B/M	-0.01 (-0.08)	-0.40 (-1.39)	0.01 (0.10)	-0.01 (-0.10)	0.00 (0.01)	-0.08 (-0.56)	-0.45*** (-2.77)	-0.18 (-1.24)	0.03 (0.18)	0.10 (0.48)
<i>Panel B: HML loading</i>										
Low B/M	0.09 (1.13)	-0.09 (-1.23)	-0.41*** (-8.97)	-0.39*** (-8.37)	-0.47*** (-6.99)	-0.30*** (-4.55)	-0.37*** (-4.69)	-0.43*** (-7.74)	-0.30*** (-6.24)	-0.00 (-0.04)
2	0.38*** (5.65)	0.28*** (5.21)	-0.10** (-2.56)	0.03 (0.45)	-0.35*** (-6.32)	-0.03 (-0.28)	0.00 (0.05)	-0.12 (-1.62)	-0.06 (-1.01)	-0.03 (-0.25)
3	0.54*** (8.96)	0.38*** (5.53)	0.15* (1.73)	0.13** (2.31)	-0.41*** (-4.75)	0.38*** (4.01)	0.19* (1.65)	0.04 (0.75)	0.08 (1.17)	-0.30*** (-2.91)
High B/M	0.73*** (9.18)	0.83*** (5.07)	0.39*** (4.59)	0.33*** (2.65)	-0.40*** (-3.19)	0.64*** (6.09)	0.56*** (5.08)	0.49*** (5.86)	0.32*** (3.53)	-0.32** (-2.51)
<i>Panel C: RMW loading</i>										
Low B/M	0.49*** (5.31)	-0.03 (-0.42)	-0.30*** (-6.27)	-0.32*** (-4.39)	-0.81*** (-7.47)	0.35*** (3.86)	-0.12 (-1.51)	0.01 (0.25)	0.35*** (3.29)	0.00 (0.01)
2	0.44*** (6.07)	0.18*** (2.86)	0.01 (0.09)	0.14 (1.62)	-0.30*** (-4.09)	0.24*** (2.88)	0.09 (1.22)	0.26*** (3.24)	0.25*** (3.64)	0.02 (0.14)
3	0.34*** (4.62)	0.03 (0.46)	0.08 (1.28)	0.05 (0.86)	-0.29** (-2.45)	0.21*** (2.64)	0.07 (0.80)	0.14 (1.26)	0.25*** (3.23)	0.04 (0.42)
High B/M	0.15 (1.56)	0.12 (0.54)	-0.16** (-2.06)	-0.24** (-2.30)	-0.39*** (-3.11)	0.00 (0.03)	0.11 (0.82)	0.11 (1.09)	0.11 (1.19)	0.10 (0.96)
<i>Panel D: CMA loading</i>										
Low B/M	-0.15* (-1.79)	-0.06 (-0.56)	-0.03 (-0.43)	0.04 (0.47)	0.19 (1.53)	0.16 (1.23)	-0.38** (-2.29)	-0.12 (-1.35)	0.34** (2.39)	0.17 (1.05)
2	0.01 (0.06)	-0.04 (-0.45)	0.25*** (2.79)	0.19** (2.17)	0.18 (1.44)	0.26** (1.97)	0.11 (1.02)	0.18* (1.87)	0.30*** (3.11)	0.04 (0.25)
3	0.02 (0.24)	0.09 (0.80)	0.26** (2.21)	0.28*** (3.18)	0.26* (1.72)	0.17 (1.30)	0.15 (0.82)	0.36** (2.47)	0.48*** (5.28)	0.31* (1.91)
High B/M	0.08 (0.62)	0.17 (0.89)	0.30** (2.10)	0.30* (1.71)	0.21 (1.06)	0.16 (1.10)	-0.01 (-0.07)	-0.09 (-0.72)	0.08 (0.56)	-0.08 (-0.50)

A further result specific to the Size-B/M- χ portfolios concerns the CMA loadings in Panel D. When we include B/M in the portfolio sorts, which is closely related to the inverse of average q , CMA loadings tend to increase with χ , especially among small firms. This indicates that slow-adjusting firms invest more conservatively than quick-adjusting firms. These results are intuitive and consistent with the investment- q relationship: for a given valuation signal, higher adjustment costs should be associated with more conservative investment behavior. Once again, the pattern is relatively clearer among small firms but weaker and less systematic among large firms, suggesting that adjustment costs are most consequential for firms earlier in their life cycle.

6.3.4.3 Size-Inv- χ Test Portfolios

Table 10 reports FF5 alphas and factor loadings for the 32 Size-Inv- χ portfolios, while a full comparison across benchmark models is provided in Appendix Table A.14. Once again, the findings echo the previous results: adjustment costs matter most for small firms, although in this case significant alphas are present at both ends of the adjustment-cost distribution, that is, among both low- χ and high- χ firms.

Appendix Table A.14 reports model alphas under CAPM, FF5, and FF5 + SMQ. The model comparison remains similar to the previous results. SMQ adds little explanatory power beyond FF5, and in some cases even performs marginally worse among large firms. CAPM again produces relatively few significant alphas and this time becomes comparable to the FF5 model in its performance, which appears interesting at first glance. However, since the traditional investment anomaly remains visible, we interpret this result cautiously. Specifically, CAPM appears to leave more investment-related structure unexplained, while FF5 leaves more adjustment-cost-related structure unexplained. The two models can therefore look relatively comparable in aggregate, even though they fail along different dimensions. As before, because FF5 provides the clearest view of the adjustment-cost effect, we concentrate on those results in this section.

Turning to Table 10, the adjustment-cost effect is again considerably clearer among small firms than large firms. FF5 alphas increase broadly monotonically across χ quartiles within most investment sorts. In addition, significant alphas now appear at both χ extremes. At the lowest χ quartile, small-firm alphas are consistently negative across all investment groups, at -0.27% ($t = -1.85$), -0.26% ($t = -2.49$), -0.16% ($t = -1.79$), and -0.42% ($t = -3.12$), while at the highest χ quartile, alphas turn positive for most groups, with values of 0.25% ($t = 2.13$), 0.33% ($t = 2.64$), and 0.13% ($t = 1.51$), alongside a remaining negative estimate of -0.22% ($t = -2.29$). Furthermore, the high-minus-low χ spreads are significant for the lower investment sorts, at 0.52% ($t = 2.49$), 0.59% ($t = 3.08$), and 0.29% ($t = 2.90$) for the low-investment, investment-2, and investment-3 sorts, respectively. Taken together, this once again reinforces the interpretation that adjustment-cost heterogeneity is most economically relevant among small firms.

Table 10: FF5 alphas, HML loadings, and RMW loadings for Size–Inv– χ portfolios

US equities, July 1982–December 2024. This table reports selected time-series regression results for the $2 \times 4 \times 4$ Size–Inv– χ portfolios under the Fama–French five-factor model. At the end of each June, firms are assigned to two size groups using the NYSE median and then independently sorted into four investment portfolios and four adjustment-cost χ portfolios using NYSE quartile breakpoints. Panels A–C report FF5 intercepts, HML loadings, and RMW loadings, respectively. The $H - L$ column reports the high-minus-low χ spread within each investment quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: FF5 alpha</i>										
Low Inv	-0.27* (-1.85)	-0.21 (-1.14)	0.09 (0.71)	0.25** (2.13)	0.52** (2.49)	-0.17 (-1.31)	-0.27** (-2.35)	0.10 (0.88)	0.02 (0.16)	0.19 (1.18)
2	-0.26** (-2.49)	-0.12 (-1.23)	0.06 (0.79)	0.33*** (2.64)	0.59*** (3.08)	-0.19 (-1.57)	-0.19 (-1.41)	0.05 (0.51)	-0.24* (-1.73)	-0.05 (-0.28)
3	-0.16* (-1.79)	0.01 (0.10)	0.09 (1.23)	0.13 (1.51)	0.29*** (2.90)	-0.16 (-1.50)	-0.14 (-1.13)	0.09 (1.05)	-0.12 (-1.18)	0.04 (0.27)
High Inv	-0.42*** (-3.12)	-0.26* (-1.84)	-0.08 (-0.91)	-0.22** (-2.29)	0.20 (1.35)	-0.16 (-1.15)	0.23 (1.28)	0.37*** (2.69)	0.07 (0.58)	0.23 (1.48)
<i>Panel B: HML loading</i>										
Low Inv	0.40*** (4.83)	0.33*** (3.52)	-0.30*** (-4.29)	-0.25*** (-3.94)	-0.65*** (-7.68)	0.18 (1.65)	-0.07 (-0.65)	-0.28*** (-2.66)	-0.09 (-1.33)	-0.27* (-1.80)
2	0.44*** (5.21)	0.17*** (2.84)	0.03 (0.47)	-0.10 (-1.20)	-0.54*** (-3.63)	0.20*** (3.17)	0.02 (0.23)	-0.18** (-2.55)	-0.01 (-0.08)	-0.20* (-1.90)
3	0.37*** (4.95)	0.13*** (2.65)	-0.10** (-2.15)	0.02 (0.32)	-0.35*** (-6.78)	0.07 (0.55)	0.06 (0.76)	-0.21*** (-3.29)	-0.06 (-0.98)	-0.13 (-0.94)
High Inv	0.18*** (3.71)	0.13* (1.95)	-0.27*** (-5.72)	-0.17 (-1.64)	-0.35*** (-4.01)	-0.05 (-0.58)	-0.14 (-1.11)	-0.30*** (-5.54)	-0.27*** (-5.91)	-0.23** (-2.25)
<i>Panel C: RMW loading</i>										
Low Inv	0.31*** (3.01)	-0.32*** (-3.11)	-0.53*** (-6.96)	-0.56*** (-8.33)	-0.87*** (-6.56)	0.25*** (2.72)	0.19 (1.57)	0.16 (1.31)	0.21*** (3.10)	-0.04 (-0.37)
2	0.34*** (4.00)	0.24*** (5.33)	0.01 (0.09)	-0.12 (-1.23)	-0.46*** (-2.78)	0.21** (2.29)	0.17** (2.31)	0.21*** (3.74)	0.43*** (4.37)	0.22 (1.35)
3	0.54*** (7.15)	0.17*** (3.21)	0.09 (1.52)	0.11* (1.81)	-0.43*** (-5.97)	0.41*** (4.23)	0.09 (1.31)	0.13** (2.06)	0.34*** (3.95)	-0.06 (-0.48)
High Inv	0.31*** (3.83)	0.02 (0.29)	-0.23*** (-4.34)	-0.07 (-0.63)	-0.39*** (-4.36)	-0.04 (-0.60)	-0.33*** (-4.94)	-0.22*** (-2.87)	0.19** (2.25)	0.22* (1.93)

The factor loadings in Panels B and C reproduce the same structure as before. Specifically, HML loadings decline as χ increases for both small and large firms, whereas RMW loadings only seem to decline for small firms. Ultimately, the consistent decline in RMW among small firms indicates that a χ -related risk premium may be closely associated with profitability exposure. From a q -theoretic perspective, small firms may be earlier along their saddle path and further away from their optimal capital stock, making the speed of adjustment more consequential for their ability to generate realized operating profitability. By contrast, large firms may be better able to absorb the risks associated with a low speed of adjustment, as they already have more installed capital, current cash flows, and production capacity. Therefore, differences in adjustment costs may be less consequential for their operating profitability.

6.3.5 Partial Conclusion

The multidimensional portfolio sorts clarify the economic role of adjustment costs. In particular, the effect of adjustment costs is most important for small firms: from a q -theoretic perspective, these firms are earlier along their saddle path, for which the ability to convert growth opportunities into actual operating profit is particularly important. The first evidence was provided by the two-dimensional 5×5 Size- χ sort, which showed that returns not captured by the five-factor model increase with adjustment costs, except for the slowest-moving firms where the size effect is concentrated. The result is intuitive because slow adjustment is especially risky for small firms, as they typically have less installed capital and therefore rely more on their growth opportunities to generate future operating profitability. For large firms, this risk is less pronounced, since they already have more capital in production and more established operating profits. The second evidence is provided by the three-dimensional $2 \times 4 \times 4$ χ -related portfolio sorts, which consistently seem to confirm the same story. The adjustment-cost effect is concentrated among small firms and appears to be closely associated with operating profitability. This further confirms the earlier findings: slow-moving small firms are less able to realize their growth opportunities and are therefore less profitable, whereas quick-moving small firms react more efficiently and are therefore more profitable and less risky. Hence, despite the weak pattern in the one-dimensional sorts, adjustment costs appear to be embedded in the five-factor dimensions in a manner that is closely aligned with the q -theory.

6.4 Robustness Checks

The empirical analysis thus far relies on the industry-level estimate of the adjustment-cost parameter, χ . While this choice is motivated by prior literature, it is not neutral, since χ is not directly observed and its estimate depends on the research design and may be sensitive to the level of aggregation. We therefore introduce two alternative specifications, firm-level χ and major-group χ , and assess whether the main findings hold across these constructions. The analysis proceeds in three steps, examining the SMQ factor properties, the Size- χ alpha patterns, and the high-minus-low χ spreads in the three-dimensional portfolio sorts.

6.4.1 Robustness of SMQ and Decile Portfolio Alphas

Table 11 examines the robustness of SMQ factor properties and χ -sorted decile alphas across three alternative χ specifications. Across all constructions, the SMQ factor exhibits a small and statistically insignificant unconditional mean return, confirming the factor's weak unconditional performance. Incremental pricing information relative to FF5 is generally preserved at the firm and industry levels, though it diminishes under the major-group aggregation.

The characterization of SMQ as a weak unconditional return premium is fully robust: across all three constructions, the mean excess return is small and statistically indistinguishable from zero, confirming that the long-short adjustment-cost strategy does not generate a compelling unconditional mean return regardless of the level at which χ is estimated. However, the evidence for incremental pricing information relative to FF5 remains strong for the firm-level construction but weakens under the major-group aggregation. Specifically, the firm-level construction produces a significant FF5 alpha of 0.24% ($t = 3.22$), the industry-level construction yields 0.32% ($t = 3.19$), while the major-group construction produces a positive but insignificant alpha of 0.18% ($t = 1.37$).

The factor loadings provide further insight. The negative and significant loading on HML persists across all three constructions, while the negative loading on RMW is also present but becomes statistically insignificant under the major-group specification. Consistent with the alpha results, the negative RMW loading is significant only when the FF5 alpha is significant, suggesting that the adjustment-cost effect is closely linked to profitability exposure. Together, these patterns confirm that the growth and weak-profitability tilt of high- χ firms is a stable feature of the adjustment cost effect rather than an artifact of the specific χ construction, although not fully robust across specifications.

Finally, the decile alpha evidence in Panel C is broadly consistent across specifications. Across all three constructions, low- χ portfolios, $P1$ – $P2$, exhibit significant negative alphas, while higher- χ portfolios, particularly $P7$ – $P8$, show significant positive alphas. The overall pattern is consistently non-monotonic across specifications, and no construction produces a clear or systematic progression in FF5 alphas from $P1$ to $P10$.

6.4.2 Robustness of Size- χ Alphas

Table 12 reports FF5 alphas for the 25 Size- χ portfolios across three alternative χ constructions. The central pattern is relatively stable, although not fully robust. Across specifications, FF5 alphas generally tend to increase with higher χ , most high-minus-low χ spreads are positive, and three of the five spreads are statistically significant at least at the 10% significance level.

Examining within-size-group patterns, FF5 alphas seem to move from negative to positive as χ increases, consistent with the hypothesis that higher adjustment costs are associated with higher expected returns relative to the five-factor model. However, the statistical significance of each individual portfolio alpha is less consistent across specifications. A notable nuance is that the baseline industry-level specification shows particu-

Table 11: Robustness of SMQ and χ -sorted decile portfolios

US equities, July 1982–December 2024. This table compares three value-weighted SMQ (slow-minus-quick) constructions based on alternative estimates of the adjustment-cost parameter χ : a firm-level measure, an SIC 4-digit industry measure, and an SIC 2-digit major-group measure. Panel A reports full-sample properties of the corresponding SMQ factors, including FF5 alpha. Panel B reports FF5 spanning regressions in which each SMQ factor is regressed on $R_M - R_f$, SMB, HML, RMW, and CMA. The intercept is reported as α , while the remaining coefficients are factor loadings. Panel C reports FF5 alphas for the corresponding χ -sorted decile portfolios, where P1 denotes the lowest- χ decile and P10 the highest- χ decile. Mean returns and alphas are monthly percentages. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	Firm-level χ	Industry χ	Major-group χ
<i>Panel A: SMQ factor properties</i>			
Mean excess return	0.12 (1.49)	0.09 (0.75)	0.06 (0.45)
CAPM alpha	0.07 (0.83)	0.07 (0.58)	0.09 (0.62)
FF3 alpha	0.16** (2.16)	0.20* (1.89)	0.17 (1.36)
FF5 alpha	0.24*** (3.22)	0.32*** (3.19)	0.18 (1.37)
Std. Dev	1.60	2.32	2.60
Sharpe ratio	0.27	0.13	0.08
<i>Panel B: FF5 alpha and factor loadings of alternative SMQ factors</i>			
α	0.24*** (3.22)	0.32*** (3.19)	0.18 (1.37)
$R_M - R_f$	0.00 (0.23)	-0.06** (-2.03)	-0.05 (-1.41)
SMB	-0.03 (-1.18)	-0.06 (-1.38)	-0.21*** (-3.33)
HML	-0.29*** (-7.33)	-0.44*** (-7.19)	-0.36*** (-3.97)
RMW	-0.19*** (-4.92)	-0.31*** (-3.70)	-0.10 (-0.77)
CMA	0.04 (0.60)	0.09 (0.86)	0.16 (1.12)
<i>Panel C: FF5 alphas for χ-sorted decile portfolios</i>			
P1	-0.03 (-0.46)	-0.21** (-2.12)	-0.39** (-2.25)
P2	-0.26* (-1.93)	-0.12 (-1.31)	-0.04 (-0.19)
P3	-0.15 (-1.07)	-0.24 (-1.42)	-0.02 (-0.17)
P4	-0.13 (-1.19)	-0.01 (-0.08)	0.23 (1.56)
P5	-0.11 (-0.94)	0.01 (0.09)	-0.06 (-0.44)
P6	0.03 (0.25)	-0.02 (-0.25)	0.00 (0.02)
P7	0.10 (1.03)	0.20* (1.93)	0.23* (1.79)
P8	0.22** (2.15)	0.13 (1.05)	-0.06 (-0.50)
P9	-0.10 (-1.20)	-0.12 (-1.14)	-0.05 (-0.50)
P10	0.08 (1.13)	-0.14 (-1.64)	-0.18 (-1.27)

Table 12: Robustness of FF5 alphas across Size- χ portfolios

US equities, July 1982–December 2024. This table reports FF5 alphas for the 25 Size- χ portfolios across alternative estimates of the adjustment-cost parameter χ . Panel A uses firm-level χ , Panel B uses SIC 4-digit industry-level χ , and Panel C uses SIC 2-digit major-group χ . In each panel, the final column reports the high-minus-low ($H-L$) alpha from a direct time-series regression of the high-minus-low χ spread return within each size quintile. Alphas are reported in monthly percent. Newey–West t -statistics with 12 lags are shown in parentheses below the estimates where available. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	χ					
Size	Low	2	3	4	High	H-L
<i>Panel A: Firm-level χ</i>						
Small	0.06 (0.49)	0.02 (0.20)	0.12 (1.13)	0.08 (0.52)	0.35*** (2.73)	0.29** (2.09)
2	-0.26*** (-3.12)	-0.24** (-2.00)	0.01 (0.12)	0.00 (0.04)	-0.06 (-0.53)	0.21* (1.80)
3	-0.10 (-0.91)	-0.19* (-1.68)	-0.19** (-2.05)	-0.02 (-0.19)	-0.04 (-0.42)	0.06 (0.60)
4	-0.14* (-1.81)	-0.24** (-2.07)	0.03 (0.35)	-0.09 (-0.94)	0.09 (1.29)	0.23** (2.43)
Big	-0.09 (-1.11)	-0.09 (-0.73)	-0.02 (-0.19)	0.23** (2.38)	0.04 (0.61)	0.13 (1.21)
<i>Panel B: Industry-level χ</i>						
Small	-0.13 (-0.74)	-0.26* (-1.67)	-0.10 (-1.02)	0.32*** (2.79)	0.24** (2.08)	0.37* (1.67)
2	-0.29** (-2.54)	-0.33** (-2.03)	-0.23*** (-2.65)	0.29*** (3.63)	-0.10 (-1.04)	0.19 (1.49)
3	-0.29*** (-2.65)	-0.16 (-1.05)	-0.16 (-1.43)	0.04 (0.58)	0.17* (1.88)	0.46*** (4.12)
4	-0.17* (-1.91)	-0.22 (-1.25)	-0.12 (-1.36)	0.30*** (3.57)	0.02 (0.20)	0.19* (1.77)
Big	-0.13 (-1.33)	-0.04 (-0.27)	0.05 (0.75)	0.24*** (2.73)	-0.18** (-2.12)	-0.05 (-0.45)
<i>Panel C: Major-group χ</i>						
Small	-0.46** (-1.97)	0.01 (0.09)	-0.02 (-0.26)	0.30* (1.69)	0.22 (1.53)	0.68** (2.34)
2	-0.44** (-2.45)	-0.17 (-1.58)	-0.07 (-0.72)	0.08 (0.95)	0.04 (0.38)	0.48** (2.37)
3	-0.37** (-2.48)	-0.09 (-0.78)	-0.14 (-1.37)	0.20** (2.39)	0.01 (0.08)	0.38** (2.36)
4	-0.25* (-1.87)	0.14 (0.82)	-0.02 (-0.23)	0.07 (0.81)	-0.07 (-0.60)	0.19 (1.11)
Big	-0.14 (-1.12)	0.21* (1.68)	0.03 (0.42)	0.12 (1.05)	-0.10 (-0.93)	0.05 (0.28)

larly strong and significant alphas in the $\chi = 4$ quintile, a pattern that becomes weaker under both the firm-level and major-group constructions, suggesting that some of the intermediate- χ variation is less stable across specifications. Hence, consistent with the main results, FF5 alphas tend to increase with higher χ , albeit in a less consistent and more dispersed manner across specifications.

Turning to the high-minus-low χ spreads, the small-firm portfolios generally exhibit positive FF5 alphas, consistent with adjustment-cost heterogeneity being more relevant for these firms. The spreads are statistically significant or weakly significant across constructions: 0.29% ($t = 2.09$) for firm-level χ , 0.37% ($t = 1.67$) for industry-level χ , and 0.68% ($t = 2.34$) for major-group χ . In contrast, the spreads for the largest firm quintile are small and statistically insignificant across specifications, and even negative for the industry-level χ construction. These findings suggest that the χ -related alpha pattern is concentrated among smaller firms rather than larger firms.

6.4.3 Robustness of High-Minus-Low χ Alphas in Three-Dimensional Sorts

Table 13 provides the most comprehensive robustness evidence in this section, examining the high-minus-low χ spreads across the Size-OP- χ , Size-B/M- χ , and Size-Inv- χ portfolio families under all three χ constructions. The results consistently indicate a positive χ spread concentrated among small firms, with the strongest effects observed for firms with low operating profitability, low book-to-market, and low investment.

Panel A shows FF5 alphas for the three-dimensional Size-OP- χ portfolio families, with the most robust results observed for the major-group χ construction. Specifically, for both industry and major-group χ , high-minus-low alpha spreads are largest for small, low-operating-profitability firms: 0.79% ($t = 3.20$) for industry and 1.18% ($t = 3.26$) for major-group. These alphas tend to decrease as operating profitability rises, consistent with the interpretation that adjustment-cost frictions are most relevant for small firms that have not yet converted growth opportunities into earnings. In contrast, for firm-level χ , the high-minus-low alpha for low-operating-profitability firms is smaller and insignificant, and the monotonic decline with operating profitability is not observed. Instead, the high-minus-low spreads are observed to be statistically significant in the high operating profitability quartiles. Hence, robustness seems stronger for major-group and industry χ rather than for firm-level χ .

Table 13: Robustness of high-minus-low χ spreads across 32-portfolio sorts

US equities, July 1982–December 2024. This table reports FF5 alphas for high-minus-low χ spread portfolios across the 32 Size-B/M- χ , Size-OP- χ , and Size-Inv- χ portfolio families. Within each size group and characteristic quartile, the high-minus-low portfolio is long the highest- χ quartile and short the lowest- χ quartile. Columns compare three alternative estimates of the adjustment-cost parameter χ : firm-level χ , SIC 4-digit industry-level χ , and SIC 2-digit major-group χ . All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Characteristic quartile	Small			Big		
	Firm-level χ	Industry χ	Major group χ	Firm-level χ	Industry χ	Major group χ
<i>Panel A: Size-OP-χ</i>						
Low OP	0.31 (1.60)	0.79*** (3.20)	1.18*** (3.26)	0.38** (2.12)	0.26 (1.27)	0.64** (2.42)
2	0.11 (0.76)	0.60*** (4.08)	0.58*** (2.79)	-0.00 (-0.02)	0.11 (0.61)	0.27 (1.17)
3	0.35** (2.28)	-0.01 (-0.07)	0.03 (0.19)	0.25 (1.47)	0.07 (0.39)	0.13 (0.70)
High OP	0.31* (1.88)	0.09 (0.69)	0.17 (1.40)	0.04 (0.35)	-0.06 (-0.36)	0.00 (0.01)
<i>Panel B: Size-B/M-χ</i>						
Low B/M	0.35*** (2.62)	0.48*** (2.99)	0.69*** (3.57)	0.12 (0.96)	0.05 (0.33)	0.25 (1.53)
2	0.15 (1.19)	0.30*** (2.70)	0.27* (1.67)	0.14 (0.97)	-0.01 (-0.05)	0.09 (0.62)
3	0.23 (1.44)	0.55*** (2.63)	0.53** (2.02)	-0.07 (-0.33)	0.17 (0.96)	0.24 (1.27)
High B/M	0.38 (1.18)	0.00 (0.01)	0.55 (1.46)	0.19 (0.78)	0.10 (0.48)	0.16 (0.63)
<i>Panel C: Size-Inv-χ</i>						
Low Inv	0.43** (2.27)	0.52** (2.49)	0.74** (2.20)	0.09 (0.56)	0.19 (1.18)	0.08 (0.44)
2	0.04 (0.39)	0.59*** (3.08)	0.74*** (3.56)	0.09 (0.68)	-0.05 (-0.28)	0.22 (1.03)
3	0.43*** (2.95)	0.29*** (2.90)	0.49*** (3.12)	0.05 (0.34)	0.04 (0.27)	0.05 (0.28)
High Inv	0.33** (2.09)	0.20 (1.35)	0.36* (1.90)	-0.04 (-0.17)	0.23 (1.48)	0.40* (1.80)

Panel B provides a similar but more robust story for the Size-B/M- χ portfolio families. High-minus-low χ spreads for small, low-B/M firms show positive and statistically significant FF5 alphas across all three χ specifications: 0.35% ($t = 2.62$) for firm-level, 0.48% ($t = 2.99$) for industry, and 0.69% ($t = 3.57$) for major-group χ . Hence, the adjustment cost effect seems to persist among low-B/M firms, consistent with the interpretation that firms with high adjustment costs remain more growth-like for longer periods. Yet, major-group χ still appears the most robust: it produces two significant and one weakly significant FF5 alpha across all value quartiles, similar to industry χ , which has three significant spreads out of four value quartiles. In contrast, firm-level χ is less robust, with only one significant spread out of the four value quartiles.

Lastly, Panel C provides the most robust story for the Size-Inv- χ portfolio families. All three χ constructions produce positive FF5 alpha spreads for the high-minus-low χ portfolios, with three out of four small-firm investment cells significant across the investment quartiles. This pattern demonstrates strong robustness for the Size-Inv- χ portfolio families across the different specifications considered in our paper.

6.4.4 Partial Conclusion

The robustness analysis broadly supports the main findings, although the evidence is not uniform across all specifications. The weak unconditional mean excess return of SMQ remains consistent across all χ specifications, while evidence of incremental FF5 pricing information persists for both the firm-level and industry-level specifications, although in different areas. For the firm-level specification, the χ -related alpha pattern is visible in the SMQ factor, which earns a significant FF5 alpha, and in the high-minus-low χ spreads for the Size- χ test portfolios. However, its robustness is less visible in the high-minus-low χ spreads for the Size-OP- χ , Size-B/M- χ , and Size-Inv- χ test portfolios. For the major-group specification, the SMQ factor does not earn a significant FF5 alpha, yet the pattern appears to persist in the high-minus-low χ spreads for the Size- χ , Size-OP- χ , Size-B/M- χ , and Size-Inv- χ test portfolios. Viewed together, we interpret these results as relatively robust evidence of an adjustment cost effect, although this interpretation should be treated with caution since the exact FF5 alpha pattern varies across specifications.

6.5 Section Conclusion

The empirical evidence suggests a nuanced role for the q-theory of investment relative to the five-factor model. Although the adjustment-cost dimension does not generate a clear unconditional risk premium in excess returns and only marginally improves overall model performance, it provides a coherent signal relative to the full five-factor model. Ultimately, the evidence suggests that adjustment-cost frictions contain some relevant information in the cross-section of expected returns. Similarly, support for an augmented six-factor model including SMQ remains limited.

6.5.1 Hypothesis 1: A Positive Adjustment-Cost Risk Premium

For Hypothesis 1, which predicts that a tradable factor constructed from heterogeneity in adjustment costs earns a positive risk premium, the evidence points to conditional pricing information rather than a clear unconditional risk premium.

The one-dimensional evidence suggests that adjustment costs do not generate a strong standalone excess return, but that SMQ contains return variation not fully captured by the five-factor model. Excess returns, Sharpe ratios, and decile alphas are non-monotonic, and SMQ has no meaningful unconditional premium. However, SMQ earns a statistically significant FF5 alpha of 0.32% per month, indicating that the adjustment-cost parameter, χ , contains pricing information not fully captured by the five factors. The factor-spanning regressions support this interpretation: SMQ loads significantly and negatively on HML and RMW, and its inclusion weakens the value-profitability relationship. Thus, SMQ does not appear to be an independent pricing factor in the conventional sense, but rather captures part of the adjustment-cost structure underlying the value-profitability dimension. From a q-theoretic perspective, this is consistent with the idea that quicker-adjusting firms respond more efficiently to economic shocks, allowing valuation signals to translate more rapidly into realized capital, cash flows, and operating profitability.

The multidimensional portfolio sorts make this q-theoretical mechanism clearer. Across the Size- χ sort and the three-dimensional χ -related portfolio families, the adjustment-cost effect is concentrated among small firms. In particular, HML loadings decline with χ for both small and large firms, indicating that slow-adjusting firms exhibit more growth-like factor exposure, while RMW loadings decline systematically with χ mainly among small firms. This pattern is intuitive and in line with the q-theory of investment. Small firms may be earlier along their saddle path, and hence further from their optimal capital stock, which makes them more dependent on the speed of adjustment to translate investment opportunities into realized profitability. Large firms, by contrast, have more installed capital, current cash flows, and production capacity, making slow adjustment less consequential for their operating profitability.

The robustness checks largely reinforce this pattern. Across alternative χ specifications, SMQ continues to lack a strong unconditional premium, while the FF5 alpha and portfolio-spread evidence remain broadly intact, particularly among small firms. More specifically, the FF5 alpha of SMQ is strongest for the firm- and industry-level constructions, whereas high-minus-low χ spreads for small firms are most persistent in the industry- and major-group specifications. This suggests that the adjustment-cost effect is not merely an artifact of our baseline estimation.

Taken together, the evidence supports a positive adjustment-cost risk premium relative to the five-factor model, even though SMQ does not earn a strong unconditional premium. The main result is therefore not that speed of adjustment is priced as a standalone factor, but that it captures a q-theoretic risk premium not fully spanned by the five-factor model. As such, we confirm Hypothesis 1 relative to the five-factor model and argue that the evidence seems consistent with some of the economic mechanisms embedded in the q-theory of investment. At the same time, we do not confirm Hypothesis 1 in

unconditional terms, nor relative to the other benchmark asset pricing models.

6.5.2 Hypothesis 2: Adding SMQ to the Fama–French Five–Factor Model

Turning to Hypothesis 2, the evidence is less convincing. Augmenting the five-factor model with SMQ does not lead to a systematic improvement in empirical asset-pricing performance. The central finding is instead one of limited incremental value: while SMQ expands the efficient frontier, model performance improves modestly for selected χ -sorted portfolio families, while the broader set of test assets remains largely unaffected.

Based on the broader model-performance statistics, the incremental pricing contribution of SMQ appears limited. Although the six-factor model often reduces the GRS statistic relative to FF5, the improvement is small and much less pronounced than the earlier gain from FF3 to FF5. At the same time, average absolute alphas and $\bar{R}^2$ remain largely unchanged, indicating that our six-factor model does not materially improve pricing performance relative to the five-factor model. Meanwhile, the clearest gains appear in portfolios sorted directly on χ , where SMQ is naturally more likely to matter because the test assets are formed using the same underlying adjustment-cost dimension. These improvements are therefore limited from a holistic perspective.

The individual portfolio alphas point in the same direction. Adding SMQ reduces some alpha estimates for the Size- χ , Size-OP- χ , Size-B/M- χ , and Size-Inv- χ portfolios, but these reductions are neither broad nor consistent across the test portfolios. A stronger result would require the adjustment-cost effect to become less visible, or at least to be absorbed more systematically by the six-factor model. Instead, the alpha patterns largely remain, suggesting that the six-factor model only partially absorbs the adjustment-cost effect. Thus, when evaluated against the hypothesis, its incremental explanatory power in the χ -related portfolio sorts remains modest rather than convincing.

Altogether, the evidence provides only limited support for the second hypothesis. The SMQ factor appears to capture a relevant adjustment-cost dimension, which expands the efficient frontier, but its contribution to empirical model performance is only marginal. The results therefore suggest limited support for the claim that a six-factor model meaningfully improves upon the Fama–French five-factor model. On this basis, we do not find sufficient evidence to confirm Hypothesis 2.

7 Discussion of Limitations

This section discusses the main limitations of the study and their implications for the interpretation of the results. These limitations concern four areas: the theoretical assumptions linking adjustment frictions to asset pricing, the empirical estimation and measurement of the adjustment-cost parameter, the practical implementation of SMQ and mean–variance optimization, and the external validity of the findings beyond the sample and setting considered. Together, these limitations define the boundaries within which the results should be interpreted.

7.1 Theoretical Assumptions

The theoretical framework in this paper primarily stems from the simplifying assumptions underlying q -theory of investment and the exact theoretical link between adjustment frictions and risk premia. The purpose of this subsection is thus to clarify these assumptions and explain how they affect the interpretation of the results.

7.1.1 Lack of a Fully Unified Asset-Pricing model

A central theoretical limitation is that the framework developed in this study remains partial equilibrium in nature. The q -theory of investment provides a clear account of how firms adjust capital and how firm value responds to shocks in the presence of adjustment frictions. Similarly, the consumption-based framework explains in general terms how risk premia arise from the covariance between payoffs and the stochastic discount factor. However, the analysis does not develop a fully unified general-equilibrium model that maps differences in firms' speed of adjustment directly into differences in expected returns. As a result, the theoretical framework herein only provides a plausible economic motivation for why adjustment frictions may matter for asset pricing. Nevertheless, it does not pin down with confidence whether firms with slower adjustment should earn higher or lower risk premia, nor whether the relevant effect operates primarily through systematic risk or valuation levels. This may help explain why an adjustment cost factor, such as SMQ, does not necessarily earn a significant unconditional risk premium, even if adjustment frictions remain economically relevant for firm behavior and valuation.

7.1.2 Functional Form Assumptions

The estimation of the adjustment cost parameter relies on strong assumptions about the functional form of adjustment costs and the resulting investment– q relationship. In particular, the empirical specification assumes quadratic adjustment costs, which implies a linear relationship between the investment rate and q . This assumption is standard in the q -theory literature and yields a tractable linear specification that can be estimated using regression methods (Abel & Eberly, 1994). However, there is a very real risk that it may not accurately reflect firms' true adjustment behavior. In practice, it might be the case that adjustment costs are non-linear, asymmetric, or vary across firms and over time (Cooper & Haltiwanger, 2006). For example, firms may face different costs when increasing versus decreasing capital (Zhang, 2005), or may adjust investment in a lumpy rather than gradual manner (Doms & Dunne, 1998). If the true relationship between investment and q deviates from the assumed linear form, the estimated adjustment cost parameter χ may be mismeasured and fail to accurately reflect the underlying economic mechanism. Ultimately, these considerations imply that the estimated adjustment cost parameter χ should be interpreted as an empirical approximation of adjustment frictions rather than a direct measure of the true underlying adjustment costs.

7.1.3 Average q as a Proxy for Marginal q

In addition to assumptions about functional form, the empirical implementation relies on average q as a proxy for the theoretically relevant marginal q , following the Hayashian theorem (Hayashi, 1982). Taken strictly, this equivalence depends on a set of restrictive assumptions that are unlikely to hold fully in practice. First, firms are assumed to be price takers, so firm value reflects competitive markets rather than market power or economic rents. Second, the production function is assumed to be linearly homogeneous in capital and labor, implying constant returns to scale. Third, adjustment costs are assumed to be linearly homogeneous in investment and capital, meaning that they depend only on the investment rate rather than on firm size itself. In reality, these assumptions are unlikely to hold exactly. For example, firms do hold certain market power, exhibit economies or diseconomies of scale, and adjustment costs may depend on experience from past investments, which break down these assumptions. These departures undoubtedly weaken the equivalence between average q and marginal q .

7.2 Empirical Limitations

The main empirical limitation of this study is that the adjustment-cost parameter χ is not directly observed, but estimated. As a result, the findings depend on several research design choices, including the estimation window, the definitions of investment and q , the regression specification, and industry pooling. Since these choices may affect both the χ estimates and the relative ordering of firms, this section clarifies these empirical limitations for transparency.

7.2.1 Choice of Estimation Window

One of the most important limitations concerns the choice of estimation window used to obtain the adjustment-cost parameter, as it directly affects both precision and economic relevance. As noted in the research design, the window length involves a trade-off. For example, while a shorter window, such as 10 years, may better reflect recent conditions, it relies on fewer observations, leading to greater estimation noise and therefore less precise and less stable estimates. In contrast, a longer window, such as 30 years, increases the number of observations and improves statistical precision, but incorporates older data that may be less representative of current firm behavior. Furthermore, it requires a longer burn-in period, which in this example would delay the first available estimates to 1992, thereby further shortening the effective sample period for our empirical asset-pricing tests. A closely related consideration is the choice of an expanding window, which uses all available data up to time t , thereby maximizing sample size, but also relies on older observations. Hence, an expanding window estimation may involve the same concerns as a longer rolling-window estimation. In this sense, there is a clear trade-off between using the most recent observations and having more observations to obtain more stable estimates. Clearly, these design choices each offer their own advantages and disadvantages, yet they are crucial for the final estimates. Since our results rely on only

one of these research designs, they remain limited to that specific design.

7.2.2 Alternative Variable Specifications for Investment rate and q

Our empirical results rest on the specific variable definitions used to measure the investment rate and q , which represents an additional limitation. As emphasized in the literature review, several alternative specifications of investment and q have been proposed. While we follow the most conventional investment- q variables from the investment literature, alternative operationalizations could plausibly produce different χ estimates, portfolio rankings, and thereby SMQ factor returns. For example, one could construct the investment- q relationship using variables closer to the Fama–French factors, with investment measured as asset growth and q proxied by the exact inverse of book-to-market (Fama & French, 2015; Xing, 2008). This would make the adjustment-cost estimates more directly comparable to the investment and value dimensions in the Fama–French model. Another example is to follow the full approach in Peters and Taylor (2017) by using total investment and total q , which incorporate intangible capital such as R&D, software, brand capital, and organizational capital. This may be particularly important for intangible-intensive firms, where physical capital alone may understate the relevant capital stock and distort both q and the investment rate. Thus, our results should be interpreted conditional on the chosen variable specification, and future research could examine whether the findings are robust to alternative definitions of investment and q .

7.2.3 Measurement Error in Average q

Our results rely on the assumption that the estimated χ parameters preserve sufficient relative information to distinguish between slow- and quick-adjusting firms for ranking purposes. A central concern in this regard is measurement error in average q , which enters the regression used to estimate the speed of adjustment and may generate implausibly large or unstable adjustment-cost estimates (Erickson & Whited, 2000, 2006; Hall, 2004; Peters & Taylor, 2017). This limitation is not necessarily fatal for our empirical design, since our interpretation does not require χ to be a precise structural estimate. Instead, we rely on the weaker assumption that the cross-sectional ranking contains economically relevant information. The more important concern arises if measurement error varies systematically across firms or industries. In that case, the χ ranking may reflect differences in measurement quality rather than economically meaningful variation in adjustment frictions, which represents an important limitation of our research design. Since we cannot confidently assess whether measurement errors differ systematically across firms and industries, we cannot fully separate rankings driven by measurement error from rankings driven by true differences in adjustment costs.

7.2.4 Regression Specification and Omitted Variables

Another limitation is related to the regression specification for the adjustment cost parameter. Our baseline estimates use fixed-effects regressions, in line with Peters and

Taylor (2017), but other specifications are possible. For example, one could estimate a pure investment- q relation without fixed effects, similar to Summers (1981), which is closest to theory but may suffer from omitted-variable bias. Alternatively, one could include cash flow to account for financing constraints, as emphasized by Fazzari, Hubbard and Petersen (1988), but this may involve controlling for variables that are not directly motivated by theory. Since the estimated adjustment cost parameter may differ across specifications, and since no specification can be identified as unambiguously correct, our results are naturally bounded by the research design we employ.

7.2.5 Trade-off Between Noise Reduction and Industry Exposure

An additional concern relates to our baseline industry-level estimation and the potential for aggregation to induce unintended industry exposure. This reflects a possible trade-off, as more aggregated estimation reduces noise but comes at the cost of losing firm-level variation (Belo et al., 2022). While adjustment costs are likely related to industry characteristics, industry-level estimation assigns identical values to all firms within a given industry, thereby overlooking potentially meaningful within-industry variation. For example, manufacturing firms likely exhibit higher adjustment costs than firms in less capital-intensive sectors such as software, but differences still exist across firms within each industry. By assigning a common value to all firms within each industry, portfolio sorting may primarily reflect differences across industries rather than variation within them. This can result in portfolios that resemble positions such as going long manufacturing and short software, thereby concentrating exposure at the industry level. From a portfolio perspective, such concentration implies less diversification than may be assumed. Accordingly, the SMQ factor and test portfolios could partly reflect industry-level exposure rather than purely firm-level differences in adjustment frictions, a concern that becomes more pronounced at higher levels of aggregation. This consideration motivates the robustness tests conducted using both firm-level and major-group level estimation, although the limitation remains important to acknowledge.

7.3 Practical Implementations

While the SMQ strategy provides economic insight into adjustment-cost heterogeneity and its asset pricing implications, several practical frictions limit its direct implementation. To be specific, transaction costs, short-selling constraints, small-firm liquidity concerns, and unstable mean-variance weights suggest that both the SMQ and, more importantly, the SMQ-augmented tangency portfolio would be difficult to implement in practice.

7.3.1 Transaction Costs of Long-Short Portfolios

The SMQ factor is, by construction, a long-short portfolio that requires taking both long and short positions and annual reconstitution of the underlying portfolios as firms are reclassified across adjustment-cost groups based on updated estimates of χ . This is poten-

tially costly because the strategy requires trading in both legs of the portfolio, including short-selling and ongoing position adjustments due to changes in portfolio membership. These features have important implications for the practical implementation of a potential SMQ strategy. In particular, given the long-short nature of the portfolio, portfolio reconstitution may generate transaction costs, which could weaken the abnormal return of the SMQ factor relative to the Fama-French Five-Factor model in practice. This concern is consistent with Novy-Marx and Velikov (2016), who show that the gross returns of many anomaly-based trading strategies are substantially reduced once trading costs are taken into account. Thus, while the SMQ factor appears to earn a positive and significant five-factor alpha, its practical role remains less clear. In this sense, the results should be interpreted primarily as evidence of return-relevant information related to adjustment frictions, rather than as direct evidence of an easily exploitable trading strategy.

7.3.2 Small Firms and Illiquidity

Furthermore, the empirical results for the adjustment-cost effect are strongest among small-cap stocks, which are typically associated with greater illiquidity, wider bid-ask spreads, and lower trading volumes. This raises additional implementation concerns, as the practical implementation of strategies concentrated in small firms has long concerned scholars. For example, Fama and French (2008) emphasize that anomaly results based on equal-weighted returns may place disproportionate weight on micro-cap stocks, which account for only a small share of total market capitalization and are costly to trade. This is one reason why value-weighted returns are often preferable to equal-weighted returns when assessing the economic relevance of return patterns. Similarly, Novy-Marx and Velikov (2016) show that anomaly returns are often stronger among smaller stocks, but that these stocks are also substantially more expensive to trade. In particular, they show that high-turnover strategies – defined as strategies that frequently buy and sell stocks, with each side of the portfolio being fully traded more than five times per year – are especially vulnerable to transaction costs, with net Sharpe ratios for micro-caps becoming negative for several anomalies. Since the adjustment-cost effect documented in this paper appears most concentrated among small firms, these insights raise important concerns about practical implementability. Consequently, the gross return patterns documented in this study may be substantially eroded once trading costs, market impact, and potential short-sale constraints are taken into account.

7.3.3 Short-Leg Concentration and Implementation Frictions

Building on a similar intuition, many of the positive and significant high-minus-low χ spreads documented in this paper are primarily driven by the short legs. This is important because short-leg performance may reflect not only compensation for risk, but also mispricing that is difficult to arbitrage away when short-sale constraints bind. In this context, Stambaugh, Yu, and Yuan (2012) show that the profitability of several anomaly-based long-short strategies is concentrated on the short side, especially following periods of high investor sentiment. Their interpretation is that high sentiment can lead to overpricing, while short-sale constraints prevent rational investors from fully correcting this

overpricing. As a result, subsequent anomaly profits arise largely because the overpriced short-leg stocks earn low future returns. This perspective is relevant for our results because many of the high-minus-low χ spreads appear to reflect low subsequent returns among low- χ firms, rather than unusually high returns among high- χ firms. The documented spreads may therefore partly reflect short-side mispricing and limits to arbitrage, rather than returns that are easily attainable in practice. This adds an important practical limitation to the paper: if the empirical performance of the χ -based strategy depends heavily on shorting low- χ firms, then shorting costs, limited short availability, and other implementation frictions may substantially erode the gross return patterns documented above.

7.3.4 The Implementation Ability of a Tangency Portfolio

The last practical limitation worth highlighting concerns the implementation of the efficient frontier and tangency portfolio. Our SMQ factor consistently shows weak unconditional mean excess returns, but despite this lack of unconditional performance, it loads negatively on HML and RMW. This is exactly where the practical relevance of SMQ lies: it serves as a hedge against value- and profitability-related risks and thereby expands the efficient frontier. However, this value depends on whether investors can translate the estimated frontier into an implementable trading strategy, which is not straightforward and gives rise to important practical limitations. In practice, past information does not necessarily reflect the future, and one would need to use these tangency portfolio weights in a real-life trading strategy and evaluate their performance through backtesting. This creates an important limitation because the mean-variance framework has been shown to be unstable and difficult to implement. First, changes in optimal portfolio weights have been shown to often be unstable and quite extreme in practice (Michaud, 1989). Second, the tangency portfolio has often been shown to underperform both the market portfolio and the naive 1/N portfolio in out-of-sample backtesting exercises (DeMiguel et al., 2009). Since the contribution of our research lies in the fact that SMQ expands the efficient frontier rather than earning a strong unconditional return, these concerns are especially important. If investors cannot implement the efficient frontier and allocate weights across factors effectively, the practical implications of SMQ remain limited.

7.4 Generalizability

To enhance comparability with Fama and French (2015), we fully replicate the five factors to ensure consistency with the SMQ factor. However, this design choice also limits the generalizability. Specifically, our analysis relies solely on U.S. data between 1982 and 2024, and the findings may therefore not extend to other countries or time periods.

First, our results are specific to the sample period used for the empirical asset-pricing tests, which spans from 1982 to 2024. This period includes several important macroeconomic events, including the post-1980s disinflation, the long decline in interest rates, the dot-com crash, the global financial crisis, the post-2008 low-rate environment, and the post-COVID inflation shock. This matters because abnormal returns are often sample-

period specific. For example, McLean and Pontiff (2016) show that many return anomalies weaken when tested outside their original sample periods and often decline after publication. Hence, the strength of the SMQ factor documented here may reflect the specific historical period examined, rather than a stable relation that necessarily generalizes across time.

Second, the analysis is based exclusively on U.S. firms, which limits the extent to which the results can be generalized internationally. Although the five-factor model has been shown to perform reasonably well in international settings, its implications differ across regions. To be specific, Fama and French (2017) find that profitability and investment effects are more evident in North America, Europe, and Asia Pacific, but weaker in Japan, where the value effect appears more pronounced. This supports the view that U.S.-based results should not automatically be interpreted as globally generalizable. The SMQ factor is unlikely to be an exception in this respect and may therefore differ across regions. As a result, the relationship between adjustment costs and expected returns documented here may not generalize directly to non-U.S. markets.

Third, our sample is restricted to firms listed on the NYSE, AMEX, and NASDAQ. While these U.S. exchanges cover a large and economically important share of publicly traded firms, the exclusion of firms listed on smaller or less regulated markets may limit the applicability of the results to the broader universe of firms. This is particularly relevant given that the SMQ strategy is most pronounced among small firms. Thus, the results should be interpreted as evidence from a specific segment of the public U.S. equity market, rather than as evidence that necessarily extends to all U.S. firms.

Together, these considerations imply that while the empirical design ensures comparability, the results should be interpreted with caution when extending them beyond the U.S. setting, the selected exchanges, and the specific time period studied.

8 Future Research

While this paper contributes to the literature on production-based asset pricing by linking adjustment-cost heterogeneity to cross-sectional returns, it also leaves several academic questions open. This section outlines key avenues for future research, including stronger theoretical foundations, time-varying risk premia, alternative estimation choices, and the role of intangible capital.

8.1 Theoretical Foundations

The theoretical framework employed in this paper draws on the traditional graduate textbook treatments of q-theory of investment and consumption-based asset pricing separately, relying on an intuitive rather than formally derived link between the two. While this serves as a productive starting point, it also represents an inherent limitation. Future research could prioritize the development of a true general-equilibrium model that formally connects the q-theory of investment with the stochastic discount factor, clearly

spelling out the asset-pricing implications of adjustment costs within a unified framework. Although the production-based asset-pricing literature has taken meaningful steps in this direction, to our knowledge, no study has yet provided such a framework. Establishing this theoretical bridge would substantially strengthen the interpretability of empirical findings such as those presented here, and would allow researchers to derive sharper predictions about how adjustment costs should manifest in expected returns across the size, value, profitability, and investment dimensions.

8.2 Alternative Estimation Windows

A natural direction for future research concerns the choice of estimation window used to obtain the adjustment-cost parameter, which involves a fundamental trade-off between capturing time-varying shifts in adjustment costs and maintaining statistical precision. While the baseline specification adopts a 20-year rolling window to balance these considerations, alternative window lengths may yield different χ estimates and portfolio implications. Shorter windows may better reflect current economic conditions but produce less stable estimates, whereas longer windows improve precision but rely on older data that may be less representative of current firm behavior. Future research could therefore systematically assess the sensitivity of the χ estimates and the resulting factor returns to alternative rolling-window lengths, or alternatively to an expanding-window design. By providing evidence from alternative research designs, future work could further strengthen the robustness of the adjustment-cost effect.

8.3 The Role of Intangible Capital

A further avenue for future research concerns the role of intangible capital in production-based asset pricing. Our framework focuses exclusively on physical capital, yet a growing literature suggests that intangible capital constitutes an increasing share of firm value. For example, Peters and Taylor (2017) extend q-theory of investment by introducing total q and total investment, both of which incorporate intangible capital. Similarly, Belo, Gala, Salomao and Vitorino (2022) develop a production-based asset-pricing model that decomposes adjustment costs across different types of capital, including intangible capital, which they find to be an important input for firm value. Since both studies highlight the relevance of intangible capital within a q-theoretic setting, a natural extension would be to examine whether incorporating intangible capital strengthens or refines the adjustment-cost effect documented here. Future research could therefore explore whether intangible capital alters the estimated adjustment-cost parameters and helps explain the cross-sectional return patterns documented in this paper.

8.4 Extensions to Other Empirical Settings

A natural avenue for future research is to examine whether the adjustment-cost patterns documented in this paper extend beyond the present empirical setting. Since the analysis

is restricted to U.S. listed equities from July 1982 to December 2024, future work could first test whether adjustment-cost heterogeneity is also related to expected returns in international equity markets. This would be particularly relevant because the empirical performance of the Fama–French factors differs across regions (Fama & French, 2017), suggesting that the role of χ may also vary across markets. A second avenue is to examine whether the results differ across macro-financial environments. The sample period contains several distinct episodes, including the Great Moderation, the global financial crisis, the post-2008 low-interest-rate period, the COVID-19 shock, and the subsequent inflation and monetary tightening cycle (Wheelock, 2021). Future research could therefore test whether adjustment-cost effects are stronger during periods of high uncertainty, tight financing conditions, or large aggregate shocks, where firms’ ability to adjust capital may be especially important. By expanding the empirical scope of the analysis, future research may provide deeper insight into whether adjustment costs capture a setting-specific return pattern or a more general economic mechanism shaping firms’ adjustment dynamics and expected returns.

9 Conclusion

This paper is motivated by the idea that the Fama–French five-factor model may reflect the same underlying economic mechanisms as the q-theory of investment. Building on this perspective, the paper examined the extent to which adjustment frictions are related to cross-sectional differences in expected stock returns and whether this dimension provides additional insight into the economic interpretation of the five-factor model.

The empirical evidence supports the view that the five-factor model may reflect economic content related to the q-theory of investment. More specifically, while the results are not as strong as initially anticipated, they point to a consistent and economically coherent pattern. The adjustment-cost factor does not generate a clear standalone excess return, but it does earn a positive risk premium relative to the five-factor model, suggesting that adjustment frictions contain pricing information not fully captured by the five-factor model. The test portfolio evidence provides further insight into where this information is located: the adjustment-cost effect is most persistent among small firms, where higher adjustment costs are associated with more growth-like factor exposure and weaker profitability exposure. In contrast, the evidence for augmenting the five-factor model with SMQ is less convincing. Adding the adjustment-cost factor improves model performance only modestly, suggesting that SMQ captures a relevant economic dimension, but not enough to constitute a clear improvement over the existing model.

In our view, the persistent small-firm evidence has an intuitive q-theoretical interpretation. Small firms are more likely to be earlier in their saddle-path adjustment process and further from their optimal capital stock. Because these firms have less installed capital, weaker current production capacity, and fewer internal resources, their ability to convert investment opportunities into realized profitability depends more critically on the *speed* at which they can adjust. For large firms, adjustment costs may be less central

because they already have substantial installed capital, established production capacity, and more stable operating profitability. Hence, although the evidence presented in this paper is less pronounced than expected, the empirical pattern remains coherent and economically meaningful from the perspective of q-theory.

As with any empirical study, the results should be interpreted in light of several limitations, which also point to future avenues for research. First, the theoretical argument does not derive a unified general-equilibrium model, but builds on separate q-theoretical and asset-pricing mechanisms. Future research could therefore link adjustment frictions more formally to the stochastic discount factor and thereby expected returns. Second, the estimated adjustment-cost parameter depends on assumptions about the functional form, the investment–q relation, average q, estimation windows, estimation approach, and variable definitions. Testing alternative specifications could strengthen robustness. Finally, the adjustment-cost factor is a long–short portfolio requiring monthly rebalancing, with the strongest effects among small firms, where transaction costs, liquidity constraints, and short-selling frictions are most severe. The results should therefore be interpreted primarily as evidence of an economic mechanism underlying the five-factor model, rather than as evidence of an easily implementable investment strategy.

In conclusion, the main contribution of this paper is not that the adjustment-cost factor emerges as a dominant new factor, but that adjustment-cost heterogeneity may help clarify some of the economic mechanisms underlying the Fama–French Five-Factor Model. In particular, adjustment frictions provide insight into how firms translate valuation signals into future operating profitability. Based on these findings, we believe that the economic mechanisms embedded in q-theory of investment may provide a useful perspective for further developing the theoretical interpretation of the Fama–French Five-Factor Model.

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A Tables

A.1 Sample Characteristics and Portfolio Counts

Table A.1: Summary statistics for firm characteristics

This table reports summary statistics for the firm characteristics used in the empirical analysis. Panel A reports statistics for the full sample period from July 1962 to December 2024 and excludes the adjustment-cost measures, since these are only used in the asset-pricing analysis. Panel B reports statistics for the asset-pricing sample from July 1982 to December 2024 and includes the three alternative adjustment-cost measures: firm-level χ , industry χ , and major-group χ . All variables are measured at the firm-month level. All reported statistics are rounded to two digits.

Variable	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
<i>Panel A: Full sample, July 1962–December 2024</i>							
B/M	0.97	2.28	0.00	0.32	0.62	1.10	576.77
OP	0.12	7.83	-1,144.83	0.07	0.21	0.33	1,676.00
Inv	2.41	369.46	-1.00	-0.01	0.08	0.22	150,262.00
I/K	0.34	6.69	-0.96	0.06	0.12	0.23	1,686.00
q	5.42	23.22	-229.25	0.43	1.11	3.65	1,974.35
<i>Panel B: Asset-pricing sample, July 1982–December 2024</i>							
B/M	0.90	2.46	0.00	0.29	0.56	0.99	576.77
OP	0.09	8.84	-1,144.83	0.04	0.19	0.31	1,676.00
Inv	3.01	415.84	-1.00	-0.02	0.08	0.23	150,262.00
I/K	0.36	6.74	-0.96	0.06	0.11	0.23	1,686.00
q	6.57	25.90	-229.25	0.56	1.45	4.74	1,974.35
Firm-level χ	-1.52×10^{16}	4.14×10^{18}	-1.10×10^{21}	4.04	21.06	64.88	1.78×10^{19}
Industry χ	-3.61×10^{10}	6.65×10^{12}	-1.22×10^{15}	24.44	50.71	95.85	661,739.20
Major-group χ	67.54	624.28	-107,117.80	39.79	63.17	93.63	9,619.95

Table A.2: Average number of firms in 25 test portfolios

US equities, July 1982–December 2024. This table reports the average number of unique firms per portfolio cell for the four sets of 25 test portfolios formed using NYSE breakpoints: Size–B/M, Size–OP, Size–Inv, and Size– χ . For each month, the number of unique firms is counted within each portfolio cell, and the reported value is the time-series average across months. Entries are rounded to the nearest whole number. Each block reports a 5×5 sort, where rows denote size portfolios and columns denote characteristic portfolios.

	Size–B/M					Size–OP				
	Low	2	3	4	High	Low	2	3	4	High
Small	417	292	331	425	696	1102	336	302	210	185
2	165	129	134	114	79	163	122	130	110	91
3	126	99	88	70	45	80	83	92	92	80
4	113	81	64	53	34	51	60	71	82	80
Big	116	69	51	42	25	31	48	55	77	92

	Size–Inv					Size– χ				
	Low	2	3	4	High	Low	2	3	4	High
Small	735	330	300	327	531	282	323	313	459	384
2	106	99	106	126	197	86	86	91	124	105
3	59	70	82	96	129	57	58	60	84	65
4	46	63	69	81	93	47	43	49	63	57
Big	35	61	71	72	70	36	40	41	58	51

Table A.3: Average number of firms in 32 test portfolios

US equities, July 1982–December 2024. This table reports the time-series average number of unique firms per portfolio cell for six sets of $2 \times 4 \times 4$ test portfolios formed using NYSE breakpoints. Firm counts are computed monthly within each cell and rounded to the nearest whole number. In each panel, the left block reports small firms and the right block reports big firms. Rows denote quartiles of the first non-size characteristic, while columns denote quartiles of the second, both ordered from low (Q1) to high (Q4). Panels A–F sort on Size–B/M–OP, Size–B/M–Inv, Size–OP–Inv, Size–OP– χ , Size–B/M– χ , and Size–Inv– χ , respectively.

	Small				Big			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
<i>Panel A: Size–B/M–OP</i>								
Low B/M	369	121	175	316	42	34	62	123
Q2	176	138	193	134	29	53	77	47
Q3	219	187	153	69	49	66	51	23
High B/M	402	141	73	42	102	51	19	8
<i>Panel B: Size–B/M–Inv</i>								
Low B/M	173	118	188	534	31	35	61	138
Q2	129	138	192	190	39	48	58	61
Q3	162	172	176	125	50	52	49	39
High B/M	323	164	112	74	69	54	38	20
<i>Panel C: Size–OP–Inv</i>								
Low OP	523	210	160	302	63	49	45	68
Q2	91	148	170	181	44	51	53	58
Q3	66	131	191	210	37	48	56	68
High OP	82	103	146	245	44	41	52	67
<i>Panel D: Size–OP–χ</i>								
Low OP	157	196	277	375	29	29	46	35
Q2	90	100	118	114	32	32	43	34
Q3	94	100	107	110	37	40	48	41
High OP	114	111	107	115	40	41	45	43
<i>Panel E: Size–B/M–χ</i>								
Low B/M	153	185	244	313	41	48	83	68
Q2	97	108	130	146	36	40	49	43
Q3	93	99	114	124	34	32	31	25
High B/M	119	123	132	143	29	22	18	17
<i>Panel F: Size–Inv–χ</i>								
Low Inv	125	152	193	244	36	35	40	38
Q2	101	102	119	128	34	30	35	32
Q3	105	106	129	136	35	31	42	37
High Inv	145	168	194	245	39	48	68	51

A.2 Supplementary Factor Evidence

Table A.4: Five-factor spanning regressions, excl. the adjustment cost factor

US equities, July 1982–December 2024. This table reports time-series regressions in which each Fama–French five-factor model factor is regressed on the remaining FF5 factors. The factor set consists of the market excess return ($R_M - R_f$), SMB (small minus big), HML (high minus low), RMW (robust minus weak) and CMA (conservative minus aggressive). At the end of each June, firms are sorted based on the lagged characteristic using NYSE breakpoints. The intercept is reported as α , slope coefficients are reported as factor loadings, and the final column reports the adjusted R^2 . All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	α	$R_M - R_f$	SMB	HML	RMW	CMA	$\bar{R}^2$
$R_M - R_f$	1.09*** (6.60)		0.22* (1.80)	0.09 (0.48)	-0.42** (-2.17)	-0.82*** (-3.14)	0.16
SMB	0.20 (1.31)	0.09** (2.12)		0.15 (1.33)	-0.57*** (-4.01)	-0.10 (-0.70)	0.21
HML	-0.23* (-1.71)	0.02 (0.47)	0.09 (1.29)		0.36*** (3.10)	1.18*** (16.68)	0.56
RMW	0.51*** (4.66)	-0.09** (-2.00)	-0.29*** (-4.24)	0.31*** (2.73)		-0.47*** (-3.48)	0.30
CMA	0.27*** (4.24)	-0.08*** (-3.35)	-0.02 (-0.67)	0.44*** (13.12)	-0.21*** (-3.45)		0.57

A.3 Alpha Comparisons Across Asset-Pricing Models

Table A.5: Alphas for Size-B/M portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports intercepts (α) in monthly percent from time-series regressions for the 25 Size-B/M portfolios formed using NYSE breakpoints. Panel A reports CAPM alphas, Panel B reports FF5 alphas, and Panel C reports FF5 + SMQ alphas based on the SIC 4-digit industry-level estimates of χ . In each panel, the final column reports the high-minus-low ($H - L$) alpha spread across B/M portfolios within each size quintile, while the bottom row reports the small-minus-big ($S - B$) alpha spread within each B/M quintile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	B/M					
Size	Low	2	3	4	High	H-L
Panel A: CAPM						
Small	-0.81*** (-3.66)	-0.02 (-0.12)	0.02 (0.11)	0.26 (1.27)	0.31 (1.14)	1.12*** (3.90)
2	-0.38*** (-2.64)	0.04 (0.25)	0.20 (1.19)	0.32* (1.90)	-0.06 (-0.24)	0.33 (1.18)
3	-0.28** (-2.31)	0.11 (0.82)	0.12 (0.76)	0.19 (1.12)	0.11 (0.52)	0.39 (1.55)
4	-0.06 (-0.49)	0.07 (0.57)	0.05 (0.28)	0.16 (0.91)	0.15 (0.73)	0.21 (0.80)
Big	0.07 (0.81)	0.11 (1.55)	0.15 (1.36)	0.09 (0.52)	0.11 (0.55)	0.04 (0.15)
S-B	-0.88*** (-3.63)	-0.13 (-0.67)	-0.13 (-0.75)	0.17 (0.93)	0.20 (0.77)	
Panel B: FF5						
Small	-0.23** (-2.15)	0.21*** (2.67)	0.03 (0.37)	0.26*** (2.82)	0.21* (1.78)	0.44*** (3.09)
2	-0.04 (-0.62)	0.04 (0.69)	0.04 (0.54)	0.14** (2.42)	-0.24*** (-3.04)	-0.20** (-2.00)
3	0.02 (0.29)	0.03 (0.47)	-0.07 (-0.95)	-0.05 (-0.60)	-0.13 (-1.53)	-0.15 (-1.46)
4	0.19** (2.14)	-0.07 (-1.06)	-0.18** (-2.26)	-0.04 (-0.49)	-0.08 (-0.95)	-0.27** (-2.38)
Big	0.07 (1.45)	-0.04 (-0.70)	-0.04 (-0.64)	-0.10 (-1.35)	-0.04 (-0.50)	-0.12 (-1.16)
S-B	-0.30*** (-2.79)	0.25** (2.48)	0.07 (0.77)	0.36*** (3.37)	0.26* (1.74)	
Panel C: FF5 + SMQ						
Small	-0.25** (-2.31)	0.17** (2.39)	0.00 (0.00)	0.24** (2.51)	0.21* (1.73)	0.46*** (3.54)
2	-0.05 (-0.73)	0.04 (0.65)	0.07 (0.90)	0.15** (2.56)	-0.24*** (-2.92)	-0.20** (-2.04)
3	0.02 (0.38)	0.03 (0.42)	-0.04 (-0.63)	-0.05 (-0.57)	-0.11 (-1.11)	-0.13 (-1.15)
4	0.19** (2.18)	-0.07 (-0.96)	-0.16** (-2.00)	-0.03 (-0.45)	-0.08 (-0.98)	-0.27** (-2.51)
Big	0.05 (0.86)	-0.05 (-0.80)	-0.02 (-0.31)	-0.10 (-1.30)	-0.10 (-1.09)	-0.15 (-1.38)
S-B	-0.30*** (-2.65)	0.22** (2.22)	0.02 (0.24)	0.34*** (2.73)	0.31** (2.04)	

Table A.6: Alphas for Size–OP portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports intercepts (α) in monthly percent from time-series regressions for the 25 Size–OP portfolios formed using NYSE breakpoints. Panel A reports CAPM alphas, Panel B reports FF5 alphas, and Panel C reports FF5 + SMQ alphas based on the SIC 4-digit industry-level estimates of χ . In each panel, the final column reports the high-minus-low ($H - L$) alpha spread across OP portfolios within each size quintile, while the bottom row reports the small-minus-big ($S - B$) alpha spread within each OP quintile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	OP					
Size	Low	2	3	4	High	H-L
Panel A: CAPM						
Small	-0.44** (-2.03)	0.17 (0.91)	0.39 (1.55)	0.16 (0.74)	0.10 (0.47)	0.54*** (2.78)
2	-0.39** (-2.55)	-0.07 (-0.45)	0.14 (0.88)	0.20 (1.16)	0.15 (0.75)	0.54*** (2.68)
3	-0.43*** (-3.01)	0.00 (0.02)	0.04 (0.29)	0.11 (0.77)	0.21 (1.23)	0.64*** (2.69)
4	-0.36*** (-2.93)	0.05 (0.42)	0.17 (1.33)	0.11 (0.78)	0.15 (1.16)	0.51*** (2.73)
Big	-0.38*** (-2.75)	-0.04 (-0.71)	-0.08 (-1.12)	0.07 (1.32)	0.21*** (3.01)	0.58*** (3.35)
S-B	-0.07 (-0.24)	0.21 (1.08)	0.47* (1.86)	0.09 (0.40)	-0.10 (-0.42)	
Panel B: FF5						
Small	-0.05 (-0.50)	0.11 (1.34)	0.25* (1.96)	-0.02 (-0.24)	-0.03 (-0.41)	0.01 (0.11)
2	-0.03 (-0.40)	-0.19** (-2.23)	-0.00 (-0.03)	0.01 (0.22)	-0.11 (-1.45)	-0.08 (-0.83)
3	0.01 (0.09)	-0.05 (-0.61)	-0.11* (-1.69)	-0.06 (-0.76)	-0.05 (-0.60)	-0.06 (-0.61)
4	0.02 (0.19)	0.07 (0.83)	0.11 (1.44)	-0.08 (-1.07)	-0.01 (-0.11)	-0.02 (-0.21)
Big	0.03 (0.34)	0.01 (0.25)	-0.11* (-1.73)	0.02 (0.39)	0.05 (0.92)	0.01 (0.11)
S-B	-0.08 (-0.52)	0.09 (0.99)	0.36*** (2.83)	-0.04 (-0.43)	-0.08 (-0.76)	
Panel C: FF5 + SMQ						
Small	-0.09 (-1.05)	0.13 (1.50)	0.25** (1.99)	-0.01 (-0.15)	-0.02 (-0.25)	0.07 (0.69)
2	-0.04 (-0.59)	-0.17** (-2.14)	0.02 (0.40)	0.05 (0.87)	-0.08 (-1.05)	-0.04 (-0.33)
3	-0.00 (-0.02)	-0.04 (-0.57)	-0.09 (-1.37)	-0.03 (-0.42)	-0.03 (-0.41)	-0.03 (-0.33)
4	0.02 (0.21)	0.07 (0.81)	0.10 (1.38)	-0.07 (-1.05)	0.00 (0.00)	-0.02 (-0.17)
Big	0.08 (0.87)	0.03 (0.50)	-0.11* (-1.67)	0.03 (0.56)	0.01 (0.26)	-0.07 (-0.56)
S-B	-0.18 (-1.26)	0.10 (1.04)	0.36*** (3.06)	-0.05 (-0.46)	-0.03 (-0.32)	

Table A.7: Alphas for Size–Inv portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports intercepts (α) in monthly percent from time-series regressions for the 25 Size–Inv portfolios formed using NYSE breakpoints. Panel A reports CAPM alphas, Panel B reports FF5 alphas, and Panel C reports FF5 + SMQ alphas based on the SIC 4-digit industry-level estimates of χ . In each panel, the final column reports the low-minus-high ($L - H$) alpha spread across Inv portfolios within each size quintile, while the bottom row reports the small-minus-big ($S - B$) alpha spread within each Inv quintile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	Inv					
Size	Low	2	3	4	High	L-H
Panel A: CAPM						
Small	-0.01 (-0.05)	0.15 (0.86)	0.23 (1.22)	0.11 (0.57)	-0.51*** (-2.89)	0.50*** (3.18)
2	-0.04 (-0.26)	0.12 (0.78)	0.21 (1.28)	0.16 (1.04)	-0.36** (-2.51)	0.32** (2.27)
3	0.06 (0.39)	0.17 (1.28)	0.16 (1.20)	0.05 (0.39)	-0.27** (-2.26)	0.33** (2.49)
4	-0.04 (-0.26)	0.15 (1.20)	0.14 (1.29)	0.16 (1.18)	-0.14 (-1.16)	0.10 (0.61)
Big	0.32** (2.46)	0.17** (2.05)	0.13* (1.73)	0.09 (1.11)	-0.11 (-0.99)	0.42** (2.09)
S-B	-0.33 (-1.15)	-0.02 (-0.09)	0.10 (0.56)	0.02 (0.09)	-0.41* (-1.67)	
Panel B: FF5						
Small	0.24* (1.80)	0.12* (1.82)	0.15** (2.00)	0.14* (1.74)	-0.24** (-2.48)	0.48*** (3.16)
2	-0.04 (-0.51)	-0.07 (-0.97)	0.08 (1.30)	0.10 (1.49)	-0.17** (-2.42)	0.13 (1.42)
3	0.00 (0.03)	-0.00 (-0.05)	0.00 (0.01)	-0.03 (-0.42)	-0.06 (-0.86)	0.06 (0.70)
4	-0.22*** (-2.60)	-0.07 (-0.96)	-0.03 (-0.48)	0.09 (1.18)	0.17* (1.78)	-0.40*** (-2.97)
Big	0.02 (0.20)	-0.09* (-1.71)	-0.02 (-0.39)	0.03 (0.33)	0.15** (2.29)	-0.13 (-1.23)
S-B	0.22 (1.21)	0.20** (2.41)	0.17* (1.75)	0.11 (1.23)	-0.40*** (-3.40)	
Panel C: FF5 + SMQ						
Small	0.21 (1.60)	0.09 (1.47)	0.12 (1.60)	0.14* (1.75)	-0.25** (-2.42)	0.46*** (3.20)
2	-0.02 (-0.25)	-0.06 (-0.89)	0.06 (1.04)	0.10 (1.59)	-0.15** (-2.22)	0.13 (1.41)
3	0.03 (0.35)	-0.01 (-0.14)	-0.00 (-0.05)	-0.01 (-0.13)	-0.06 (-0.88)	0.09 (0.97)
4	-0.18** (-2.20)	-0.06 (-0.82)	-0.02 (-0.38)	0.09 (1.19)	0.16* (1.79)	-0.34*** (-2.85)
Big	0.00 (0.05)	-0.07 (-1.45)	-0.03 (-0.56)	-0.03 (-0.37)	0.15** (2.33)	-0.15 (-1.32)
S-B	0.20 (1.13)	0.16** (2.04)	0.16 (1.48)	0.17* (1.72)	-0.40*** (-3.35)	

Table A.8: Alphas for Size- χ portfolios across asset pricing models

US equities: monthly returns, July 1982–December 2024. This table reports intercepts (α , in monthly percent) from time-series regressions for the 25 Size- χ portfolios formed using NYSE breakpoints. Panel A reports CAPM alphas, Panel B reports FF5 alphas, and Panel C reports FF5 alphas augmented with the SIC value-weighted slow-minus-quick factor (FF5 + SMQ). In each panel, the final column reports the high-minus-low ($H - L$) alpha spread across χ portfolios within each size quintile, while the bottom row reports the small-minus-big ($S - B$) alpha spread within each χ quintile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	χ					
Size	Low	2	3	4	High	H-L
<i>Panel A: CAPM</i>						
Small	0.02 (0.06)	-0.28 (-1.08)	-0.18 (-0.85)	-0.04 (-0.21)	-0.07 (-0.33)	-0.09 (-0.35)
2	-0.05 (-0.28)	-0.22 (-1.02)	-0.21 (-1.30)	-0.01 (-0.05)	-0.12 (-0.88)	-0.07 (-0.49)
3	-0.08 (-0.50)	-0.11 (-0.59)	-0.09 (-0.62)	-0.18 (-1.46)	0.15 (1.07)	0.23* (1.70)
4	-0.00 (-0.03)	-0.22 (-1.20)	-0.01 (-0.11)	0.06 (0.47)	0.10 (0.86)	0.10 (0.91)
Big	0.11 (1.08)	-0.14 (-1.02)	0.08 (0.90)	0.15 (1.28)	0.05 (0.55)	-0.06 (-0.57)
S-B	-0.10 (-0.38)	-0.14 (-0.48)	-0.26 (-0.99)	-0.20 (-0.76)	-0.12 (-0.52)	
<i>Panel B: FF5</i>						
Small	-0.13 (-0.74)	-0.26* (-1.67)	-0.10 (-1.02)	0.32*** (2.79)	0.24** (2.08)	0.37* (1.67)
2	-0.29** (-2.54)	-0.33** (-2.03)	-0.23*** (-2.65)	0.29*** (3.63)	-0.10 (-1.04)	0.19 (1.49)
3	-0.29*** (-2.65)	-0.16 (-1.05)	-0.16 (-1.43)	0.04 (0.58)	0.17* (1.88)	0.46*** (4.12)
4	-0.17* (-1.91)	-0.22 (-1.25)	-0.12 (-1.36)	0.30*** (3.57)	0.02 (0.20)	0.19* (1.77)
Big	-0.13 (-1.33)	-0.04 (-0.27)	0.05 (0.75)	0.24*** (2.73)	-0.18** (-2.12)	-0.05 (-0.45)
S-B	-0.00 (-0.01)	-0.23 (-1.51)	-0.15 (-1.30)	0.09 (0.74)	0.42*** (3.07)	
<i>Panel C: FF5 + SMQ</i>						
Small	0.01 (0.06)	-0.08 (-0.54)	-0.09 (-0.85)	0.28** (2.46)	0.09 (0.98)	0.08 (0.44)
2	-0.15 (-1.59)	-0.13 (-1.00)	-0.20** (-2.34)	0.25*** (3.25)	-0.19** (-2.24)	-0.04 (-0.44)
3	-0.19* (-1.91)	0.05 (0.37)	-0.14 (-1.25)	-0.01 (-0.19)	0.07 (0.70)	0.26*** (2.77)
4	-0.08 (-0.91)	-0.02 (-0.13)	-0.12 (-1.36)	0.25*** (3.15)	-0.03 (-0.32)	0.05 (0.53)
Big	0.02 (0.22)	0.16 (1.18)	0.07 (1.01)	0.17* (1.91)	-0.33*** (-3.18)	-0.34*** (-2.95)
S-B	-0.01 (-0.04)	-0.24 (-1.52)	-0.16 (-1.32)	0.10 (0.91)	0.42*** (2.72)	

Table A.9: Alphas for Size-B/M-OP portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size-B/M-OP portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the high-minus-low OP alpha within each B/M quartile, while the bottom row reports the high-minus-low B/M alpha within each OP quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

OP →	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: CAPM alpha</i>										
Low B/M	-0.92*** (-3.93)	-0.29* (-1.72)	-0.09 (-0.59)	0.05 (0.30)	0.97*** (3.60)	-0.40* (-1.69)	-0.03 (-0.15)	0.05 (0.47)	0.12 (1.51)	0.52** (2.12)
2	-0.40** (-2.01)	0.11 (0.61)	0.23 (1.35)	0.30 (1.29)	0.70*** (2.90)	-0.24* (-1.72)	-0.01 (-0.09)	0.12 (1.22)	0.29*** (2.73)	0.53*** (2.79)
3	-0.11 (-0.57)	0.21 (1.11)	0.45** (2.13)	0.36 (1.48)	0.47** (2.34)	0.01 (0.11)	0.12 (0.81)	0.18 (1.15)	0.25 (1.41)	0.24 (1.35)
High B/M	-0.08 (-0.33)	0.22 (0.91)	0.56* (1.67)	0.24 (0.59)	0.32 (1.06)	0.03 (0.20)	0.12 (0.61)	0.32 (1.57)	-0.06 (-0.21)	-0.10 (-0.36)
H-L	0.84*** (2.59)	0.51* (1.94)	0.65** (2.24)	0.19 (0.59)		0.44 (1.18)	0.15 (0.46)	0.27 (0.98)	-0.18 (-0.58)	
<i>Panel B: FF5 alpha</i>										
Low B/M	-0.19* (-1.87)	-0.09 (-0.79)	-0.08 (-1.14)	-0.12** (-2.06)	0.07 (0.60)	0.35** (1.97)	0.25* (1.66)	0.14 (1.63)	-0.01 (-0.18)	-0.36* (-1.96)
2	-0.09 (-1.04)	-0.06 (-0.65)	-0.02 (-0.25)	-0.04 (-0.57)	0.05 (0.41)	-0.05 (-0.41)	-0.14 (-1.64)	-0.09 (-1.31)	0.01 (0.09)	0.06 (0.37)
3	-0.04 (-0.58)	0.02 (0.26)	0.19** (2.46)	0.01 (0.10)	0.05 (0.38)	-0.01 (-0.11)	-0.14 (-1.29)	-0.10 (-1.07)	-0.04 (-0.28)	-0.03 (-0.19)
High B/M	-0.10 (-0.86)	-0.08 (-0.84)	0.24 (1.28)	-0.09 (-0.43)	0.01 (0.03)	-0.08 (-1.11)	-0.15 (-1.40)	0.07 (0.54)	-0.30 (-1.17)	-0.22 (-0.77)
H-L	0.09 (0.64)	0.01 (0.07)	0.32 (1.63)	0.03 (0.12)		-0.43** (-2.08)	-0.40** (-2.14)	-0.07 (-0.51)	-0.29 (-1.09)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low B/M	-0.27*** (-2.73)	-0.10 (-0.88)	-0.06 (-0.81)	-0.09 (-1.63)	0.18 (1.58)	0.35** (2.00)	0.26 (1.63)	0.15* (1.69)	-0.06 (-0.86)	-0.42** (-2.24)
2	-0.14 (-1.45)	-0.04 (-0.44)	-0.01 (-0.08)	-0.01 (-0.12)	0.13 (0.99)	-0.01 (-0.04)	-0.16* (-1.89)	-0.13* (-1.67)	0.04 (0.38)	0.05 (0.29)
3	-0.04 (-0.55)	-0.00 (-0.06)	0.21*** (2.80)	0.04 (0.48)	0.09 (0.63)	0.02 (0.18)	-0.15 (-1.50)	-0.04 (-0.45)	-0.01 (-0.04)	-0.02 (-0.12)
High B/M	-0.08 (-0.69)	-0.06 (-0.62)	0.20 (1.12)	-0.09 (-0.41)	-0.01 (-0.02)	-0.07 (-0.93)	-0.19* (-1.84)	0.04 (0.33)	-0.28 (-1.09)	-0.21 (-0.72)
H-L	0.18 (1.25)	0.04 (0.25)	0.26 (1.35)	-0.00 (-0.01)		-0.43** (-2.05)	-0.45** (-2.26)	-0.10 (-0.74)	-0.22 (-0.82)	

Table A.10: Alphas for Size–B/M–Inv portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size–B/M–Inv portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the low-minus-high Inv alpha within each B/M quartile, while the bottom row reports the high-minus-low B/M alpha within each Inv quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Inv →	Small					Big				
	1	2	3	4	L-H	1	2	3	4	L-H
<i>Panel A: CAPM alpha</i>										
Low B/M	-0.30 (-1.46)	0.11 (0.71)	0.07 (0.47)	-0.46*** (-3.32)	0.16 (1.19)	0.25** (2.08)	0.15 (1.48)	0.17* (1.85)	-0.10 (-0.68)	0.35* (1.77)
2	-0.03 (-0.17)	0.13 (0.81)	0.22 (1.24)	0.11 (0.58)	-0.14 (-0.86)	0.23** (2.23)	0.14* (1.74)	0.06 (0.66)	-0.09 (-0.74)	0.32** (2.17)
3	0.17 (0.82)	0.25 (1.45)	0.38* (1.92)	0.10 (0.43)	0.08 (0.39)	0.15 (1.17)	0.20 (1.59)	0.10 (0.73)	-0.00 (-0.02)	0.16 (1.26)
High B/M	0.11 (0.37)	0.20 (0.83)	-0.00 (-0.01)	0.22 (0.72)	-0.11 (-0.56)	0.18 (0.92)	0.02 (0.08)	0.14 (0.79)	0.14 (0.71)	0.04 (0.24)
H-L	0.41 (1.26)	0.09 (0.42)	-0.07 (-0.32)	0.68** (2.39)		-0.07 (-0.33)	-0.13 (-0.68)	-0.02 (-0.11)	0.24 (0.79)	
<i>Panel B: FF5 alpha</i>										
Low B/M	0.04 (0.34)	0.15* (1.86)	0.09* (1.73)	-0.19*** (-3.46)	0.23* (1.76)	-0.01 (-0.05)	-0.11 (-1.30)	0.05 (0.78)	0.27*** (3.22)	-0.28** (-2.14)
2	-0.02 (-0.21)	-0.07 (-0.98)	0.05 (0.71)	-0.03 (-0.34)	0.01 (0.09)	-0.04 (-0.39)	-0.02 (-0.21)	-0.10 (-1.19)	-0.18 (-1.63)	0.14 (1.00)
3	0.14 (1.30)	0.02 (0.28)	0.17*** (2.61)	-0.03 (-0.24)	0.16 (0.93)	-0.22** (-2.46)	-0.01 (-0.07)	-0.09 (-1.02)	0.01 (0.09)	-0.23* (-1.81)
High B/M	-0.07 (-0.50)	-0.01 (-0.09)	-0.18 (-1.52)	0.25 (1.18)	-0.32 (-1.59)	-0.07 (-0.76)	-0.18** (-2.07)	-0.01 (-0.08)	0.10 (0.65)	-0.17 (-0.97)
H-L	-0.11 (-0.58)	-0.16 (-1.08)	-0.27** (-1.99)	0.44* (1.92)		-0.07 (-0.53)	-0.08 (-0.73)	-0.06 (-0.59)	-0.18 (-0.95)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low B/M	0.01 (0.06)	0.11 (1.42)	0.09 (1.59)	-0.18*** (-3.66)	0.19 (1.54)	-0.08 (-0.80)	-0.11 (-1.33)	-0.00 (-0.05)	0.27*** (3.08)	-0.35*** (-2.59)
2	-0.03 (-0.29)	-0.07 (-1.08)	0.05 (0.77)	-0.02 (-0.29)	-0.00 (-0.01)	-0.03 (-0.32)	-0.01 (-0.11)	-0.10 (-1.30)	-0.22* (-1.94)	0.19 (1.37)
3	0.14 (1.27)	-0.01 (-0.11)	0.17*** (2.69)	-0.01 (-0.11)	0.15 (0.81)	-0.20** (-2.23)	0.03 (0.32)	-0.07 (-0.76)	-0.02 (-0.26)	-0.17 (-1.44)
High B/M	-0.04 (-0.29)	0.01 (0.11)	-0.20 (-1.54)	0.15 (0.69)	-0.20 (-0.92)	-0.06 (-0.54)	-0.19** (-2.23)	-0.08 (-0.84)	0.09 (0.58)	-0.14 (-0.79)
H-L	-0.05 (-0.27)	-0.10 (-0.67)	-0.28* (-1.91)	0.34 (1.47)		0.02 (0.19)	-0.08 (-0.73)	-0.08 (-0.71)	-0.18 (-0.97)	

Table A.11: Alphas for Size–OP–Inv portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size–OP–Inv portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the low-minus-high Inv alpha within each OP quartile, while the bottom row reports the high-minus-low OP alpha within each Inv quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Inv →	Small					Big				
	1	2	3	4	L-H	1	2	3	4	L-H
<i>Panel A: CAPM alpha</i>										
Low OP	-0.33 (-1.45)	-0.06 (-0.36)	-0.33* (-1.77)	-1.10*** (-5.73)	0.77*** (3.87)	-0.05 (-0.37)	-0.21* (-1.77)	-0.07 (-0.79)	-0.30** (-2.33)	0.25 (1.09)
2	0.14 (0.70)	0.19 (1.17)	0.17 (1.01)	-0.26 (-1.54)	0.40*** (2.81)	0.20 (1.47)	-0.09 (-0.83)	0.11 (0.78)	-0.22** (-2.23)	0.41** (2.40)
3	0.45** (2.01)	0.30* (1.89)	0.29* (1.82)	-0.01 (-0.04)	0.45*** (3.15)	0.36*** (2.83)	0.22* (1.85)	0.02 (0.28)	-0.07 (-0.65)	0.43** (2.27)
High OP	0.37* (1.87)	0.22 (1.09)	0.25 (1.25)	-0.03 (-0.17)	0.40*** (3.12)	0.36*** (3.41)	0.25** (2.26)	0.17* (1.94)	-0.00 (-0.04)	0.37* (1.94)
H-L	0.70*** (2.92)	0.28 (1.57)	0.58*** (3.08)	1.07*** (4.62)		0.42** (2.49)	0.46*** (2.78)	0.24* (1.77)	0.30* (1.87)	
<i>Panel B: FF5 alpha</i>										
Low OP	0.03 (0.26)	0.17** (2.03)	-0.11 (-1.01)	-0.48*** (-4.59)	0.51*** (3.12)	-0.11 (-0.93)	-0.14 (-1.29)	0.12 (1.46)	0.17** (2.10)	-0.27* (-1.72)
2	-0.07 (-0.64)	-0.01 (-0.14)	0.08 (1.21)	-0.20** (-2.14)	0.13 (0.99)	-0.14* (-1.72)	-0.22** (-2.20)	0.00 (0.02)	0.02 (0.23)	-0.16 (-1.21)
3	0.13 (1.21)	0.07 (0.90)	0.14** (2.18)	-0.04 (-0.46)	0.17 (1.25)	0.01 (0.07)	0.04 (0.36)	-0.12 (-1.48)	0.10 (1.00)	-0.10 (-0.80)
High OP	0.02 (0.22)	-0.13* (-1.86)	-0.03 (-0.51)	-0.11** (-1.97)	0.14 (1.21)	0.01 (0.11)	-0.05 (-0.62)	-0.03 (-0.46)	0.15 (1.10)	-0.14 (-0.73)
H-L	-0.01 (-0.04)	-0.30*** (-2.64)	0.07 (0.59)	0.37*** (3.65)		0.12 (0.74)	0.09 (0.59)	-0.15 (-1.34)	-0.02 (-0.12)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low OP	0.02 (0.15)	0.11 (1.25)	-0.13 (-1.27)	-0.50*** (-5.30)	0.52*** (3.23)	-0.05 (-0.47)	-0.09 (-0.87)	0.14* (1.65)	0.15* (1.89)	-0.20 (-1.30)
2	-0.02 (-0.22)	-0.02 (-0.30)	0.07 (0.93)	-0.18** (-2.00)	0.16 (1.25)	-0.13 (-1.53)	-0.20** (-2.02)	-0.06 (-0.52)	0.02 (0.20)	-0.15 (-1.07)
3	0.13 (1.08)	0.07 (0.85)	0.15** (2.54)	-0.01 (-0.17)	0.14 (0.96)	-0.00 (-0.02)	0.06 (0.53)	-0.15** (-1.99)	0.12 (1.12)	-0.12 (-0.94)
High OP	0.07 (0.61)	-0.11 (-1.49)	0.01 (0.13)	-0.08 (-1.48)	0.15 (1.27)	-0.04 (-0.44)	-0.07 (-0.88)	-0.07 (-0.83)	0.13 (0.98)	-0.17 (-0.94)
H-L	0.05 (0.24)	-0.22* (-1.80)	0.14 (1.13)	0.42*** (4.39)		0.01 (0.07)	0.02 (0.11)	-0.20 (-1.61)	-0.02 (-0.12)	

Table A.12: Alphas for Size-OP- χ portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size-OP- χ portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the high-minus-low χ alpha within each OP quartile, while the bottom row reports the high-minus-low OP alpha within each χ quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: CAPM alpha</i>										
Low OP	-0.70*** (-3.06)	-0.78*** (-2.75)	-0.49** (-2.24)	-0.43* (-1.83)	0.27 (0.95)	-0.10 (-0.55)	-0.51** (-2.35)	-0.35** (-2.12)	0.00 (0.03)	0.10 (0.44)
2	-0.19 (-0.86)	-0.17 (-0.81)	-0.05 (-0.28)	0.18 (0.93)	0.37** (2.26)	-0.05 (-0.30)	-0.06 (-0.38)	-0.03 (-0.31)	0.05 (0.38)	0.09 (0.57)
3	0.19 (0.96)	0.18 (0.87)	0.05 (0.30)	0.08 (0.46)	-0.11 (-0.80)	0.07 (0.48)	-0.16 (-1.03)	0.14 (1.07)	0.20 (1.53)	0.14 (0.83)
High OP	0.12 (0.56)	0.02 (0.08)	0.03 (0.19)	0.07 (0.38)	-0.05 (-0.37)	0.21 (1.49)	0.02 (0.15)	0.32*** (2.75)	0.09 (0.70)	-0.12 (-0.74)
H-L	0.81*** (3.76)	0.80*** (3.44)	0.52** (2.21)	0.50** (2.05)		0.31 (1.28)	0.53*** (2.21)	0.67*** (3.34)	0.08 (0.35)	
<i>Panel B: FF5 alpha</i>										
Low OP	-0.68*** (-3.76)	-0.44** (-2.01)	0.07 (0.72)	0.11 (0.90)	0.79*** (3.20)	-0.01 (-0.05)	-0.26 (-1.35)	0.13 (0.97)	0.26* (1.86)	0.26 (1.27)
2	-0.46*** (-4.24)	-0.21 (-1.48)	0.02 (0.20)	0.14 (1.32)	0.60*** (4.08)	-0.24 (-1.37)	0.12 (0.64)	0.12 (1.18)	-0.13 (-1.06)	0.11 (0.61)
3	-0.07 (-0.55)	0.09 (0.75)	0.04 (0.42)	-0.08 (-0.98)	-0.01 (-0.07)	-0.11 (-0.90)	0.12 (0.73)	0.11 (0.90)	-0.04 (-0.37)	0.07 (0.39)
High OP	-0.23** (-2.01)	-0.21* (-1.89)	-0.10 (-1.16)	-0.14 (-1.60)	0.09 (0.69)	-0.17 (-1.48)	-0.10 (-0.84)	0.28*** (2.83)	-0.23* (-1.94)	-0.06 (-0.36)
H-L	0.44** (2.39)	0.23 (1.34)	-0.17 (-1.47)	-0.26* (-1.65)		-0.16 (-0.74)	0.16 (0.81)	0.15 (0.83)	-0.48** (-2.48)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low OP	-0.50*** (-3.01)	-0.19 (-0.93)	0.07 (0.75)	-0.12 (-1.12)	0.37* (1.81)	0.23 (1.40)	-0.08 (-0.41)	0.13 (0.96)	0.14 (0.97)	-0.08 (-0.46)
2	-0.36*** (-3.52)	-0.08 (-0.60)	-0.02 (-0.25)	0.09 (0.88)	0.45*** (3.27)	-0.04 (-0.24)	0.24 (1.25)	0.06 (0.62)	-0.23* (-1.83)	-0.19 (-1.08)
3	0.04 (0.39)	0.18 (1.49)	0.04 (0.44)	-0.11 (-1.34)	-0.16 (-1.21)	0.04 (0.37)	0.26 (1.51)	0.08 (0.67)	-0.16 (-1.34)	-0.21 (-1.21)
High OP	-0.12 (-1.18)	-0.08 (-0.78)	-0.11 (-1.38)	-0.17* (-1.91)	-0.06 (-0.45)	-0.07 (-0.54)	-0.01 (-0.07)	0.22** (2.17)	-0.37*** (-2.79)	-0.30* (-1.72)
H-L	0.38* (1.95)	0.12 (0.63)	-0.18* (-1.69)	-0.05 (-0.35)		-0.29 (-1.35)	0.07 (0.35)	0.09 (0.52)	-0.51** (-2.48)	

Table A.13: Alphas for Size-B/M- χ portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size-B/M- χ portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the high-minus-low χ alpha within each B/M quartile, while the bottom row reports the high-minus-low B/M alpha within each χ quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: CAPM alpha</i>										
Low B/M	-0.24 (-1.46)	-0.33* (-1.86)	-0.25 (-1.56)	-0.23 (-1.37)	0.01 (0.07)	0.03 (0.23)	-0.25 (-1.51)	0.09 (0.65)	0.14 (1.29)	0.11 (0.75)
2	0.10 (0.50)	0.06 (0.31)	0.06 (0.36)	0.22 (1.21)	0.12 (0.96)	0.09 (0.79)	-0.04 (-0.32)	0.19* (1.67)	0.09 (0.86)	0.01 (0.06)
3	-0.11 (-0.48)	0.23 (0.86)	0.12 (0.69)	0.26 (1.07)	0.37* (1.92)	0.06 (0.36)	0.05 (0.33)	0.20 (1.40)	0.25* (1.79)	0.19 (1.25)
High B/M	0.24 (0.78)	-0.12 (-0.30)	0.09 (0.36)	0.01 (0.03)	-0.24 (-0.92)	0.16 (0.80)	-0.24 (-1.08)	-0.02 (-0.11)	0.19 (0.99)	0.03 (0.13)
H-L	0.48* (1.82)	0.21 (0.68)	0.34 (1.48)	0.24 (1.11)		0.13 (0.53)	0.01 (0.03)	-0.11 (-0.40)	0.05 (0.25)	
<i>Panel B: FF5 alpha</i>										
Low B/M	-0.40*** (-3.35)	-0.20* (-1.80)	0.08 (1.13)	0.08 (0.92)	0.48*** (2.99)	-0.10 (-0.86)	0.03 (0.25)	0.24*** (2.62)	-0.05 (-0.48)	0.05 (0.33)
2	-0.17 (-1.59)	-0.04 (-0.38)	0.07 (0.82)	0.13 (1.56)	0.30*** (2.70)	-0.10 (-0.85)	-0.12 (-0.83)	0.05 (0.42)	-0.10 (-0.93)	-0.01 (-0.05)
3	-0.38*** (-2.84)	0.14 (0.83)	0.02 (0.16)	0.17 (1.37)	0.55*** (2.63)	-0.21 (-1.41)	-0.08 (-0.50)	0.01 (0.05)	-0.03 (-0.26)	0.17 (0.96)
High B/M	-0.01 (-0.08)	-0.40 (-1.39)	0.01 (0.10)	-0.01 (-0.10)	0.00 (0.01)	-0.08 (-0.56)	-0.45*** (-2.77)	-0.18 (-1.24)	0.03 (0.18)	0.10 (0.48)
H-L	0.38* (1.92)	-0.20 (-0.82)	-0.07 (-0.45)	-0.10 (-0.62)		0.02 (0.13)	-0.48** (-2.55)	-0.41** (-2.50)	0.08 (0.43)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low B/M	-0.27*** (-2.66)	-0.06 (-0.53)	0.07 (1.03)	-0.07 (-0.79)	0.20 (1.56)	0.08 (0.64)	0.13 (1.07)	0.19** (1.98)	-0.20 (-1.64)	-0.27* (-1.69)
2	-0.07 (-0.72)	0.07 (0.65)	0.04 (0.41)	0.07 (0.85)	0.14 (1.38)	0.08 (0.75)	-0.06 (-0.47)	0.02 (0.18)	-0.21* (-1.78)	-0.29** (-2.07)
3	-0.29** (-2.41)	0.27* (1.71)	-0.03 (-0.30)	0.07 (0.58)	0.35** (2.07)	-0.03 (-0.23)	0.08 (0.47)	-0.06 (-0.48)	-0.09 (-0.66)	-0.05 (-0.30)
High B/M	0.14 (0.88)	-0.24 (-0.84)	-0.03 (-0.22)	-0.11 (-0.88)	-0.25 (-1.21)	0.07 (0.59)	-0.33** (-2.13)	-0.26* (-1.84)	-0.06 (-0.34)	-0.13 (-0.62)
H-L	0.41* (1.94)	-0.18 (-0.72)	-0.10 (-0.67)	-0.04 (-0.26)		-0.00 (-0.01)	-0.47** (-2.43)	-0.45*** (-2.71)	0.14 (0.70)	

Table A.14: Alphas for Size-Inv- χ portfolios across asset pricing models

US equities, July 1982–December 2024. This table reports results for the 32 Size-Inv- χ portfolios formed using NYSE breakpoints. Panels A–C report intercepts (α , in monthly percent) from time-series regressions on the CAPM, the FF5, and FF5 + SMQ based on the SIC 4-digit industry-level χ estimates, respectively. The final column in each size group reports the high-minus-low χ alpha within each Inv quartile, while the bottom row reports the high-minus-low Inv alpha within each χ quartile. All reported t -statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$\chi \rightarrow$	Small					Big				
	1	2	3	4	H-L	1	2	3	4	H-L
<i>Panel A: CAPM alpha</i>										
Low Inv	0.06 (0.26)	-0.20 (-0.76)	-0.10 (-0.46)	0.02 (0.10)	-0.04 (-0.16)	0.18 (1.18)	0.02 (0.12)	0.34** (2.27)	0.26** (2.48)	0.09 (0.55)
2	-0.02 (-0.10)	0.08 (0.41)	0.06 (0.39)	0.31 (1.53)	0.32* (1.91)	0.02 (0.15)	-0.01 (-0.06)	0.19* (1.89)	0.08 (0.59)	0.06 (0.37)
3	0.14 (0.69)	0.07 (0.35)	0.07 (0.42)	0.13 (0.83)	-0.01 (-0.07)	0.08 (0.69)	-0.11 (-0.88)	0.15 (1.52)	0.11 (1.00)	0.02 (0.16)
High Inv	-0.34* (-1.86)	-0.40* (-1.89)	-0.40** (-2.56)	-0.43** (-2.59)	-0.09 (-0.53)	-0.24** (-2.02)	-0.28 (-1.37)	-0.03 (-0.21)	0.08 (0.63)	0.32** (2.13)
L-H	0.40*** (2.70)	0.19 (1.06)	0.30* (1.66)	0.46*** (3.10)		0.42** (2.44)	0.30 (1.05)	0.38* (1.72)	0.19 (1.35)	
<i>Panel B: FF5 alpha</i>										
Low Inv	-0.27* (-1.85)	-0.21 (-1.14)	0.09 (0.71)	0.25** (2.13)	0.52** (2.49)	-0.17 (-1.31)	-0.27** (-2.35)	0.10 (0.88)	0.02 (0.16)	0.19 (1.18)
2	-0.26** (-2.49)	-0.12 (-1.23)	0.06 (0.79)	0.33*** (2.64)	0.59*** (3.08)	-0.19 (-1.57)	-0.19 (-1.41)	0.05 (0.51)	-0.24* (-1.73)	-0.05 (-0.28)
3	-0.16* (-1.79)	0.01 (0.10)	0.09 (1.23)	0.13 (1.51)	0.29*** (2.90)	-0.16 (-1.50)	-0.14 (-1.13)	0.09 (1.05)	-0.12 (-1.18)	0.04 (0.27)
High Inv	-0.42*** (-3.12)	-0.26* (-1.84)	-0.08 (-0.91)	-0.22** (-2.29)	0.20 (1.35)	-0.16 (-1.15)	0.23 (1.28)	0.37*** (2.69)	0.07 (0.58)	0.23 (1.48)
L-H	0.16 (1.29)	0.05 (0.31)	0.17 (1.03)	0.48*** (3.72)		-0.01 (-0.09)	-0.50** (-2.22)	-0.27 (-1.47)	-0.06 (-0.41)	
<i>Panel C: FF5 + SMQ alpha</i>										
Low Inv	-0.11 (-0.94)	-0.01 (-0.07)	0.05 (0.43)	0.10 (0.99)	0.21 (1.44)	-0.04 (-0.27)	-0.21* (-1.74)	0.03 (0.23)	-0.09 (-0.80)	-0.05 (-0.31)
2	-0.21** (-2.21)	-0.03 (-0.33)	0.03 (0.37)	0.21* (1.94)	0.41*** (2.73)	-0.02 (-0.17)	-0.12 (-0.86)	0.04 (0.41)	-0.35** (-2.36)	-0.33* (-1.69)
3	-0.04 (-0.51)	0.11 (0.92)	0.05 (0.69)	0.07 (0.84)	0.11 (1.28)	-0.01 (-0.09)	-0.09 (-0.75)	0.03 (0.31)	-0.26** (-2.14)	-0.25 (-1.63)
High Inv	-0.27** (-2.55)	-0.10 (-0.78)	-0.05 (-0.54)	-0.36*** (-3.63)	-0.09 (-0.80)	0.03 (0.25)	0.37** (2.11)	0.31** (2.31)	-0.09 (-0.64)	-0.12 (-0.75)
L-H	0.16 (1.30)	0.09 (0.52)	0.10 (0.64)	0.46*** (3.73)		0.07 (0.46)	-0.58*** (-2.67)	-0.28 (-1.63)	-0.01 (-0.06)	

A.4 Fama–French Five-Factor Alphas and Factor Loadings

Table A.15: Five-factor alphas and loadings for Size- χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the 5×5 Size- χ portfolios using the Fama–French Five-Factor model. At the end of each June, firms are independently sorted into five size portfolios and five portfolios based on the SIC 4-digit industry-level χ estimate, where 1 denotes the lowest- χ portfolio and 5 the highest- χ portfolio. The table reports intercepts (α_i , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i					b_i				
	1	2	3	4	5	1	2	3	4	5
Small	-0.13 (-0.74)	-0.26* (-1.67)	-0.10 (-1.02)	0.32*** (2.79)	0.24** (2.08)	1.02*** (36.91)	1.04*** (32.09)	1.03*** (35.00)	0.98*** (39.58)	0.96*** (26.15)
2	-0.29** (-2.54)	-0.33** (-2.03)	-0.23*** (-2.65)	0.29*** (3.63)	-0.10 (-1.04)	1.11*** (40.02)	1.15*** (28.63)	1.13*** (38.44)	1.03*** (42.52)	1.04*** (41.50)
3	-0.29*** (-2.65)	-0.16 (-1.05)	-0.16 (-1.43)	0.04 (0.58)	0.17* (1.88)	1.11*** (50.14)	1.16*** (39.94)	1.11*** (38.85)	1.07*** (53.23)	1.04*** (30.78)
4	-0.17* (-1.91)	-0.22 (-1.25)	-0.12 (-1.36)	0.30*** (3.57)	0.02 (0.20)	1.10*** (37.87)	1.16*** (28.59)	1.12*** (36.65)	1.04*** (52.94)	1.04*** (33.60)
Big	-0.13 (-1.33)	-0.04 (-0.27)	0.05 (0.75)	0.24*** (2.73)	-0.18** (-2.12)	0.98*** (32.66)	0.99*** (28.66)	0.99*** (42.17)	1.05*** (43.07)	0.93*** (33.54)
$\chi \rightarrow$	s_i					h_i				
	1	2	3	4	5	1	2	3	4	5
Small	1.10*** (19.29)	1.28*** (25.42)	1.11*** (17.43)	1.24*** (21.84)	1.14*** (25.63)	0.30*** (4.24)	0.31*** (3.94)	0.17** (2.02)	-0.36*** (-6.51)	-0.10 (-1.38)
2	0.97*** (21.62)	1.05*** (19.79)	0.95*** (15.66)	1.00*** (24.67)	0.87*** (14.17)	0.31*** (5.23)	0.21*** (3.06)	0.06 (0.69)	-0.32*** (-7.79)	-0.03 (-0.26)
3	0.63*** (14.51)	0.82*** (20.58)	0.62*** (9.15)	0.68*** (20.89)	0.70*** (22.41)	0.39*** (5.96)	0.18*** (2.80)	0.12 (1.14)	-0.27*** (-6.03)	-0.07 (-1.46)
4	0.35*** (7.79)	0.50*** (8.09)	0.35*** (5.95)	0.36*** (11.21)	0.33*** (6.28)	0.28*** (5.15)	0.18* (1.74)	0.01 (0.11)	-0.31*** (-6.42)	0.02 (0.20)
Big	-0.07 (-1.47)	-0.06 (-1.12)	-0.22*** (-5.84)	-0.25*** (-8.53)	-0.10** (-2.46)	0.08 (1.38)	0.00 (0.04)	-0.10 (-1.61)	-0.38*** (-11.18)	-0.08* (-1.89)
$\chi \rightarrow$	r_i					c_i				
	1	2	3	4	5	1	2	3	4	5
Small	0.17* (1.88)	-0.08 (-0.89)	-0.15** (-2.26)	-0.56*** (-8.63)	-0.50*** (-6.40)	0.14 (1.07)	0.01 (0.09)	0.03 (0.21)	0.26*** (3.85)	0.07 (0.74)
2	0.43*** (5.05)	0.23*** (2.79)	0.06 (0.56)	-0.35*** (-5.53)	0.01 (0.10)	-0.02 (-0.19)	0.01 (0.07)	0.09 (0.74)	0.05 (0.61)	0.09 (0.95)
3	0.38*** (4.76)	0.11 (1.58)	0.16 (1.49)	-0.19*** (-3.90)	0.08 (1.19)	-0.17* (-1.87)	-0.05 (-0.43)	-0.03 (-0.21)	-0.04 (-0.44)	0.00 (0.00)
4	0.25*** (3.11)	-0.07 (-0.67)	0.16 (1.47)	-0.24*** (-4.96)	0.18* (1.86)	-0.06 (-0.64)	0.00 (0.02)	0.15 (1.11)	-0.07 (-0.77)	0.02 (0.22)
Big	0.28*** (4.38)	-0.17 (-1.56)	0.09 (1.49)	-0.01 (-0.30)	0.31*** (3.62)	0.27*** (2.74)	-0.10 (-0.61)	0.00 (0.04)	0.07 (0.92)	0.34*** (3.18)

Table A.16: Five-factor alphas and loadings for small Size–OP– χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the small-firm 4×4 OP– χ portfolios under the Fama–French Five-Factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the small-firm group, firms are independently sorted into four operating profitability (OP) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 small-firm portfolios. Rows correspond to OP quartiles, with Low OP denoting the lowest operating profitability quartile and High OP the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low OP	-0.68*** (-3.76)	-0.44** (-2.01)	0.07 (0.72)	0.11 (0.90)	1.15*** (23.83)	1.23*** (24.93)	1.09*** (33.49)	1.02*** (26.32)
2	-0.46*** (-4.24)	-0.21 (-1.48)	0.02 (0.20)	0.14 (1.32)	1.11*** (39.09)	1.12*** (36.37)	1.01*** (36.93)	0.98*** (46.47)
3	-0.07 (-0.55)	0.09 (0.75)	0.04 (0.42)	-0.08 (-0.98)	1.04*** (34.55)	1.06*** (46.66)	1.03*** (49.17)	1.00*** (41.23)
High OP	-0.23** (-2.01)	-0.21* (-1.89)	-0.10 (-1.16)	-0.14 (-1.60)	1.10*** (37.07)	1.12*** (40.68)	1.05*** (39.46)	1.08*** (37.32)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low OP	1.06*** (15.24)	1.07*** (15.46)	1.03*** (28.21)	1.10*** (20.93)	0.28** (2.52)	0.17 (1.32)	-0.45*** (-9.28)	-0.45*** (-7.41)
2	0.85*** (15.05)	0.89*** (16.32)	0.94*** (19.87)	0.88*** (24.77)	0.46*** (6.12)	0.31*** (4.37)	-0.06 (-1.01)	0.01 (0.20)
3	0.90*** (13.26)	1.09*** (26.67)	1.00*** (25.87)	0.78*** (14.15)	0.30*** (3.29)	0.19*** (3.03)	-0.04 (-0.95)	0.07 (0.91)
High OP	0.93*** (16.72)	1.09*** (18.71)	0.94*** (22.57)	0.86*** (12.83)	0.38*** (4.96)	0.28*** (4.99)	0.04 (0.69)	0.20** (2.25)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low OP	-0.21* (-1.94)	-0.84*** (-7.11)	-0.88*** (-17.70)	-0.85*** (-9.08)	0.20 (1.28)	0.23 (1.23)	0.17* (1.65)	0.19 (1.64)
2	0.38*** (4.46)	0.06 (0.81)	-0.02 (-0.36)	0.13* (1.74)	0.02 (0.21)	-0.09 (-0.73)	0.05 (0.49)	0.08 (1.06)
3	0.51*** (5.75)	0.24*** (3.91)	0.13** (2.58)	0.35*** (4.59)	-0.06 (-0.56)	-0.06 (-0.50)	0.07 (0.91)	0.06 (0.82)
High OP	0.73*** (8.02)	0.57*** (8.46)	0.43*** (7.91)	0.52*** (7.25)	-0.17* (-1.88)	-0.19** (-2.00)	-0.10 (-1.44)	-0.15* (-1.95)

Table A.17: Five-factor alphas and loadings for big Size–OP– χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the big-firm 4×4 OP– χ portfolios under the Fama–French five-factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the big-firm group, firms are independently sorted into four operating profitability (OP) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 big-firm portfolios. Rows correspond to OP quartiles, with Low OP denoting the lowest operating profitability quartile and High OP the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low OP	-0.01 (-0.05)	-0.26 (-1.35)	0.13 (0.97)	0.26* (1.86)	1.15*** (20.87)	1.21*** (20.31)	1.12*** (35.68)	1.03*** (25.03)
2	-0.24 (-1.37)	0.12 (0.64)	0.12 (1.18)	-0.13 (-1.06)	1.07*** (21.24)	1.05*** (24.66)	1.03*** (28.32)	1.00*** (37.34)
3	-0.11 (-0.90)	0.12 (0.73)	0.11 (0.90)	-0.04 (-0.37)	0.97*** (29.57)	0.99*** (22.52)	1.03*** (27.85)	0.90*** (30.11)
High OP	-0.17 (-1.48)	-0.10 (-0.84)	0.28*** (2.83)	-0.23* (-1.94)	0.93*** (21.74)	1.01*** (37.43)	1.03*** (39.39)	0.88*** (24.96)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low OP	0.09 (1.23)	0.14 (1.21)	-0.04 (-0.48)	0.05 (0.84)	0.35** (2.32)	0.10 (0.80)	-0.07 (-0.74)	-0.15** (-2.10)
2	0.12** (2.22)	0.05 (0.66)	0.07 (1.16)	0.04 (0.86)	-0.08 (-0.97)	-0.21* (-1.82)	-0.14** (-2.07)	-0.02 (-0.26)
3	0.09 (1.20)	-0.03 (-0.46)	-0.14*** (-2.78)	-0.07* (-1.71)	0.21** (2.38)	0.18** (2.06)	-0.20*** (-3.38)	-0.17*** (-3.92)
High OP	0.05 (0.99)	-0.01 (-0.14)	-0.18*** (-3.94)	-0.08 (-1.38)	0.03 (0.38)	-0.27*** (-4.04)	-0.41*** (-7.61)	-0.14** (-2.39)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low OP	-0.35*** (-2.95)	-0.56*** (-3.81)	-0.70*** (-5.87)	-0.39*** (-4.09)	-0.08 (-0.45)	-0.02 (-0.11)	-0.44*** (-3.36)	-0.05 (-0.35)
2	0.17** (2.09)	-0.35** (-2.38)	-0.22*** (-2.87)	0.28*** (3.64)	0.45*** (3.14)	0.17 (0.90)	-0.01 (-0.07)	0.16* (1.77)
3	0.20*** (3.86)	-0.25** (-2.56)	0.24*** (4.14)	0.40*** (4.24)	0.08 (0.67)	-0.72*** (-3.24)	-0.10 (-0.76)	0.33*** (2.84)
High OP	0.59*** (7.54)	0.29*** (4.82)	0.22*** (4.20)	0.46*** (4.98)	0.29** (2.26)	0.17 (1.53)	0.13 (1.48)	0.42*** (3.78)

Table A.18: Five-factor alphas and loadings for small Size-B/M- χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the small-firm 4×4 B/M- χ portfolios under the Fama–French Five-Factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the small-firm group, firms are independently sorted into four book-to-market (B/M) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 small-firm portfolios. Rows correspond to B/M quartiles, with Low B/M denoting the lowest book-to-market quartile and High B/M the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low B/M	-0.40*** (-3.35)	-0.20* (-1.80)	0.08 (1.13)	0.08 (0.92)	1.08*** (39.05)	1.14*** (43.73)	1.01*** (45.09)	1.00*** (37.35)
2	-0.17 (-1.59)	-0.04 (-0.38)	0.07 (0.82)	0.13 (1.56)	1.06*** (43.67)	1.06*** (47.46)	1.03*** (53.64)	1.00*** (40.59)
3	-0.38*** (-2.84)	0.14 (0.83)	0.02 (0.16)	0.17 (1.37)	1.06*** (40.07)	1.07*** (40.53)	1.11*** (39.04)	1.04*** (35.79)
High B/M	-0.01 (-0.08)	-0.40 (-1.39)	0.01 (0.10)	-0.01 (-0.10)	1.17*** (41.82)	1.31*** (12.93)	1.16*** (28.68)	1.15*** (20.90)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low B/M	0.90*** (15.11)	1.07*** (24.30)	1.03*** (25.75)	0.98*** (25.41)	0.09 (1.13)	-0.09 (-1.23)	-0.41*** (-8.97)	-0.39*** (-8.37)
2	0.89*** (21.62)	0.94*** (21.14)	0.99*** (18.79)	0.84*** (19.39)	0.38*** (5.65)	0.28*** (5.21)	-0.10** (-2.56)	0.03 (0.45)
3	0.96*** (19.40)	1.08*** (24.65)	0.89*** (16.53)	1.04*** (24.36)	0.54*** (8.96)	0.38*** (5.53)	0.15* (1.73)	0.13** (2.31)
High B/M	1.00*** (18.09)	1.32*** (11.67)	1.01*** (16.66)	0.97*** (14.61)	0.73*** (9.18)	0.83*** (5.07)	0.39*** (4.59)	0.33*** (2.65)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low B/M	0.49*** (5.31)	-0.03 (-0.42)	-0.30*** (-6.27)	-0.32*** (-4.39)	-0.15* (-1.79)	-0.06 (-0.56)	-0.03 (-0.43)	0.04 (0.47)
2	0.44*** (6.07)	0.18*** (2.86)	0.01 (0.09)	0.14 (1.62)	0.01 (0.06)	-0.04 (-0.45)	0.25*** (2.79)	0.19** (2.17)
3	0.34*** (4.62)	0.03 (0.46)	0.08 (1.28)	0.05 (0.86)	0.02 (0.24)	0.09 (0.80)	0.26** (2.21)	0.28*** (3.18)
High B/M	0.15 (1.56)	0.12 (0.54)	-0.16* (-2.06)	-0.24** (-2.30)	0.08 (0.62)	0.17 (0.89)	0.30** (2.10)	0.30* (1.71)

Table A.19: Five-factor alphas and loadings for big Size-B/M- χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the big-firm 4×4 B/M- χ portfolios under the Fama–French five-factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the big-firm group, firms are independently sorted into four book-to-market (B/M) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 big-firm portfolios. Rows correspond to B/M quartiles, with Low B/M denoting the lowest book-to-market quartile and High B/M the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low B/M	-0.10 (-0.86)	0.03 (0.25)	0.24*** (2.62)	-0.05 (-0.48)	1.06*** (21.35)	1.06*** (33.69)	1.06*** (36.93)	0.87*** (31.39)
2	-0.10 (-0.85)	-0.12 (-0.83)	0.05 (0.42)	-0.10 (-0.93)	0.94*** (22.42)	1.03*** (32.45)	1.00*** (29.48)	0.96*** (27.04)
3	-0.21 (-1.41)	-0.08 (-0.50)	0.01 (0.05)	-0.03 (-0.26)	1.00*** (27.44)	0.99*** (22.35)	1.06*** (34.98)	1.03*** (28.80)
High B/M	-0.08 (-0.56)	-0.45*** (-2.77)	-0.18 (-1.24)	0.03 (0.18)	1.13*** (29.57)	1.09*** (30.48)	1.12*** (31.34)	1.07*** (34.90)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low B/M	0.05 (0.77)	-0.02 (-0.45)	-0.19*** (-5.70)	-0.09** (-1.99)	-0.30*** (-4.55)	-0.37*** (-4.69)	-0.43*** (-7.74)	-0.30*** (-6.24)
2	0.03 (0.42)	0.04 (0.80)	-0.01 (-0.30)	-0.04 (-0.83)	-0.03 (-0.28)	0.00 (0.05)	-0.12 (-1.62)	-0.06 (-1.01)
3	0.02 (0.34)	0.07 (1.00)	0.03 (0.55)	0.10** (2.34)	0.38*** (4.01)	0.19* (1.65)	0.04 (0.75)	0.08 (1.17)
High B/M	0.10** (2.00)	0.14 (1.57)	0.21*** (3.68)	0.20*** (3.20)	0.64*** (6.09)	0.56*** (5.08)	0.49*** (5.86)	0.32*** (3.53)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low B/M	0.35*** (3.86)	-0.12 (-1.51)	0.01 (0.25)	0.35*** (3.29)	0.16 (1.23)	-0.38** (-2.29)	-0.12 (-1.35)	0.34** (2.39)
2	0.24*** (2.88)	0.09 (1.22)	0.26*** (3.24)	0.25*** (3.64)	0.26** (1.97)	0.11 (1.02)	0.18* (1.87)	0.30*** (3.11)
3	0.21*** (2.64)	0.07 (0.80)	0.14 (1.26)	0.25*** (3.23)	0.17 (1.30)	0.15 (0.82)	0.36** (2.47)	0.48*** (5.28)
High B/M	0.00 (0.03)	0.11 (0.82)	0.11 (1.09)	0.11 (1.19)	0.16 (1.10)	-0.01 (-0.07)	-0.09 (-0.72)	0.08 (0.56)

Table A.20: Five-factor alphas and loadings for small Size–Inv– χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the small-firm 4×4 Inv– χ portfolios under the Fama–French five-factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the small-firm group, firms are independently sorted into four investment (Inv) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 small-firm portfolios. Rows correspond to investment quartiles, with Low Inv denoting the lowest investment quartile and High Inv the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low Inv	-0.27*	-0.21	0.09	0.25**	1.15***	1.23***	1.12***	1.06***
	(-1.85)	(-1.14)	(0.71)	(2.13)	(31.03)	(28.82)	(39.68)	(32.16)
2	-0.26**	-0.12	0.06	0.33***	1.07***	1.06***	1.04***	0.98***
	(-2.49)	(-1.23)	(0.79)	(2.64)	(38.69)	(53.22)	(42.84)	(20.01)
3	-0.16*	0.01	0.09	0.13	1.02***	1.04***	0.98***	1.00***
	(-1.79)	(0.10)	(1.23)	(1.51)	(39.26)	(51.19)	(48.15)	(44.94)
High Inv	-0.42***	-0.26*	-0.08	-0.22**	1.14***	1.16***	1.07***	1.06***
	(-3.12)	(-1.84)	(-0.91)	(-2.29)	(42.30)	(37.95)	(45.52)	(32.85)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low Inv	1.07***	1.09***	1.09***	1.03***	0.40***	0.33***	-0.30***	-0.25***
	(14.74)	(16.24)	(22.16)	(24.04)	(4.83)	(3.52)	(-4.29)	(-3.94)
2	0.84***	0.95***	0.90***	1.08***	0.44***	0.17***	0.03	-0.10
	(15.22)	(28.20)	(22.41)	(13.95)	(5.21)	(2.84)	(0.47)	(-1.20)
3	0.87***	1.00***	1.03***	0.79***	0.37***	0.13***	-0.10**	0.02
	(16.08)	(25.46)	(23.49)	(16.72)	(4.95)	(2.65)	(-2.15)	(0.32)
High Inv	1.00***	1.14***	0.98***	0.93***	0.18***	0.13*	-0.27***	-0.17
	(24.69)	(24.58)	(26.26)	(13.65)	(3.71)	(1.95)	(-5.72)	(-1.64)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low Inv	0.31***	-0.32***	-0.53***	-0.56***	0.41***	0.43***	0.68***	0.55***
	(3.01)	(-3.11)	(-6.96)	(-8.33)	(3.54)	(3.42)	(7.84)	(5.29)
2	0.34***	0.24***	0.01	-0.12	0.01	0.30***	0.13	0.41***
	(4.00)	(5.33)	(0.09)	(-1.23)	(0.04)	(3.02)	(1.42)	(3.04)
3	0.54***	0.17***	0.09	0.11*	-0.03	-0.00	0.09	-0.03
	(7.15)	(3.21)	(1.52)	(1.81)	(-0.33)	(-0.02)	(1.09)	(-0.34)
High Inv	0.31***	0.02	-0.23***	-0.07	-0.20**	-0.37***	-0.24***	-0.22**
	(3.83)	(0.29)	(-4.34)	(-0.63)	(-2.21)	(-3.21)	(-3.35)	(-1.99)

Table A.21: Five-factor alphas and loadings for big Size-Inv- χ portfolios

US equities, July 1982–December 2024. This table reports time-series regression results for the big-firm 4×4 Inv- χ portfolios under the Fama–French five-factor model. At the end of each June, firms are first assigned to two size groups using the NYSE median. Within the big-firm group, firms are independently sorted into four investment (Inv) portfolios and four adjustment-cost (χ) portfolios using NYSE quartile breakpoints. The intersections yield 16 big-firm portfolios. Rows correspond to investment quartiles, with Low Inv denoting the lowest investment quartile and High Inv the highest. Columns 1 to 4 denote increasing χ quartiles. The table reports intercepts (α , in monthly percent) and slope coefficients on $R_m - R_f$, SMB, HML, RMW, and CMA. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + e_{i,t}.$$

$\chi \rightarrow$	a_i				b_i			
	1	2	3	4	1	2	3	4
Low Inv	-0.17 (-1.31)	-0.27** (-2.35)	0.10 (0.88)	0.02 (0.16)	1.00*** (28.75)	0.99*** (25.51)	1.11*** (33.74)	0.97*** (24.62)
2	-0.19 (-1.57)	-0.19 (-1.41)	0.05 (0.51)	-0.24* (-1.73)	0.98*** (26.83)	0.98*** (28.97)	0.95*** (29.77)	0.95*** (27.23)
3	-0.16 (-1.50)	-0.14 (-1.13)	0.09 (1.05)	-0.12 (-1.18)	1.02*** (17.23)	1.03*** (28.82)	1.04*** (32.31)	0.92*** (32.36)
High Inv	-0.16 (-1.15)	0.23 (1.28)	0.37*** (2.69)	0.07 (0.58)	1.08*** (30.41)	1.09*** (30.78)	1.09*** (31.78)	0.94*** (30.90)
$\chi \rightarrow$	s_i				h_i			
	1	2	3	4	1	2	3	4
Low Inv	0.01 (0.14)	-0.00 (-0.02)	-0.13** (-2.48)	-0.09* (-1.83)	0.18 (1.65)	-0.07 (-0.65)	-0.28*** (-2.66)	-0.09 (-1.33)
2	0.08 (1.21)	-0.02 (-0.35)	-0.06 (-1.28)	-0.05 (-0.80)	0.20*** (3.17)	0.02 (0.23)	-0.18** (-2.55)	-0.01 (-0.08)
3	0.09* (1.68)	0.05 (0.92)	-0.17*** (-3.63)	-0.07 (-1.25)	0.07 (0.55)	0.06 (0.76)	-0.21*** (-3.29)	-0.06 (-0.98)
High Inv	0.08 (1.51)	0.12* (1.83)	-0.02 (-0.41)	0.03 (0.63)	-0.05 (-0.58)	-0.14 (-1.11)	-0.30*** (-5.54)	-0.27*** (-5.91)
$\chi \rightarrow$	r_i				c_i			
	1	2	3	4	1	2	3	4
Low Inv	0.25*** (2.72)	0.19 (1.57)	0.16 (1.31)	0.21*** (3.10)	0.57*** (4.70)	0.68*** (4.53)	0.77*** (5.37)	0.54*** (5.01)
2	0.21** (2.29)	0.17** (2.31)	0.21*** (3.74)	0.43*** (4.37)	0.19 (1.39)	0.31** (2.24)	0.29** (2.47)	0.40*** (4.94)
3	0.41*** (4.23)	0.09 (1.31)	0.13** (2.06)	0.34*** (3.95)	0.13 (0.94)	-0.08 (-0.71)	0.17** (2.20)	0.27** (2.11)
High Inv	-0.04 (-0.60)	-0.33*** (-4.94)	-0.22*** (-2.87)	0.19** (2.25)	-0.16 (-1.20)	-0.99*** (-3.91)	-0.68*** (-5.77)	-0.01 (-0.05)

A.5 Validation of Replicated Factors and Test Portfolios

Table A.22: Validation of five-factors against Kenneth French data

US equities, July 1982–December 2024. This table compares our replicated Fama–French five factors with the corresponding factors from Kenneth French’s data library. Columns refer to the internally constructed factors. Panel A reports the average monthly return, the mean difference relative to the Kenneth French factor, and the correlation with the corresponding Kenneth French factor. The mean difference is defined as the mean excess returns from our replicated factors minus the mean excess returns from the Kenneth French Library factors. Panel B reports the outcomes from time-series regressions in which each replicated constructed factor is regressed on its Kenneth French counterpart. All reported t-statistics are based on Newey–West standard errors with 12 lags, where statistical significance is denoted *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	Replicated Fama-French factor				
	$R_M - R_f$	SMB	HML	RMW	CMA
<i>Panel A: Mean excess returns, differences, and correlations</i>					
Mean	0.79	0.06	0.18	0.38	0.21
Mean difference	0.00	0.02	-0.01	-0.01	-0.01
Factor correlation	1.00	1.00	0.96	0.97	0.96
<i>Panel B: Replication regressions on Kenneth French factors</i>					
Alpha	0.00 (-0.59)	0.00 (1.47)	0.00 (-0.08)	0.00 (0.95)	0.00 (0.35)
	1.00***	1.01***	0.94***	0.91***	0.88***
Loading	(1326.72)	(117.72)	(57.68)	(46.70)	(34.77)
R^2	1.00	0.99	0.93	0.94	0.91

Table A.23: Validation of Size-B/M portfolios against Kenneth French data

US equities, July 1982–December 2024. This table compares the replicated constructed Size-B/M 5×5 portfolios with the corresponding portfolios from Kenneth French's Data Library. For each portfolio cell, the replicated test portfolio is regressed on the matched Kenneth French portfolio mean excess return. Panel A reports the regression intercepts in monthly percent, Panel B reports the slope coefficients, Panel C reports adjusted R^2 , and Panel D reports the time-series correlation between the two return series. All reported t -statistics are based on Newey–West standard errors with 12 lags. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Size-B/M					
	Low	2	3	4	High
<i>Panel A: Alpha, % per month</i>					
Small	0.05 (0.93)	0.15** (1.99)	0.08 (1.12)	0.05 (1.00)	-0.08 (-1.41)
2	0.06 (1.17)	0.06 (1.17)	0.08** (1.98)	0.17*** (3.61)	-0.17** (-2.50)
3	0.05 (1.08)	0.06 (1.34)	0.08* (1.82)	0.06 (1.12)	-0.07 (-1.02)
4	0.02 (0.51)	0.03 (0.90)	0.03 (0.49)	0.03 (0.56)	0.10 (1.51)
Big	-0.04** (-2.49)	0.06 (1.62)	0.11* (1.68)	0.22*** (3.02)	0.15 (1.53)
<i>Panel B: Loading on Kenneth French portfolio</i>					
Small	0.98*** (46.91)	0.91*** (27.66)	0.94*** (82.49)	0.99*** (65.29)	1.09*** (50.98)
2	0.96*** (122.38)	0.96*** (71.41)	0.97*** (114.46)	0.99*** (106.27)	1.03*** (50.76)
3	0.96*** (102.34)	0.96*** (77.20)	0.99*** (76.34)	0.94*** (65.27)	0.99*** (58.77)
4	0.98*** (64.88)	0.97*** (120.20)	1.01*** (71.38)	0.95*** (66.98)	0.94*** (35.71)
Big	1.01*** (158.16)	0.95*** (68.49)	0.97*** (48.02)	0.92*** (29.49)	0.77*** (29.41)
<i>Panel C: R^2</i>					
Small	0.97	0.92	0.95	0.97	0.97
2	0.97	0.95	0.97	0.97	0.96
3	0.98	0.97	0.96	0.95	0.94
4	0.98	0.98	0.96	0.95	0.91
Big	0.99	0.94	0.93	0.92	0.82
<i>Panel D: Correlation with Kenneth French portfolio</i>					
Small	0.98	0.96	0.98	0.99	0.99
2	0.99	0.98	0.98	0.98	0.98
3	0.99	0.98	0.98	0.98	0.97
4	0.99	0.99	0.98	0.97	0.96
Big	1.00	0.97	0.96	0.96	0.91

Table A.24: Validation of Size–OP portfolios against Kenneth French data

US equities: monthly value-weighted portfolio returns, July 1983–December 2024. This table compares the internally constructed Size–OP 5×5 portfolios with the corresponding portfolios from Kenneth French’s data library. Panel A reports intercepts from portfolio-level replication regressions, where each internally constructed portfolio is regressed on its Kenneth French counterpart. Alphas are reported in basis points per month, with Newey–West t -statistics using 12 lags shown in parentheses. Panel B reports the corresponding regression loading, with Newey–West t -statistics for the null hypothesis that the loading equals zero shown in parentheses. Panel C reports the adjusted R^2 , and Panel D reports the time-series correlation between the internally constructed and Kenneth French portfolio returns. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

		Size-OP			
	Low	2	3	4	High
Panel A: Alpha, % per month					
Small	0.05* (1.77)	0.00 (-0.02)	0.13 (1.57)	0.36 (1.50)	0.21*** (4.07)
2	0.10** (2.03)	-0.01 (-0.23)	-0.03 (-0.96)	0.07** (2.26)	0.02 (0.31)
3	0.07 (1.31)	0.01 (0.35)	-0.05 (-1.56)	-0.02 (-0.56)	0.10* (1.96)
4	-0.04 (-0.75)	0.06 (1.30)	0.08* (1.89)	-0.05 (-1.10)	0.03 (1.05)
Big	0.06 (1.20)	0.07* (1.81)	-0.06 (-1.26)	-0.05 (-1.31)	0.03 (1.61)
Panel B: Loading on Kenneth French portfolio					
Small	1.00*** (125.92)	0.98*** (98.26)	1.04*** (43.94)	0.61*** (2.84)	0.97*** (55.86)
2	0.97*** (91.57)	0.96*** (78.19)	1.01*** (98.27)	0.97*** (82.79)	0.99*** (78.08)
3	0.97*** (85.63)	0.95*** (95.12)	1.00*** (99.54)	0.99*** (127.44)	0.95*** (125.18)
4	0.95*** (38.17)	0.92*** (49.44)	1.01*** (113.87)	1.00*** (117.10)	1.00*** (106.11)
Big	0.95*** (62.30)	0.90*** (44.53)	0.99*** (63.44)	1.02*** (134.12)	0.99*** (192.47)
Panel C: R ²					
Small	0.99	0.98	0.79	0.60	0.96
2	0.98	0.97	0.97	0.98	0.97
3	0.98	0.97	0.98	0.98	0.97
4	0.97	0.95	0.95	0.97	0.98
Big	0.97	0.94	0.95	0.97	0.99
Panel D: Correlation with Kenneth French portfolio					
Small	1.00	0.99	0.89	0.78	0.98
2	0.99	0.99	0.99	0.99	0.99
3	0.99	0.98	0.99	0.99	0.99
4	0.98	0.98	0.98	0.99	0.99
Big	0.98	0.97	0.97	0.99	1.00

Table A.25: Validation of Size–Inv portfolios against Kenneth French data

US equities: monthly value-weighted portfolio returns, July 1983–December 2024. This table compares the internally constructed Size–Inv 5×5 portfolios with the corresponding portfolios from Kenneth French’s data library. Panel A reports intercepts from portfolio-level replication regressions, where each internally constructed portfolio is regressed on its Kenneth French counterpart. Alphas are reported in basis points per month, with Newey–West t -statistics using 12 lags shown in parentheses. Panel B reports the corresponding regression loading, with Newey–West t -statistics for the null hypothesis that the loading equals zero shown in parentheses. Panel C reports the adjusted R^2 , and Panel D reports the time-series correlation between the internally constructed and Kenneth French portfolio returns. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

		Size-Inv			
	Low	2	3	4	High
Panel A: Alpha, % per month					
Small	-0.02 (-0.51)	-0.04 (-1.14)	-0.03 (-0.94)	0.03 (1.12)	0.13* (1.85)
2	0.00 (0.04)	-0.02 (-0.42)	-0.02 (-0.62)	0.00 (-0.12)	0.14** (2.15)
3	-0.02 (-0.40)	0.04 (1.16)	0.01 (0.41)	0.03 (1.12)	0.05 (0.77)
4	-0.08** (-1.99)	-0.01 (-0.19)	0.01 (0.34)	0.02 (0.83)	0.09* (1.65)
Big	0.11 (1.33)	-0.01 (-0.39)	0.04 (1.52)	0.00 (0.10)	0.01 (0.25)
Panel B: Loading on Kenneth French portfolio					
Small	1.01*** (133.74)	1.00*** (91.19)	0.97*** (65.45)	1.01*** (104.15)	0.97*** (88.77)
2	1.03*** (95.81)	0.98*** (103.61)	1.00*** (97.14)	0.98*** (115.11)	0.95*** (62.89)
3	0.99*** (104.72)	1.00*** (154.57)	0.98*** (160.39)	0.97*** (122.30)	0.95*** (60.27)
4	0.99*** (81.43)	0.99*** (112.96)	0.99*** (108.84)	0.98*** (125.81)	0.96*** (53.63)
Big	0.98*** (85.40)	1.01*** (112.82)	0.98*** (119.31)	0.97*** (81.64)	0.98*** (86.56)
Panel C: R ²					
Small	0.99	0.98	0.98	0.98	0.97
2	0.98	0.98	0.98	0.98	0.98
3	0.98	0.98	0.98	0.98	0.98
4	0.98	0.98	0.98	0.98	0.98
Big	0.93	0.98	0.98	0.98	0.98
Panel D: Correlation with Kenneth French portfolio					
Small	1.00	0.99	0.99	0.99	0.98
2	0.99	0.99	0.99	0.99	0.99
3	0.99	0.99	0.99	0.99	0.99
4	0.99	0.99	0.99	0.99	0.99
Big	0.96	0.99	0.99	0.99	0.99

Table A.26: Validation of Size–OP–Inv portfolios against Kenneth French data

US equities: monthly value-weighted portfolio returns, July 1983–December 2024. This table compares the internally constructed Size–OP–Inv $2 \times 4 \times 4$ portfolios with the corresponding portfolios from Kenneth French’s data library. Rows correspond to operating profitability quartiles and columns correspond to investment quartiles, reported separately for small and big firms. Panel A reports intercepts from portfolio-level replication regressions, where each internally constructed portfolio is regressed on its Kenneth French counterpart. Alphas are reported in basis points per month, with Newey–West t -statistics using 12 lags shown in parentheses. Panel B reports the corresponding regression loading, with Newey–West t -statistics for the null hypothesis that the loading equals zero shown in parentheses. Panel C reports the adjusted R^2 , and Panel D reports the time-series correlation between the internally constructed and Kenneth French portfolio returns. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Inv →	Small				Big			
	Low	2	3	High	Low	2	3	High
<i>Panel A: Alpha, % per month</i>								
Low OP	-0.03 (-0.67)	-0.06 (-1.00)	0.02 (0.39)	0.01 (0.24)	-0.01 (-0.24)	0.09 (1.44)	0.00 (-0.01)	0.12** (1.99)
2	0.05 (0.65)	0.01 (0.17)	0.02 (0.34)	0.03 (0.64)	0.09* (1.89)	-0.11* (-1.74)	0.15 (1.57)	-0.06 (-1.09)
3	0.10 (1.15)	-0.01 (-0.24)	0.02 (0.49)	0.00 (0.05)	0.09 (0.89)	0.00 (-0.00)	0.02 (0.32)	-0.02 (-0.35)
High OP	-0.06 (-0.88)	0.02 (0.39)	0.02 (0.49)	0.03 (0.75)	0.06 (1.60)	0.05 (0.81)	0.05 (1.21)	-0.03 (-0.41)
<i>Panel B: Loading on Kenneth French portfolio</i>								
Low OP	1.02*** (105.24)	0.98*** (53.92)	0.96*** (37.46)	0.95*** (62.92)	0.93*** (49.35)	0.88*** (27.03)	0.93*** (33.51)	0.93*** (41.28)
2	0.97*** (68.64)	0.98*** (60.85)	0.95*** (98.75)	0.96*** (74.01)	0.95*** (59.57)	0.99*** (48.03)	0.98*** (23.08)	0.98*** (60.67)
3	1.01*** (42.30)	0.99*** (75.92)	1.01*** (89.36)	0.98*** (83.37)	1.01*** (38.65)	0.96*** (27.13)	0.95*** (31.82)	0.97*** (41.75)
High OP	1.00*** (74.99)	0.98*** (83.45)	0.97*** (67.97)	0.97*** (152.30)	0.97*** (81.05)	0.94*** (61.76)	0.98*** (95.98)	1.00*** (58.23)
<i>Panel C: R^2</i>								
Low OP	0.98	0.95	0.96	0.97	0.95	0.93	0.92	0.97
2	0.92	0.96	0.95	0.96	0.91	0.92	0.90	0.93
3	0.92	0.96	0.97	0.97	0.85	0.88	0.91	0.93
High OP	0.95	0.96	0.96	0.98	0.96	0.92	0.96	0.96
<i>Panel D: Correlation with Kenneth French portfolio</i>								
Low OP	0.99	0.98	0.98	0.99	0.98	0.96	0.96	0.98
2	0.96	0.98	0.98	0.98	0.95	0.96	0.95	0.97
3	0.96	0.98	0.99	0.98	0.92	0.94	0.95	0.97
High OP	0.97	0.98	0.98	0.99	0.98	0.96	0.98	0.98

Table A.27: Validation of Size–B/M–Inv portfolios against Kenneth French data

US equities: monthly value-weighted portfolio returns, July 1983–December 2024. This table compares the internally constructed Size–B/M–Inv $2 \times 4 \times 4$ portfolios with the corresponding portfolios from Kenneth French’s data library. Rows correspond to book-to-market quartiles and columns correspond to investment quartiles, reported separately for small and big firms. Panel A reports intercepts from portfolio-level replication regressions, where each internally constructed portfolio is regressed on its Kenneth French counterpart. Alphas are reported in basis points per month, with Newey–West t -statistics using 12 lags shown in parentheses. Panel B reports the corresponding regression loading, with Newey–West t -statistics for the null hypothesis that the loading equals zero shown in parentheses. Panel C reports the adjusted R^2 , and Panel D reports the time-series correlation between the internally constructed and Kenneth French portfolio returns. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

Inv →	Small				Big			
	Low	2	3	High	Low	2	3	High
<i>Panel A: Alpha, % per month</i>								
Low B/M	-0.05 (-0.74)	0.06 (1.08)	0.04 (1.00)	0.11** (2.07)	0.05 (0.54)	-0.01 (-0.29)	0.04 (0.88)	-0.04 (-0.75)
2	-0.12 (-1.62)	-0.08* (-1.88)	0.07 (1.34)	0.21** (2.27)	0.14 (1.62)	0.06 (0.99)	0.07 (1.20)	-0.06 (-0.72)
3	0.09 (0.86)	0.03 (0.60)	0.02 (0.35)	0.09 (1.13)	-0.01 (-0.21)	0.09 (1.37)	0.03 (0.35)	0.29** (2.24)
High B/M	-0.04 (-0.47)	-0.06 (-0.98)	-0.07 (-0.79)	0.22 (1.46)	0.09* (1.72)	0.09* (1.73)	0.12* (1.66)	0.35*** (3.19)
<i>Panel B: Loading on Kenneth French portfolio</i>								
Low B/M	1.01*** (62.83)	0.99*** (59.88)	0.98*** (70.01)	0.93*** (66.67)	0.99*** (49.93)	0.97*** (72.68)	0.97*** (85.09)	0.97*** (47.32)
2	0.97*** (47.02)	0.99*** (69.44)	0.98*** (93.44)	0.91*** (31.77)	0.98*** (48.70)	0.94*** (38.54)	0.96*** (65.82)	0.93*** (42.24)
3	0.98*** (53.63)	1.00*** (127.68)	0.97*** (49.84)	0.96*** (48.78)	0.97*** (55.14)	0.97*** (52.02)	0.97*** (55.87)	0.93*** (23.38)
High B/M	0.99*** (21.00)	0.99*** (47.65)	0.99*** (43.06)	1.06*** (24.33)	0.94*** (58.28)	0.93*** (65.64)	0.89*** (51.88)	0.91*** (24.31)
<i>Panel C: R^2</i>								
Low B/M	0.95	0.96	0.97	0.98	0.89	0.94	0.97	0.97
2	0.91	0.95	0.96	0.90	0.85	0.88	0.93	0.87
3	0.91	0.95	0.95	0.91	0.92	0.90	0.89	0.83
High B/M	0.94	0.94	0.88	0.79	0.95	0.95	0.93	0.82
<i>Panel D: Correlation with Kenneth French portfolio</i>								
Low B/M	0.97	0.98	0.99	0.99	0.95	0.97	0.98	0.99
2	0.96	0.98	0.98	0.95	0.92	0.94	0.96	0.93
3	0.95	0.98	0.97	0.96	0.96	0.95	0.95	0.91
High B/M	0.97	0.97	0.94	0.89	0.98	0.98	0.97	0.91

Table A.28: Validation of Size–B/M–OP portfolios against Kenneth French data

US equities: monthly value-weighted portfolio returns, July 1983–December 2024. This table compares the internally constructed Size–B/M–OP $2 \times 4 \times 4$ portfolios with the corresponding portfolios from Kenneth French’s data library. Rows correspond to book-to-market quartiles and columns correspond to operating profitability quartiles, reported separately for small and big firms. Panel A reports intercepts from portfolio-level replication regressions, where each internally constructed portfolio is regressed on its Kenneth French counterpart. Alphas are reported in basis points per month, with Newey–West t -statistics using 12 lags shown in parentheses. Panel B reports the corresponding regression loading, with Newey–West t -statistics for the null hypothesis that the loading equals zero shown in parentheses. Panel C reports the adjusted R^2 , and Panel D reports the time-series correlation between the internally constructed and Kenneth French portfolio returns. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

OP →	Small				Big			
	Low	2	3	High	Low	2	3	High
<i>Panel A: Alpha, % per month</i>								
Low B/M	0.07 (1.41)	0.03 (0.48)	-0.02 (-0.30)	0.02 (0.69)	-0.06 (-0.63)	0.13 (1.44)	-0.03 (-0.49)	-0.02 (-0.91)
2	-0.10 (-1.30)	0.05 (0.78)	0.04 (0.81)	0.06 (0.77)	0.00 (0.03)	-0.07 (-1.16)	0.01 (0.20)	0.33*** (3.86)
3	0.09 (1.12)	0.09 (1.53)	0.04 (0.78)	0.22** (2.50)	0.17* (1.92)	0.04 (0.53)	0.10 (1.28)	0.27** (2.17)
High B/M	0.00 (-0.04)	0.01 (0.14)	0.20* (1.72)	0.08 (0.37)	0.19*** (3.46)	0.05 (0.72)	0.09 (0.83)	0.57** (2.32)
<i>Panel B: Loading on Kenneth French portfolio</i>								
Low B/M	0.97*** (105.87)	0.96*** (69.13)	0.99*** (89.41)	0.97*** (151.78)	0.88*** (28.41)	0.92*** (46.14)	0.99*** (71.00)	1.00*** (148.66)
2	0.95*** (74.67)	0.92*** (56.98)	1.00*** (53.76)	0.97*** (42.44)	0.89*** (29.33)	0.95*** (45.90)	0.98*** (66.68)	0.86*** (28.29)
3	0.96*** (58.06)	0.97*** (58.82)	1.00*** (59.83)	0.89*** (33.91)	0.94*** (31.55)	0.97*** (54.71)	0.95*** (32.13)	0.86*** (18.96)
High B/M	1.03*** (37.75)	0.98*** (52.59)	0.97*** (26.74)	0.91*** (22.18)	0.87*** (38.87)	0.94*** (46.95)	0.95*** (42.42)	0.44*** (2.89)
<i>Panel C: R^2</i>								
Low B/M	0.98	0.94	0.97	0.99	0.90	0.88	0.93	0.98
2	0.93	0.94	0.96	0.94	0.81	0.89	0.92	0.77
3	0.92	0.96	0.95	0.87	0.84	0.90	0.87	0.77
High B/M	0.94	0.93	0.89	0.77	0.94	0.90	0.80	0.39
<i>Panel D: Correlation with Kenneth French portfolio</i>								
Low B/M	0.99	0.97	0.98	0.99	0.95	0.94	0.96	0.99
2	0.97	0.97	0.98	0.97	0.90	0.94	0.96	0.88
3	0.96	0.98	0.97	0.94	0.92	0.95	0.93	0.88
High B/M	0.97	0.96	0.94	0.88	0.97	0.95	0.90	0.63